SMART POULTRY FARMING: IOT- BASED ENVIRONMENTAL QUALITY INDEX MONITORING WITH MACHINE LEARNING FOR OPTIMIZING CHICKEN GROWTH IN NASUGBU, BATANGAS

A Capstone Project Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology
Specialized In Service Management Program

LORD CYRIL O. DE LUNA
SHENELYN MIRANDA
TRINA P. ARUTA

JOCELYN O. PADALLAN
JUNE 2026

APPROVAL SHEET

The Capstone Project entitled "**SMART POULTRY FARMING: IOT-BASED ENVIRONMENTAL QUALITY INDEX MONITORING WITH MACHINE LEARNING FOR OPTIMIZING CHICKEN GROWTH IN NASUGBU, BATANGAS"**, prepared and submitted by **Trina P. Aruta, Lord Cyril O. De Luna** and **Shenelyn Miranda** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program, is hereby recommended for approval and acceptance.

JOCELYN O. PADALLAN
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____.

MIKE PHILIP T. RAMOS
Member

SHERWIN B. SAPIN
Member

JHOCO S. MILLANES
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, Ed. D.
Chairperson, Research and Development Services

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGMENT

Deepest gratitude is offered to Almighty God for granting wisdom, strength, and perseverance throughout the completion of this project. His guidance and blessings made the successful realization of this study possible.

Sincere appreciation is extended to the Aruta family, especially Mr. Jimmy Aruta, for his invaluable assistance in constructing the chicken pen and for his dedication in caring for the chickens from the first day until harvest. Gratitude is also given to Tita Gladys for generously allowing the chicken coop to be established and maintained beside her home.

Heartfelt thanks are conveyed to the De Luna family, particularly Ms. Roderica De Luna, for warmly welcoming the team into their home and providing a comfortable place to stay. Her delicious meals, genuine hospitality, and constant motivation greatly inspired continued effort toward achieving the project's goals.

Gratitude is likewise expressed to the Miranda family, especially Ms. Sheila Miranda, for allowing the use of her home as a workspace, providing comfort, and serving meals that sustained productivity throughout the project. Special thanks are also given to Ms. Paulina Miranda for her invaluable guidance, unwavering support, and financial assistance, which significantly contributed to the successful completion of this study.

Sincere appreciation is given to the research adviser, Ms. Jocelyn O. Padallan, for her guidance, constructive feedback, and encouragement that helped shape the direction of this research. Acknowledgment is also extended to the Faculty of the College of Computer Studies, Laguna State Polytechnic

University–Los Baños Campus, for the knowledge and skills imparted throughout the academic journey.

Grateful recognition is given to the panelists: Mr. Mike Philip T. Ramos, Subject Specialist, for his expertise and insightful recommendations; Dr. Sherwin B. Sapin, Statistician, for his valuable guidance in the accurate analysis and interpretation of data; and Mr. Jhoco S. Millanes, Technical Panelist, for his technical advice, constructive feedback, and continuous support in improving the system.

Acknowledgment is also extended to Mr. Jefferson L. Lerios, Research Implementing Unit Head, for his leadership, support, and encouragement throughout the development of the study.

Appreciation is extended to friends who provided continuous support throughout the process. Special thanks are given to Michaela for her constant companionship during paperwork, to Chynna for her guidance and encouragement in improving the project, and to Kuya Fyke for his time, free consultations, and valuable suggestions that helped enhance and refine the system.

Finally, special thanks are extended to Mr. and Mrs. Mercado, owners of Mercado's Poultry Farm in Nasugbu, Batangas, for their trust and cooperation in allowing the testing and implementation of the system.

THE RESEARCHERS

DEDICATION

This humble work is lovingly dedicated to all the individuals who supported the researchers throughout their academic journey.

The researchers express their deepest gratitude to their parents and families for their unconditional love, sacrifices, and constant encouragement. Their unwavering faith provided the strength needed to overcome challenges and turn hardships into success.

Sincere appreciation is extended to the mentors and professors whose guidance and wisdom played a vital role in shaping the researchers not only academically but also as individuals prepared for the professional world.

Endless gratitude is also given to their beloved partners Kate Restar, Mark Anthony Marin, and John Michael Bariring for their love, patience, and unwavering support. Their encouragement and belief served as a constant source of motivation and inspiration throughout the journey.

Above all, this work is humbly offered to Almighty God, the ultimate source of strength, wisdom, and guidance. Through His grace and blessings, the successful completion of this research was made possible.

T.P.A.
L.C.O.D.L.
S.M.

ABSTRACT

This study was developed to help poultry farmers address the challenges of maintaining healthy farm conditions. Changes in temperature, humidity, and ammonia levels often cause stress and disease among chickens, leading to high mortality and lower productivity. To respond to this problem, an IoT-based system was designed using sensors that record environmental data in real time. The collected data are processed by a microcontroller, which calculates an Environmental Quality Index (EQI). A machine learning model then analyzes the patterns and predicts the most suitable conditions for chicken growth. The system also includes a web-based dashboard that shows real-time readings, trends, and alerts to help farmers make better decisions. The system was tested at Mercado's Poultry Farm in Nasugbu, Batangas, and was evaluated using the Technology Acceptance Model (TAM) and ISO 25010 software quality standards. Findings showed that the system is functional, reliable, and user-friendly, proving its potential to reduce poultry losses and improve farm efficiency. Additionally, the system minimizes manual monitoring by providing automated data collection and continuous environmental assessment, helping farmers save time and reduce human error. Its predictive capability enables early detection of unfavorable conditions, allowing timely adjustments that support healthier chicken growth. The implementation also demonstrated that integrating IoT and machine learning can enhance decision-making and streamline farm operations. Furthermore, the study highlights the practicality of adopting technology-driven solutions in local poultry settings, offering farmers a useful tool to optimize productivity. Through these innovations, the system supports improved farm management practices and contributes to the broader goal of creating more efficient and sustainable agricultural environments.

Keywords: *Smart poultry farming, Internet of Things (IoT), Environmental Quality Index (EQI), Machine Learning*

TABLE OF CONTENTS

Preliminaries	***Page***
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	v
Abstract	vi
Table of Contents	vii
List of Figures	ix
List of Tables	x
Definition of Terms	xii

CHAPTER

I	**INTRODUCTION**	
	Project Context	1
	Project Objective	3
	Project Purpose	4
	Scope and Limitation	4
	Conceptual Framework	8
II	**REVIEW OF RELATED LITERATURE AND STUDIES**	
	Related Literatures	10
	Synthesis and Relevance of the Studies	18
	Literature Matrix	20
III	**METHODOLOGY**	
	Project Design	21
	System Architecture	23
	Schematic	23
	Smart Poultry Farming Prototype	25
	Evolutionary Prototyping Model	26
	Block Diagram	29
	Project Locale	30
	Model Development	31
	Software Development	35
	Materials	36
	Software	37
	Project Development	41
	Agile Scrum Methodology	42
	Scope	42
	Design	43
	Product Backlogs	43
	Sprint Planning for Admin	46
	Daily Scrum	48
	Sprint Execution	48
	Sprint Review Meeting	48

Sprint Retrospective 49
Project Testing and Evaluation 49
Software Testing 49
Unit Testing 51
Integration Testing 52
System Testing 52
Acceptance Testing 53
DHT22 Temperature and Humidity Sensor Calibration 54
Ammonia Sensors Calibrations 54
Environmental Quality Requirements 55
Target Quality Requirements 55
System Flowchart 57
Use Case Diagram 60
Evaluation Procedure 60
Population of the Study 61
Data Collection Instrument 63
Parameter Evaluation and EQI Formula 64

IV RESULTS AND DISCUSSIONS

Image of Smart Farming Sensor 67
ESP8266 Microcontroller and Relay 68
Chicken Pen Used for Implementation 69
Prediction of Environmental Condition 70
Smart Farming Web-Based System 72
Test and Evaluate the Software using ISO 25010 73
Results of Survey Answered by the Respondents 98
Result of Software Testing 106
Integration Testing Result 108
Load Testing Result 110
Compatibility Testing Result 111
Result of Stress Test 113
Environmental Data Analysis 115

V SUMMARY, CONCLUSION, AND RECOMMENDATIONS

Summary 123
Conclusion 124
Recommendations 127

REFERENCES
APPENDICES

Appendix A. Output Screenshot 133
Appendix B. Figures 139
Appendix C. Relevant Source Code 143
Appendix D. Communications Letters and Approvals 149
Appendix E. Plagiarism Results 157
Appendix F. Curriculum Vitae 160
Appendix H. Documentation 164
Appendix G. Publishable Paper 166

LIST OF FIGURES

No.	Title	Page
1	Conceptual Framework	9
2	System Architecture	23
3	Schematic Diagram of Sensor	24
4	Schematic Diagram of Relay	25
5	Prototype Design of Smart Poultry Farming	26
6	Evolutionary Prototyping Model of Smart Poultry Farm	28
7	Block Diagram	29
8	Owner of Mercado's Farm	31
9	ARIMA Model	33
10	Agile Scrum Methodology	42
11	Levels of Testing	50
12	Unit Testing	51
13	Integration Testing	52
14	System Testing	53
15	User Acceptance Testing	54
16	System Flowchart of Smart Farming System	58
17	System Flowchart of Hardware	59
18	Used Case Diagram	60
19	Quality Characteristics of ISO 25010 Standard	61
20	Smart Farming Sensor	67
21	ESP8266 Microcontroller Connected to Relay	68
22	Chicken Pen used in the implementation	69
23	Prediction Trends	71
24	System Dashboard & Prediction	73
25	Temperature, Humidity, and Ammonia Level Trends	121

LIST OF TABLES

No.	Title	Page
1	Literature Matrix	20
2	ARIMA Model Parameters	35
3	Hardware Requirements Used for Laptop/Computer	37
4	Hardware Specifications	41
5	Product Backlog for Admin	45
6	Main Product Backlog	46
7	Sprint Planning for Admin Features	47
8	Required Environmental Conditions	55
9	Threshold Levels Based on Chicken Age	56
10	EQI Threshold Levels	57
11	Respondents of the Research Project	62
12	Functional Suitability based on IT Specialist Response	74
13	Performance Efficiency based on IT Specialist Response	75
14	Compatibility based on IT Specialist Response	76
15	Interaction Capability based on IT Specialist Response	78
16	Reliability based on IT Specialist Response	79
17	Security based on IT Specialist Response	81
18	Maintability based on IT Specialist Response	82
19	Flexibility based on IT Specialist Response	83
20	Safety based on IT Specialist Response	84
21	Result of ISO 25010 based on IT Specialist Response	85
22	Functional Suitability based on Veterinarian’s Response	86
23	Performance Efficiency based on Veterinarian’s Response	87
24	Compatibility based on Veterinarian’s Response	88
25	Interaction Capability based on Veterinarian’s Response	90
26	Reliability based on Veterinarian’s Response	91
27	Security based on Veterinarian’s Response	92
28	Maintability based on Veterinarian’s Response	94
29	Flexibility based on Veterinarian’s Response	95
30	Safety based on Veterinarian’s Response	96

31	Result of ISO 25010 based on Veterinarian's Response	97
32	Perceived Usefulness based on Respondents' Response	98
33	Perceived Ease of Used based on Respondents' Response	99
34	Personal Innovation based on Respondents' Response	100
35	Social Influence based on Respondents' Response	101
36	Information Awareness based on Respondents' Response	102
37	Perceived Cost based on Respondents' Response	103
38	Willingness to Adopt based on Respondents' Response	104
39	Result of Technology Acceptance Model Response	106
40	Unit Testing for Web-Based Module	107
41	Integration Testing for Web-Based Module	108
42	Load Testing Result	110
43	Compatibility Testing Result	111
44	Stress Testing Procedure	113
45	Environmental Quality Index Data Recorded	115
46	Survivability Rate of Chicken Batch 1	116
47	Survivability Rate of Chicken Batch 2	118
48	Mortality Comparison Before and After Using Sensors	119
49	Sensor Thresholds and System Trigger Responses	122

DEFINITION OF TERMS

Actuator	A device, such as a fan or light, that the system controls to physically adjust the chicken environment based on sensor data.
Alert Status	A setting that determines whether the system should send an immediate notification warning to the farmer when critical environmental thresholds are breached.
Ammonia (NH_3)	Ammonia is a gas with a strong odor that is commonly produced in poultry houses from chicken waste. In this study, ammonia levels are measured using the MQ-137 sensor as part of computing the Environmental Quality Index (EQI).
Archive	The systematic storage of historical data, particularly records related to harvested chickens or old logs, for future analysis and compliance.
Batch Number	A unique identifier assigned to a specific group of chickens being raised together, allowing their growth data to be tracked from placement to harvest.
Dashboard	A dashboard is an interface that presents important data in visual form. In this project, it refers to the web-based platform that displays sensor readings, EQI values, and alerts for farmers.
Environmental Quality Index	EQI is a measure that summarizes different environmental factors into a single value to assess overall conditions. In this study, it combines temperature, humidity, and ammonia levels to determine if the poultry house is suitable for chicken growth.
Harvest	The action of removing chickens from the farm once they have reached their optimal market growth weight.

Humidity	Humidity is the amount of water vapor in the air, expressed as a percentage. It is measured using the DHT22 sensor in this project to help keep the poultry environment at healthy levels.
Internet Of Things (IOT)	IoT is a network of devices connected through the internet that can collect and share data. In this research, IoT is applied through sensors and the NodeMCU ESP8266 to monitor poultry conditions in real time.
ISO 25010	ISO 25010 is an international standard that defines software quality in terms of functionality, usability, and reliability. It was used in this project to assess the quality of the developed system.
Logs	The continuous, time-stamped records of environmental measurements and system activities, including sensor data and chicken counts.
Machine Learning (ML)	ML is a branch of artificial intelligence that allows computers to learn patterns from data and make predictions. In this study, it is used to analyze environmental readings and predict conditions for optimal chicken growth.
Mercado's Poultry Farm	Mercado's Poultry Farm is a poultry farm located in Nasugbu, Batangas, and served as the testing site for this study.
NodeMCU ESP8266	The NodeMCU ESP8266 is a microcontroller with Wi-Fi capability often used in IoT systems. Here, it collects sensor data, processes it, and sends it to the web-based dashboard.
Pin Code	A security code required to authenticate the user, granting access to the system's dashboard and settings.
Prediction	The output from the machine learning model that forecasts future environmental conditions or suggests optimal settings for maximizing chicken growth.

Sensor	A device that detects and converts physical environmental parameters like temperature, humidity, and gas into electrical signals for the IoT system to process.
Technology Acceptance Technology (TAM)	Data and information whose collection has just been completed and sent without any waiting time.
Temperature	Temperature is the measure of heat in the environment, expressed in degrees Celsius. It is monitored by the DHT22 sensor to ensure proper conditions for chicken comfort and growth.

CHAPTER I

INTRODUCTION

Project Context

Poultry farming continues to be a crucial part of the agricultural industry in the Philippines, supporting local economies and providing an essential source of protein to the population. In 2023, Amy and Crenpina Mercado established a small poultry farm with just 300 broiler chickens. As the farm expanded and began producing eggs and dressing chickens, it became increasingly clear that maintaining ideal environmental conditions was critical to the health and survival of their flock. Managing factors such as temperature, humidity, and gas levels became a daily challenge, especially as the number of chickens increased.

A major concern in poultry farming is the high mortality rate of chickens caused by poor environmental conditions. When temperature, humidity, and ammonia levels rise or fall beyond safe thresholds, chickens become vulnerable to stress, disease, and even death. These environmental fluctuations are often difficult to track manually, especially in larger-scale operations, leading to delayed responses and potential losses. Relying on human observation alone is not only inefficient but also unreliable, particularly when conditions can change rapidly.

To address this issue, the proposed project focuses on developing a hardware-based solution using Internet of Things (IoT) technology. The system will consist of sensors that continuously monitor temperature, humidity, and

ammonia concentrations inside the poultry house. These sensors will be connected to a microcontroller that processes and transmits data in real time. The primary goal is to provide an automated, real-time monitoring system that can detect environmental changes early and trigger appropriate responses, such as turning on ventilation fans or activating cooling systems.

Recent advancements in digital agriculture have demonstrated the transformative impact of emerging technologies such as the Internet of Things (IoT) and machine learning (ML) in enhancing poultry health and welfare. According to Ojo et al. (2022), the integration of sensor-based monitoring systems and AI-driven analytics has enabled real-time tracking of environmental conditions and early detection of diseases in poultry farming. These innovations not only improve animal welfare but also optimize productivity and farm management practices. The growing body of research in this field highlights a global shift toward smarter, data-driven approaches in agriculture, underscoring the relevance and necessity of technology adoption in modern poultry operations.

Supporting this, a study conducted at Ilocos Sur Polytechnic State College developed an IoT-based Decision Support System for poultry care that monitors environmental factors such as temperature, humidity, and ammonia level in real time. The system provides early warnings to improve poultry health and farm productivity. User evaluations showed high satisfaction, highlighting the system's effectiveness and ease of use. This research demonstrates how IoT can enhance decision-making and sustainability in poultry farming.

By using affordable, scalable IoT hardware in line with these global trends, this project aims to reduce poultry mortality, improve growing conditions, and increase the overall productivity and sustainability of poultry farms.

Project Objectives

This capstone project aims to develop an innovative and integrated solution to meet the changing needs of modern poultry farming. By leveraging Internet of Things (IoT) technology and intelligent software tools, the system is designed to optimize daily farm operations and enhance environmental monitoring. Through real-time data collection, automation, and data-driven insights, the project seeks to improve the overall efficiency, health conditions, and productivity of poultry farms. The following objectives outline the core goals and functionalities the system is intended to achieve:

1. Develop a smart IoT system to monitor temperature, humidity, and ammonia in real time, and calculate an Environmental Quality Index (EQI).

2. Implement a machine learning model using pre-trained data to predict optimal environmental conditions for chicken growth based on environmental and performance data.

3. Build a user-friendly software platform to display real-time data and EQI trends insights for better farm management.

4. Evaluate the system using TAM and ISO 25010 software product standards to verify its efficacy and quality.

Project Purpose

The purpose of this project, "Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth," is to help improve the way poultry farms are managed by using smart technology. It focuses on creating a system that helps farmers keep track of important environmental factors like temperature, humidity, and air quality using IoT sensors. By collecting this data in real time, the system calculates an Environmental Quality Index (EQI), which gives farmers a clear idea of how healthy and comfortable the environment is for their chickens. On top of that, the system uses machine learning to analyze trends and suggest the best conditions for chicken growth. This helps farmers make better decisions, reduce manual work, and take early action when needed.

Scope and Limitation

This project focuses on introducing a smart poultry farming system at Mercado's Poultry Farm in Nasugbu, Batangas. It aims to explore how using IoT devices like sensors that track temperature, humidity, and ammonia can help farmers better monitor the environment inside their poultry houses. The data collected used to calculate an Environmental Quality Index (EQI), which gives farmers a simple way to understand whether the living conditions for their chickens are healthy or need improvement. The project also includes a machine learning component that looks at patterns in the data and helps predict the best conditions for chicken growth. This should help farmers make better decisions, react quickly to changes, and ultimately raise healthier flocks with less guesswork. That said, there are a few limitations to consider. Since the project

is being tested only in Nasugbu, the results might not apply to other areas with different weather, infrastructure, or farming conditions. Also, the system depends on reliable internet and electricity, things that aren't always guaranteed in rural areas. Budget limitations may also affect how many sensors can be used, which could limit how detailed the monitoring can be. Finally, this study focuses on smaller farms and doesn't go into more advanced technologies like full-scale AI analytics or automation used in large commercial operations.

PIN Code Authentication Module. This module acts as the main security layer of the system, ensuring that only authorized users can access the smart poultry farming dashboard. To enter, users must provide a pre-set Personal Identification Number (PIN). For added protection, the system limits access to only three login attempts. This feature helps prevent unauthorized access, safeguards sensitive information, and ensures that only trained personnel can adjust settings or view operational data. Additionally, an easy-to-use interface allows users to update or change the PIN code within the settings, maintaining both convenience and security.

Dashboard Module. The dashboard serves as the central interface that provides a comprehensive overview of the poultry farm's real-time data. It displays critical information such as temperature, humidity, ammonia levels, chicken status graph and overall Environmental Quality Index (EQI) scores. The dashboard is designed to be user-friendly, offering graphical representations for better understanding and quicker decision-making. This module provides

actionable insights to the farm operator, enabling efficient monitoring and management of the environment within poultry enclosures.

Monitoring and Daily Logs Module. The Monitoring and Daily Logs Module keeps track of daily poultry records and environmental data. Based on the interface, it allows users to record key information such as the date, batch number, age stage (e.g., chick), the number of live and dead chickens, and remarks like “Routine Check.” These entries help maintain organized farm records, making it easier to monitor flock health and performance. Users can also search, print, export, or manage records directly from the dashboard. By combining manual input with system monitoring, the module ensures accurate tracking of poultry status and supports better farm management decisions.

Sensor Log Module. The Sensor Logs module provides a crucial record of environmental data, specifically displaying Temperature, Humidity, and Ammonia levels over time. As seen in the interface, users can switch between a Daily Logs view and a more detailed Sensor Logs view, which includes Date & Time stamps for granular tracking. This detailed logging is essential for verifying data accuracy and integrity and is vital for system maintenance by enabling operators to quickly identify performance anomalies like sensor malfunctions or disconnections, allowing for prompt troubleshooting to maintain system reliability.

Inventory Management Module. The Inventory Management Module allows farm personnel to manually input the number of live and dead chickens on a regular basis. This data entry is typically done daily to ensure accurate tracking of the flock. Each entry includes the number of live chickens, the

number of mortalities, and the corresponding age of the chickens in days. This module provides a real-time snapshot of the farm's poultry inventory and directly supports the Monitoring and Daily Logs Module, which references these inputs for analysis and historical record-keeping. By enabling manual inventory updates, the system ensures flexibility and accuracy, especially in situations where automatic tracking is not feasible. This module is essential for maintaining operational awareness and supporting decision-making related to feeding, harvesting, and mortality management.

Notification and Alerts Module. The Notification and Alerts Module act as the system’s real-time warning center, alerting users when temperature, humidity, or ammonia levels exceed safe limits. It provides sound and visual alerts for quick response and also sends SMS notifications when Wi-Fi is unavailable or users are away from the farm. This ensures continuous monitoring and fast action to maintain a safe poultry environment.

Settings and Configuration Module. The Settings Module allows users to manage system preferences and device configurations. It includes options to update the PIN code, control fans and lights, and manage notification and SMS alert settings. Users can also input an SMS API token, check the connection status of sensors and actuators, and configure alert settings and batch numbers. This module ensures proper system management, secure access, and continuous monitoring of both hardware and environmental conditions. In addition, the module allows users to adjust system parameters based on farm requirements and monitor device performance, helping ensure that all components remain functional and the system operates efficiently.

Conceptual Framework

The conceptual framework of the Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning illustrates the flow from sensor data collection to user interaction for effective poultry environment management. The process begins with real-time data gathered from sensors such as the MQ137 for ammonia levels and the DHT22 for temperature and humidity, which are connected to the ESP8266 microcontroller for wireless data transmission. Additional components, including the relay module for fan and light control and the I2C LCD for local display, support environmental regulation and on-site monitoring.

This system allows farmers to detect environmental changes immediately and take corrective actions to prevent stress or illness in poultry. It also reduces the reliance on manual observation and ensures continuous data tracking. The integration of IoT technology provides a scalable solution that can be expanded as farm operations grow.

Farmers can customize alert thresholds and notifications according to their specific farm requirements. Historical data storage enables trend analysis, helping to improve long-term decision-making. The framework supports multiple users, allowing collaborative farm management.

Security features ensure that only authorized personnel can access the system. Regular updates to the software and algorithms ensure that predictions remain accurate and relevant. The framework also encourages energy-efficient operations by optimizing fan and light usage. Data-driven insights from the

system can contribute to sustainable poultry farming practices, improving both productivity and animal welfare.

The system's Processing Module receives and manages this data by performing collection, IoT gateway edge processing, and EQI calculation. Machine learning algorithms are then applied to analyze conditions and provide predictive insights for maintaining optimal air quality and poultry growth. The processed results are transmitted to the web system, where they are made accessible to the user.

Through the Output Module, farmers can view real-time visualizations via a web-based dashboard, manage actuator controls for fans and lights, and receive alerts and notifications in case of abnormal conditions. The system also provides EQI predictions, enabling proactive decision-making. Overall, this framework ensures a continuous cycle of monitoring, analysis, and action, designed to support healthier poultry environments and improved productivity.

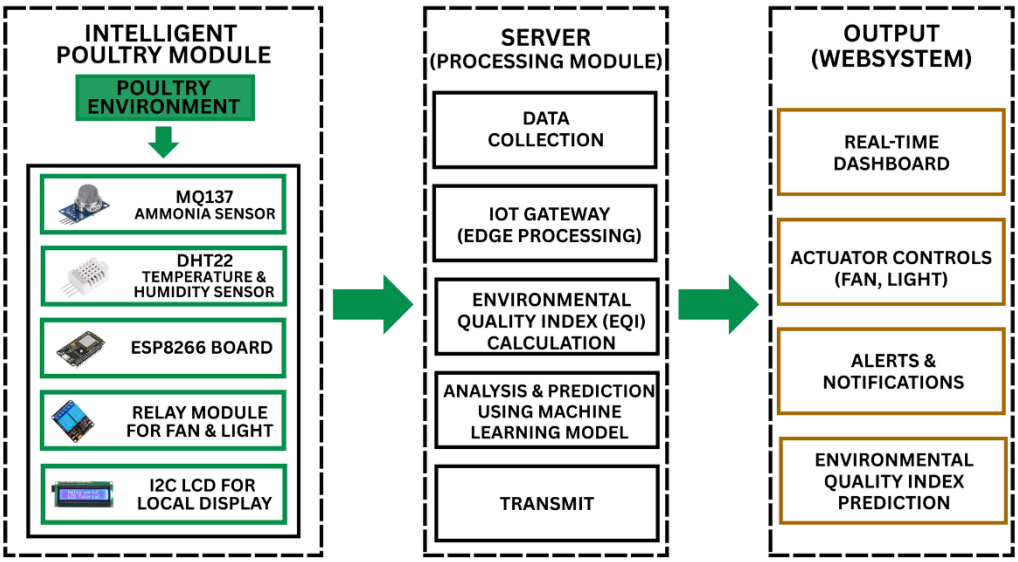


Figure 1. Conceptual Framework

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

This chapter presents a comprehensive review of related literature and studies, both local and foreign, found in reliable sources, including hardbound thesis books, online articles, blogs, and other trustworthy published materials. It will serve as the researchers' guide in conducting the study. Additionally, it connects and examines several earlier studies that will help in explaining and understanding the significance of the study.

IoT in Smart Farming and Poultry Management

The adoption of Internet of Things (IoT) in agriculture has significantly transformed traditional farming practices, enabling real-time monitoring, automation, and data-driven decision-making. (Navarro et al. ,2020), identified critical technologies such as cloud-based platforms, wireless sensors, communication protocols, and predictive analytics. Their research highlights how the Internet of Things has progressed from reactive systems to proactive and predictive analytics, enabling farmers to monitor environmental conditions, optimize resources, and improve production. Building upon this, (Digambar et al. ,2023) expanded on this by concentrating on IoT and Machine Learning (ML) applications in poultry farming. Their study highlights how important functions like feeding, watering, and environmental monitoring may be automated with the help of smart sensors and AI-driven systems. By leveraging Wireless Sensor Networks (WSN), poultry farmers can maintain optimal conditions for poultry health and production. Additionally, ML algorithms reduce human labor

and increase efficiency by predicting disease outbreaks, optimizing resource utilization, and enhancing overall poultry welfare.

Further advancing the integration of AI-IoT in poultry farming. (A.S. George & A.S.H. George ,2023) explored advanced monitoring and control systems to optimize poultry production. Their study demonstrates how intelligent systems with real-time tracking can automatically modify feeding, ventilation, lighting, and temperature in accordance with the stages of poultry growth. AI and computer vision are also helpful in the early detection of disease or injury, enabling preventative treatment and lowering death rates. The NybSys Smart Poultry Farm Automation and Monitoring System expands on the use of AI and IoT in poultry farming by providing a complete solution that turns standard poultry operations into smart, effective, and sustainable businesses. This system uses a network of sensors to continuously monitor environmental parameters in poultry houses, such as temperature, fire, light intensity, dust, and gas sensors. Data collected from these sensors are transmitted via the NybSys iG100 IoT Gateway to a cloud-based dashboard, enabling real-time monitoring and control. By automating critical functions such as climate control, feeding, and lighting, the system reduces operational costs, conserves energy, and enhances overall productivity. (NybSys, 2024).

IoT in Philippine Poultry toward Smart Environmental Monitoring

Building upon the challenges discussed in the Philippine poultry industry. One such advancement is the development of a smart poultry prototype tailored for small-scale farmers in Ilocos Norte. Simon and Palaoag (2024) introduced

a prototype designed to automatically regulate critical environmental factors within poultry houses, such as ammonia gas levels and temperature.

This system uses sensors to track temperature and humidity, turning on blowers and heat lamps as necessary. It also reduces the need for physical intervention by automating the timely distribution of food and drink. This method seeks to improve chicken productivity and quality by providing a consistent and healthy environment, which could increase small-scale farmers' profitability. Nevertheless, solutions have become increasingly significant. The Department of Science and Technology (DOST) has allocated over PHP 4.7 million to support the "ChicIoT" project, which aims to improve chicken farming using IoT-based environmental monitoring. (Arayata, 2021).

The "ChicIoT" project, a collaboration between the University of Southeastern Philippines and Tetra Consulting Corporation. which focuses on installing devices and sensors inside chicken farms to track important environmental factors like temperature, humidity, dust concentration, and light intensity. Farm owners can access the real-time data collected by these sensors through a user-friendly dashboard. Farmers may make well-informed decisions and optimize conditions to support healthier and more productive flocks because of this instant access to environmental data.

Applications of IoT and Hardware Technologies in Poultry Production

Recent advancements in poultry farming have leveraged IoT and machine learning technologies to improve farm management and animal health. Padua, Malubay, and Tandingan Jr. (2024) developed an IoT-based smart poultry farming system that uses wireless sensors and automated

controls to regulate environmental factors such as temperature and ammonia levels, enhancing poultry welfare and productivity.

Similarly, a 2020 study published in *Sensors* by an international team of researchers emphasized the integration of IoT platforms to improve data collection, traceability, and biosecurity throughout the poultry production chain, fostering better operational efficiency. Furthermore, research by Martinez et al. (2023) explored the use of machine learning on chicken behavioral data to detect early signs of disease outbreaks, providing a non-invasive and timely method for disease management in poultry farms. Collectively, these studies demonstrate the growing role of smart technologies in creating more efficient, sustainable, and health-conscious poultry farming practices.

Smart IoT Solutions for Sustainable Poultry Farming in the Philippines

With the advancement of technology, the use of the Internet of Things (IoT) in poultry farming is transforming traditional methods of raising chickens. By enabling real-time monitoring and automated control of the farm environment, this technology aims to improve poultry health, increase productivity, and reduce labor costs. Recent studies have highlighted the benefits and design of IoT-based smart poultry systems.

Intellectual Property Office of the Philippines (IPOPHL) presents an IoT-based smart poultry system that utilizes sensors to monitor environmental factors such as temperature, humidity, and ammonia levels inside the poultry house. This system automates feeding, lighting, and ventilation to maintain optimal conditions for the chickens, resulting in healthier livestock and more efficient farm operations.

Meanwhile, Ali, S., et al. (2023), published on TechRxiv, details the design and implementation of an IoT-enabled poultry farm management system. It collects sensor data to automate environmental controls and provides alerts to farmers when intervention is needed. The results demonstrate improvements in animal health, reduction in manual labor, and increased productivity.

Environmental Factors Affecting Chicken Growth

Environmental conditions such as temperature, humidity, ventilation, and air quality significantly influence poultry health and productivity. The National Research Council (1994) emphasizes that "environmental factors are generally recognized to have a major impact on the production of meat and eggs from poultry." These factors can affect feed intake, growth rates, and overall well-being of the birds. For instance, inadequate ventilation can lead to the accumulation of harmful gases like ammonia, adversely affecting respiratory health and growth performance.

Moreover, environmental stressors can compromise the immune system of chickens, making them more susceptible to diseases. A study by Lara and Rostagno (2013) highlights that "heat stress is one of the most important environmental stressors challenging poultry productions worldwide." Such stress can lead to reduced feed intake, lower weight gain, and increased mortality rates.

Importance of Temperature Monitoring in Poultry Farms

Maintaining optimal temperature is crucial for poultry health and productivity. According to the Vencomatic Group (2023), "when inside

temperatures rise above 30°C (86°F), heat stress can set in for your birds. This stress can lead to lower feed intake, decreased egg production, and even death in extreme cases." Heat stress not only affects the physiological functions of chickens but also their behavior, leading to increased panting and reduced activity.

Implementing temperature monitoring systems can help in early detection of unfavorable conditions, allowing for timely interventions. A study by Sinha et al. (2020) developed a smart temperature control system for poultry farms, concluding that "maintaining an optimal thermal environment in poultry farms enhances productivity and welfare while reducing economic losses." Such systems can automate temperature regulation, ensuring consistent and favorable conditions for poultry growth.

Role of Humidity in Poultry Health and Productivity

Humidity levels within poultry houses play a significant role in bird health and performance. High humidity can exacerbate the effects of ammonia, leading to respiratory issues. Greengage (2023) reports that "high humidity levels not only aggravate the adverse effects of ammonia but also lead to a significant imbalance between Th1 and Th2 cells, a situation detrimental to the immune response in poultry." This imbalance can result in heightened inflammation and weakened disease resistance.

Conversely, low humidity can cause dust accumulation, leading to respiratory problems. Maintaining optimal humidity levels is essential for litter quality, as excessive moisture can lead to wet litter, promoting the growth of

pathogens. Regular monitoring and control of humidity are therefore vital for ensuring poultry welfare and productivity.

Environmental Quality Index (EQI) in Livestock Farming

The Environmental Quality Index (EQI) is a comprehensive measure that assesses various environmental factors affecting health and productivity. Developed by the U.S. Environmental Protection Agency, the EQI incorporates data from five domains: air, water, land, built environment, and sociodemographic factors. According to the EPA (2014), "the EQI was designed to be used in two main ways: to represent 'environmental quality' in research designed to assess the relationship between environmental quality and human health outcomes and as a variable to account for confounding in models of environmental exposures.

In livestock farming, the EQI can help identify areas where environmental conditions may be impacting animal health and productivity. By analyzing EQI data, farmers and researchers can implement targeted interventions to improve environmental conditions, thereby enhancing livestock performance and welfare.

Comparison of Traditional vs. Smart Poultry Farming Methods

The integration of smart technologies in poultry farming has revolutionized traditional practices. A comparative study by Kumar et al. (2021) found that "the better health performance of poultry birds found in the smart management system of poultry farming as compare to the semi-automation management system and manual management system." Smart systems offer

automated control over environmental parameters, feeding, and health monitoring, leading to improved productivity and reduced labor costs

Traditional farming methods, while cost-effective in terms of initial investment, often lack the precision and efficiency of smart systems. The same study noted that manual management systems faced "major obstacles due to the limitations in use of automated management of poultry farming," resulting in poorer health performance and higher incidences of disease. Adopting smart technologies can thus enhance poultry welfare and farm profitability.

Prediction Analytics for Poultry House Environmental Conditions

The study of Ji and Teng (2025) focused on predicting temperature and humidity in poultry houses, since these factors play a big role in the health and productivity of chickens. They developed a model called the GFF-transformer, which combines a graph Fourier filter with a transformer to improve multistep prediction of environmental conditions. The results showed that this model performs better than traditional methods like LSTM and standard transformer models, making the predictions more accurate and stable. This research is important because it can help poultry farmers manage the environment inside poultry houses more effectively, leading to healthier flocks and improved efficiency in farm operations. In a similar way, prediction analytics can also be applied to human environments. For example, forecasting temperature, humidity, and air quality in workplaces, homes, or hospitals can help maintain comfort, prevent health risks, and support overall well-being. Just as poultry benefit from a controlled environment, humans also rely on accurate predictions of these conditions to avoid heat stress, respiratory problems, or discomfort,

showing that the same predictive technologies can be adapted for both animal and human needs.

In their 2024 study, Küçüktopçu, Cemek, and Simsek addressed the challenge of accurately estimating ammonia levels in poultry farms, a gas that poses risks to both animal welfare and human health. They developed hybrid models that combine machine learning techniques such as Random Forest, KNN, and ELM with wavelet transform processing to better capture fluctuating environmental signals. Their best model achieved very low prediction error and a strong correlation with actual ammonia concentrations, especially when using simple environmental inputs like temperature, humidity, and airflow. This research has broader implications beyond poultry farms, as ammonia exposure is hazardous in any poorly ventilated environment. By improving real-time prediction of ammonia levels, such techniques could help protect farm workers, nearby communities, and even people in enclosed settings such as greenhouses, waste treatment facilities, or animal handling areas by providing early warnings or enabling control systems that reduce exposure and improve air quality.

Synthesis and Relevance of the Study

Smart technology, especially IoT, is changing the way poultry farms are managed. With tools that track things like temperature, humidity, and ammonia in real time, farmers can now keep a much closer eye on the conditions their chickens live in. Studies have shown that this kind of monitoring leads to better health for the birds and helps make daily farm operations more efficient.

Many traditional poultry farms still face issues like disease outbreaks and poor air quality, which can hurt both the animals and the farm's productivity. That’s where technology makes a difference. By using IoT along with machine learning, farmers can better understand what their chickens need and spot problems before they get worse. It’s not just about collecting data it’s about turning that data into useful insights.

This study matters because it brings all of these ideas together in a system built specifically for small to mid-sized farms like Mercado’s Poultry. By combining smart sensors and data analysis, it offers a practical and affordable way to improve chicken growth and farm management. More importantly, it provides a model that other farms can follow, helping push poultry farming toward a more modern, sustainable future.

Additionally**,** the system can generate automated alerts and predictive insights, allowing farmers to respond immediately to potential issues, such as temperature spikes or rising ammonia levels. By doing so, it minimizes the risks of disease outbreaks and stress-related losses among chickens. This proactive approach also reduces labor costs and ensures a safer environment for both animals and farmworkers.

Moreover**,** the study provides a framework for other farms to adopt IoT technology in an affordable and scalable way. By documenting environmental conditions and analyzing patterns in real time, the system can help farmers make data-driven decisions that improve productivity, reduce mortality, and optimize operational efficiency.

Literature Matrix

Table 1. Literature Matrix

Author(s) & Year	Title	Purpose	Methodology	Key Findings	Relevance to Capstone
Chang & Tu (2023)	*Use of IoT Sensors to Build an Intelligent Monitoring and Control System for Poultry House Environment*	To develop an intelligent system for real-time monitoring and control of poultry house environments using IoT sensors.	Implemented IoT sensors to monitor temperature, humidity, and gas concentrations; data analyzed for environmental control.	The system effectively maintained optimal environmental conditions, enhancing poultry health and reducing disease risks.	Demonstrates the practical application of IoT in maintaining optimal poultry environments, supporting the development of an EQI system
Orakwue et al. (2021)	*IoT Based Smart Monitoring System for Efficient Poultry Farming*	To design a smart system using IoT for monitoring and controlling environmental conditions in poultry farming.	Utilized ESP32 microcontroller with sensors for temperature, humidity, and gas detection; data transmitted to cloud for analysis.	Improved monitoring led to reduced mortality rates and enhanced poultry production efficiency.	Highlights the benefits of IoT in real-time environmental monitoring, aligning with the capstone's objectives.
Singh & Kajla (2024)	*Optimizing Ammonia Concentration in Poultry Houses through Random Forest Prediction and LoRa-WAN Monitoring & Controlling System*	To optimize ammonia levels in poultry houses using machine learning and IoT technologies.	Implemented Random Forest algorithm with LoRa-WAN for real-time monitoring and control of ammonia concentrations.	Achieved significant reduction in ammonia levels, improving poultry health and farm sustainability.	Integrates machine learning with IoT for environmental control, directly supporting the capstone's focus on EQI optimization.
Alvi & Mondal (2023)	*Ammonia Gas Detection of Poultry Farms and Compost Plants by Low-Cost Smart Sensing System*	To develop a cost-effective system for detecting harmful gases in poultry farms.	Designed low-cost sensors for ammonia detection; data collected and analyzed for environmental assessment.	The system effectively identified high gas concentrations, enabling timely interventions to protect poultry health.	Supports the capstone's aim to monitor environmental quality using affordable IoT solutions.

CHAPTER III

METHODOLOGY

This study used a quantitative research approach and applied the Agile development method to build an IoT-based environmental monitoring system for Mercado's Poultry Farm in Nasugbu, Batangas. Agile allowed for flexible, sprint-based development and ongoing refinement of the system's modules, including temperature, humidity, and ammonia monitoring.

The system was built using ESP32 microcontrollers and sensors, with data visualized through a web-based dashboard. Real-time data logs and EQI calculations were tested in both controlled and actual farm conditions. Quantitative metrics like sensor accuracy, response time, and data reliability were used to evaluate performance. Notifications were also integrated to alert users when environmental conditions became unfavorable.

Project Design

The proposed Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning adopts a modular system architecture that integrates smart hardware, IoT connectivity, and web-based analytics to optimize poultry farm management. Figure 2 illustrates the overall design of the system, which is composed of the Power Management Unit, the EQI Monitoring Module, the IoT Gateway, and the Web-Based Server Module.

The Power Management Unit supplies regulated power to the system components, providing 3.3V for the ESP8266 microcontroller and 5V for relays and fans. This ensures continuous and stable operation of both the sensing and actuation mechanisms. The Environment Quality Index Monitoring Module

serves as the core unit, integrating the ESP8266 with the DHT22 temperature and humidity sensor and the MQ137 ammonia sensor. These sensors detect real-time environmental conditions and transmit the data to the microcontroller. When unsafe thresholds are reached, the ESP8266 automatically activates relays to control exhaust fans and lighting, thereby maintaining a safe and optimal environment for poultry growth. The IoT Gateway bridges the monitoring module with the server. It is composed of a transport layer, which establishes wireless connectivity through Wi-Fi, and a processing layer, which organizes sensor data before transmitting it to the backend system. The Web-Based Server Module functions as the decision-support and user interface platform. It provides real-time monitoring by displaying data on a dashboard, issues notifications when critical values are detected, and generates analytics that track environmental trends over time.

By integrating sensing hardware, IoT communication, and web-based analytics, the system architecture provides poultry farm administrators and technicians with a reliable tool for real-time monitoring and automated control. This data-driven approach supports timely interventions, improves poultry health and productivity, and promotes sustainable farming practices aligned with modern agricultural operations.

The system is scalable, allowing for additional sensors or modules to be incorporated as farm operations expand. This scalability enables farm managers to make data-driven decisions for improved productivity and sustainability. The design demonstrates reliability and adaptability to meet the evolving requirements of modern poultry farming.

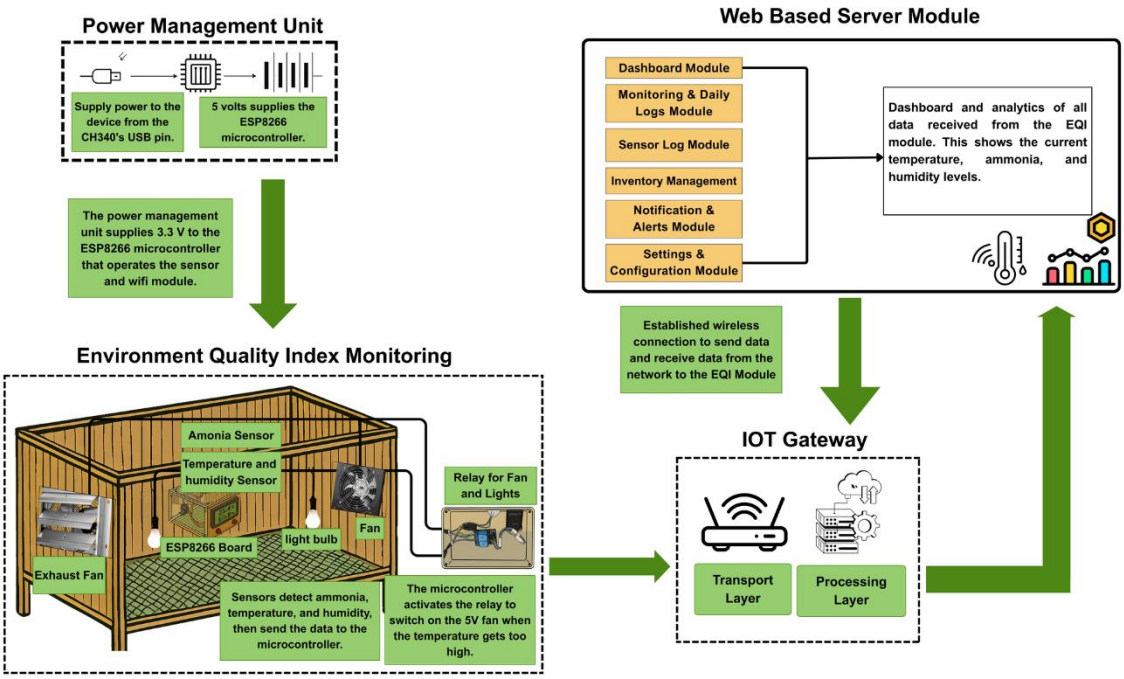


Figure 2. System Architecture

Schematic

The schematic shows the overall design of Smart Poultry Farming system, with the ESP8266 microcontroller as the main controller. It connects the different sensors, display, and relay to make the system work together. The MQ137 gas sensor is used to monitor ammonia levels, while the DHT22 sensor measures both temperature and humidity inside the poultry house. These readings are important because they directly affect the comfort and growth of the chickens.

The collected data are displayed on an I2C LCD, and added switches so that the user can choose whether to view the sensor data locally on the display or send it through Wi-Fi for remote monitoring. For automation, a relay module connected to the ESP8266. This relay controls the ventilation fan and the light bulb, both powered by 220V AC, so that the system can automatically adjust the environment based on the sensor readings.

The whole setup runs on a 5V USB supply, with the ESP8266 regulating the components that need 3.3V. By combining the sensors, display, and relay, the system can monitor the poultry house conditions in real time and automatically respond when needed. At the same time, it allows data to be sent online, where it can be processed using machine learning to generate an Environmental Quality Index (EQI) for optimizing chicken growth.

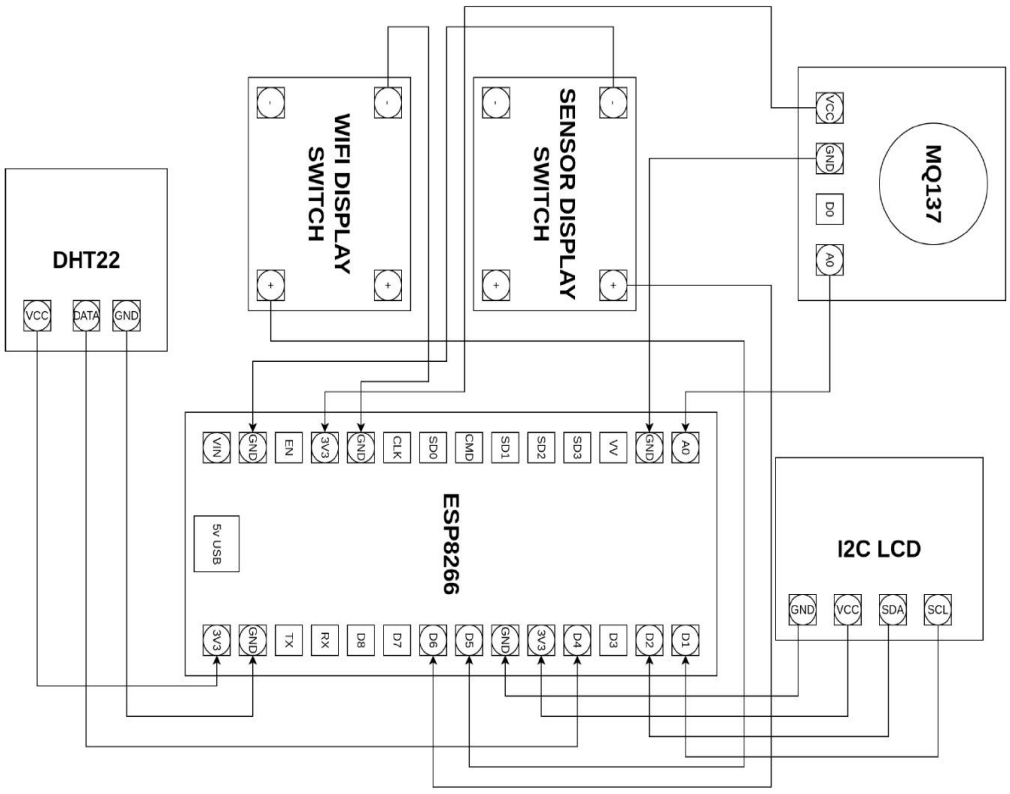


Figure 3. Schematic Diagram of Sensor

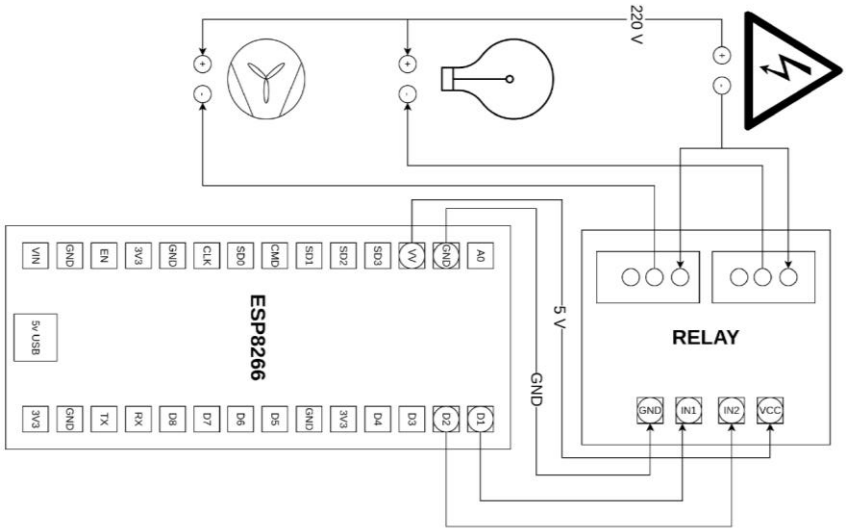


Figure 4. Schematic Diagram for Relay

Smart Poultry Farming Prototype

This prototype design presents the Smart Poultry Farming system equipped with IoT-based sensors installed inside a 4m × 1.5m × 2m poultry enclosure. These devices continuously collect environmental data such as temperature, humidity, and ammonia levels. The ESP8266 microcontroller processes and transmits the readings to the web-based server, where real-time monitoring, notifications, and EQI visualization are displayed. Machine learning analyzes these sensor patterns to automate exhaust fans, lighting, and other controls ensuring optimal environmental conditions for improved chicken growth and farm efficiency.

The system also allows farmers to access historical data to track trends and make informed management decisions. Alerts are sent to the farmer's mobile device if environmental conditions fall outside safe ranges, enabling prompt corrective actions.

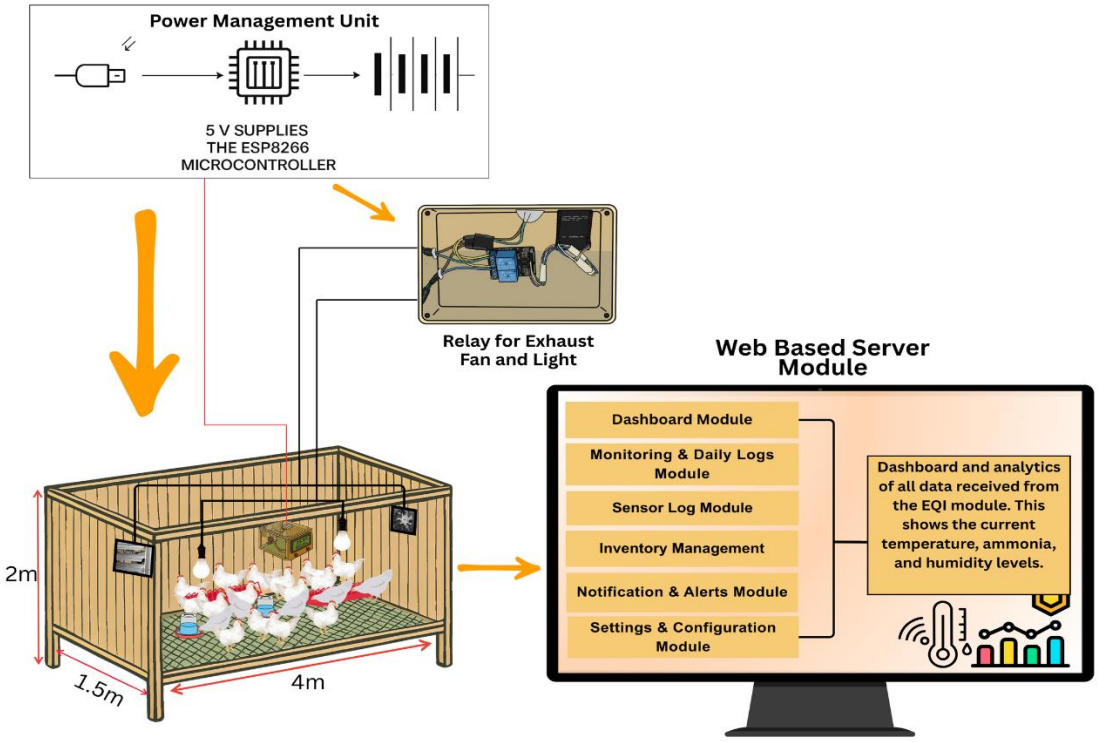


Figure 5. Prototype Design of Smart Poultry Farming

Evolutionary Prototyping Model

This diagram represents the software development methodology used in the project evolutionary prototyping. It emphasizes iterative development cycles involving requirement gathering, prototype building, customer evaluation, and refinement. This approach ensures the system evolves based on continuous feedback, leading to a more accurate and user-focused final product. The initial steps involved are the following:

Requirements Gathering. This phase focuses on identifying the key needs and expectations for the Smart Poultry Farming system. Stakeholders such as poultry farmers and agricultural experts provide insights on what environmental data is important (e.g., temperature, humidity, CO_2, ammonia) and how automation and monitoring can optimize chicken growth. These inputs guide the direction of the entire system development.

Build Prototype. Based on the initial requirements, a working prototype of the system is constructed. This includes setting up the NodeMCU ESP8266, integrating it with environmental sensors, connecting to relays for controlling fans and lights. A basic version of the database and web dashboard is also developed at this stage to test initial functionalities.

Customer Evaluation. The prototype is presented to users for real-world evaluation. Farmers or project stakeholders interact with the system and provide feedback on usability, accuracy, and performance. Their observations help identify any technical or user-experience issues that need to be addressed.

Refinement. Feedback from the customer evaluation is analyzed, and necessary improvements are made to the prototype. This could involve improving sensor accuracy, modifying control logic, updating the user interface, or adjusting the placement of components. Refinement ensures that the prototype becomes more aligned with real-world use cases.

Design. Once the concept has been refined, a more detailed and finalized system design is developed. This includes the finalized hardware configuration, software flow, data architecture, and integration of machine learning models. The design stage sets the blueprint for a complete and more stable version of the system.

Implement. With the final design in place, the system is implemented in full. Hardware is assembled and programmed, backend systems are developed, and the web dashboard is integrated with live sensor data. The Environmental Quality Index (EQI) monitoring logic and automation routines are

fully deployed. Machine learning models are incorporated to analyze environmental trends and support decision-making.

Test. The complete system undergoes rigorous testing, including unit testing of individual components and full system testing under simulated and actual poultry house conditions. The goal is to ensure reliability, real-time responsiveness, and accuracy of monitoring, automation, and machine learning predictions.

Maintain. After deployment, the system enters the maintenance phase. Regular checks are conducted to ensure sensor performance, update software components, retrain machine learning models with new data, and resolve any emerging issues. Maintenance ensures long-term reliability and adaptability of the system to changing conditions or user needs.

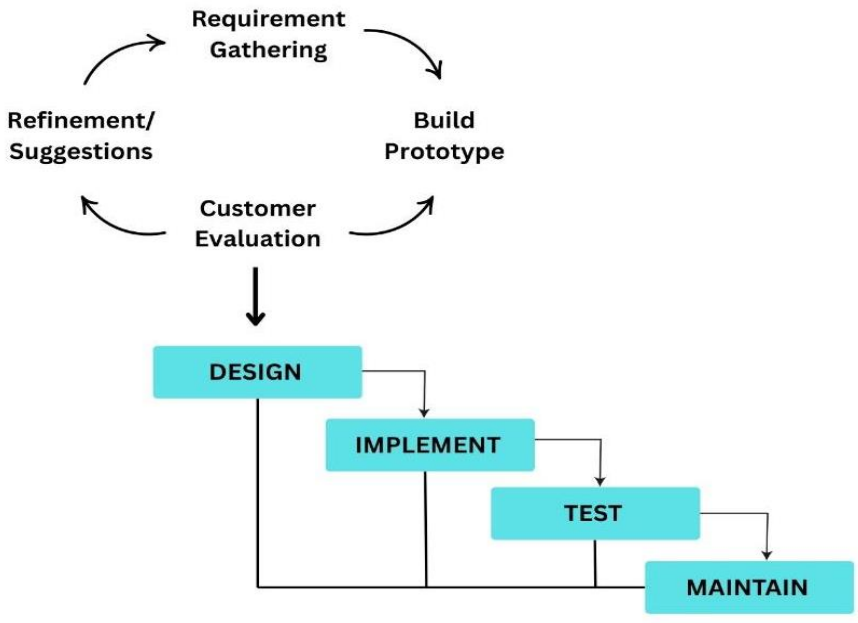


Figure 6. Evolutionary Prototyping Model of Smart Poultry Farm

Block Diagram

The core of this IoT-based monitoring system is a NodeMCU ESP8266 microcontroller. It gathers environmental data from three primary input sources: a Temperature Sensor, a Humidity Sensor, and a Ammonia Sensor. The NodeMCU then processes this information and directs it along two output paths. First, the data is stored in a Firebase Database and is then made accessible via a Web-based interface for monitoring. Second, the NodeMCU controls a Relay, which in turn manages the operation of two connected output devices: an Exhaust fan and a Light.

The system allows real-time monitoring of environmental conditions inside the poultry house. Alerts can be generated if sensor readings exceed predefined safe thresholds. Users can access the web interface remotely using a computer or mobile device. The Relay ensures automated control of the fan and light based on sensor data. Additionally, the stored data can be analyzed over time to improve environmental management and optimize chicken growth.

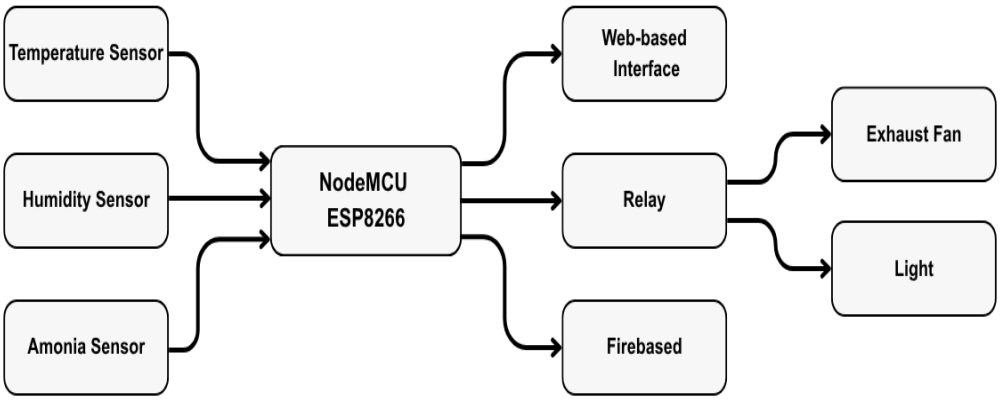


Figure 7. Block Diagram

Project Locale

The implementation of the capstone project titled "Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth" will take place at a local poultry farm located in Nasugbu, Batangas. This site was chosen for its accessibility, relevance to the study's objectives, and the farm owner's willingness to participate. During the site visit, the research team presented the project proposal and discussed its scope, expected outcomes, and benefits. The farm, with its typical rural setup, an open-sided structure made of wood and galvanized roofing provides an ideal environment for installing monitoring devices.

The chosen location offers a real-world setting for testing the system’s ability to monitor environmental factors such as temperature, humidity, ammonia levels, and air quality. Data collected from the sensors will be analyzed using machine learning algorithms to determine their influence on chicken growth. This implementation will help assess the system’s effectiveness and its practical application in a working poultry environment.

Sensors will be strategically positioned throughout the poultry house to capture accurate environmental data, and periodic inspections will be performed to ensure the devices operate correctly and provide reliable measurements. The collected data will be transmitted in real-time to a central monitoring system for analysis, and any deviations from optimal conditions will trigger alerts to the farm staff for immediate corrective action.



Figure 8. Owner of the Mercado's Farm

Model Development

The figure presents the structured workflow used in developing the ARIMA-based forecasting model for environmental quality monitoring in the Smart Poultry Farming system. This model serves as an essential component in enabling automated prediction and assessment of poultry house conditions using time-series data. The ARIMA development process consists of several sequential stages, each contributing to the creation of an accurate and efficient forecasting mechanism. These stages ensure that raw sensor data is properly processed, modeled, and converted into actionable insights that support environmental quality management. The steps are explained as follows:

Data Collection and Preparation. The first stage involves gathering raw sensor data from the IoT devices installed in the poultry house, including variables such as temperature, humidity, and ammonia levels. The collected data undergoes preprocessing tasks such as cleaning, formatting, and time-

series structuring to remove noise, handle missing values, and convert the dataset into a form suitable for forecasting. This ensures that the ARIMA model receives reliable and consistent input data.

Auto ARIMA Parameter Selection. After preparing the data, the system applies Auto-ARIMA to automatically identify the optimal parameters (p, d, q) needed for model construction. This automated process evaluates multiple parameter combinations and selects the best configuration based on statistical criteria, reducing manual tuning and improving the overall accuracy of the model.

Model Fitting. Using the selected parameters, the ARIMA model is trained on the historical time-series data. During this phase, the model learns patterns, trends, and seasonality present in the environmental conditions. Proper model fitting ensures that the ARIMA structure accurately represents the behavior of the poultry environment over time.

Experiment with Parameters. After selecting a candidate model, its parameters are fine-tuned to improve prediction accuracy. Hyperparameter tuning techniques are applied to adjust settings such as tree depth (for decision trees) or kernel type (for SVM), depending on the model used.

Forecast Generation. Once the model is fitted, it generates forecasts of future environmental conditions. These predicted values serve as the basis for estimating the Environmental Quality Index (EQI) and help in anticipating potential deviations from optimal poultry house conditions.

EQI Prediction. The forecasted sensor values are processed and translated into corresponding EQI scores, providing an evaluative measure of

environmental quality. This allows the system to classify upcoming conditions as ideal, moderate, or potentially harmful for poultry health and growth.

Dashboard Integration. The final stage involves displaying the EQI predictions and forecast results on the monitoring dashboard. This real-time visualization enables farm operators to easily interpret environmental trends and supports informed decision-making regarding ventilation, heating, and other control mechanisms.

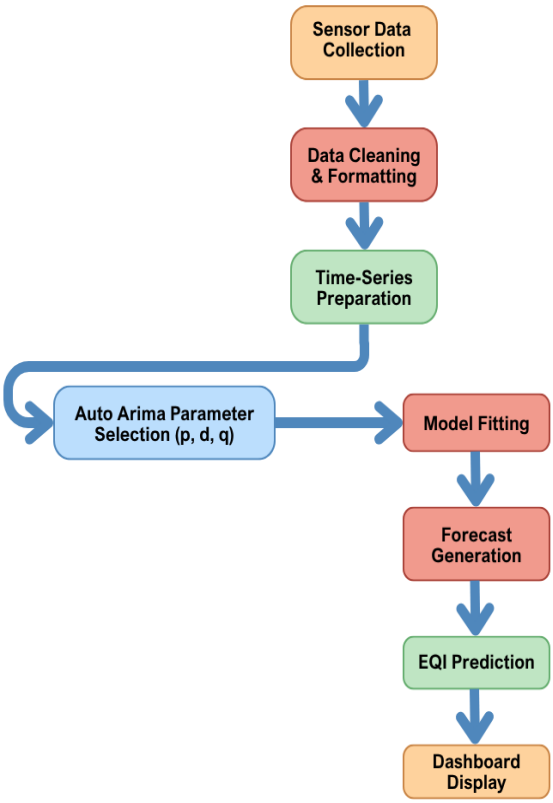


Figure 9. ARIMA Model

ARIMA Model and Dataset Description

For this study, a pre-trained ARIMA model was applied using an existing dataset of historical environmental readings. The dataset includes previously

collected values of temperature, humidity, and ammonia from poultry environments, allowing the model to learn underlying patterns and temporal variations without requiring additional on-site training. Before applying the model, the dataset underwent preprocessing operations such as noise reduction, differencing checks, and stationarity assessment to ensure suitability for time-series forecasting. This pre-existing dataset enabled the ARIMA model to capture recurring environmental cycles and typical fluctuations found in poultry operations, contributing to accurate short-term predictions, and the prediction results will be further analyzed to identify trends and improve environmental management strategies in the poultry house.

Why ARIMA Was Used for Machine Learning

ARIMA was chosen as the system's machine learning model due to its strong performance on structured, time-dependent datasets such as environmental sensor data. Since the project utilized a pre-existing dataset, ARIMA offered an efficient and highly accurate forecasting approach without the need for extensive computation or large-scale training. Compared to deep learning models like LSTM or GRU that require substantial data and processing resources, ARIMA is lightweight, interpretable, and optimized for short-term forecasting. These advantages make ARIMA particularly well-suited for IoT-based systems operating in real-world agricultural settings.

Arima Model Parameters

The ARIMA forecasting model utilized three primary parameters: p, d, and q, which were automatically selected through the Auto-ARIMA algorithm. The algorithm evaluated multiple parameter combinations and selected the

best configuration based on AIC (Akaike Information Criterion). Most environmental variables such as temperature, humidity, and ammonia were best modeled using ARIMA configurations ranging between (1,1,0) and (3,1,2) depending on the stability of the signal.

A first-order differencing (d = 1) was consistently applied to remove trends and stabilize the environmental data. Because the dataset did not exhibit strict seasonality patterns, no seasonal ARIMA (SARIMA) parameters were included. The model was trained using more than 2000 cleaned and preprocessed sensor readings, providing a robust foundation for short-term prediction of environmental trends used for EQI calculation.

Table 2. ARIMA Model Parameters

Parameter	Description	Assigned Value	Remarks
p (Auto-Regressive Order)	Number of past values used to predict the next value	1–3 (Auto-selected)	Determined by Auto-ARIMA based on sensor patterns
d (Differencing Order)	Number of times differencing is applied to achieve stationarity	1	Environmental data required first-order differencing
q (Moving Average Order)	Number of lagged forecast errors included	0–2 (Auto-selected)	Adjusts for fluctuations in humidity & ammonia
Seasonal Parameters (P, D, Q, m)	Seasonal effects in time-series	Not applied (non-seasonal model)	Daily cycle included implicitly, not forced seasonality
Training Dataset Size	Number of historical data points used	2000+ sensor readings	Pre-processed and cleaned before model fitting

Software Development

The development of the software component for the project "Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth" involved several key phases,

including planning, designing, coding, testing, and refinement. The system was built to collect and process real-time environmental data from various sensors deployed in the poultry farm, such as temperature, humidity, ammonia levels, and air quality.

To guide the development process, the Evolutionary Prototyping Model was utilized. This model supports a progressive approach to software development, where an initial prototype is created and repeatedly improved through user feedback and performance evaluation. Each iteration of the prototype integrated enhancements and corrections, leading to a more robust and functional system over time.

This approach allowed the development team to adapt quickly to challenges encountered in the field, such as sensor inaccuracies, communication delays, or data inconsistencies. With each cycle, the software was optimized not only to ensure accurate monitoring of environmental conditions but also to improve the machine learning component responsible for analyzing data and providing insights into chicken growth optimization.

Materials

This section details the hardware and software tools used in the project, highlighting the components essential for both the physical infrastructure and the development environment. These materials are chosen to meet the project's specific needs for functionality and performance.

Table 3. Hardware Requirements Used for Laptop/Computer

Requirements Used	
RAM	16GB
HDD/SSD	Samsung SSD 970 EVO Plus
CPU	500GB
GPU	AMD Ryzen 5 5500
CLOCKSPEED	NVIDIA GeForce RTX 2080 3.60Ghz

The hardware specifications for the laptop or computer used in this project are detailed in the table. These include a minimum of 16GB of RAM to ensure sufficient memory for smooth operation. The storage is provided by a Samsung SSD 970 EVO Plus with 500GB, offering fast read and write speeds to accommodate project files and software. The system is powered by an AMD Ryzen 5 5500 processor, which is capable of handling the project's processing requirements efficiently. For graphics, the laptop is equipped with an NVIDIA GeForce RTX 2080, ensuring high-performance visual processing. The system operates at a clock speed of 3.60GHz, which ensures quick processing speeds and optimal performance for complex tasks. These specifications ensure that the computer can handle the demands of the project effectively.

Software

This section describes the software tools utilized in the development of the web-based system, explaining the reasons for their selection and their key features. These tools are vital for the system’s development, ensuring efficiency and reliability in the final product. They also enable seamless integration with the IoT devices for real-time data monitoring and control.

Python. Is a popular high-level programming language known for its simplicity and readability. It’s widely used in various fields such as web development, data analysis, artificial intelligence, and IoT applications. In smart

farming and IoT projects, Python is often used to process sensor data, build machine learning models, and create backend services due to its extensive libraries and ease of integration with hardware and software tools.

Flask. Is a lightweight web framework for Python that allows developers to quickly create web applications and APIs. It is particularly useful for building backend services that handle data collection, processing, and communication between IoT devices and user interfaces. Flask helps create dashboards and web platforms to display real-time sensor data in smart poultry farming systems.

Javascript. Is a scripting language primarily used to make web pages interactive and dynamic. It runs in the browser and allows developers to create responsive elements such as buttons, forms, and live-updating content. In IoT dashboards and monitoring systems, JavaScript enables real-time visualization of environmental data and interactive controls for users.

HTML. Is the foundational language used to structure content on the web. It defines elements like headings, paragraphs, tables, and images that make up a web page. HTML provides the basic framework upon which interactive and styled elements are added using CSS and JavaScript in web-based IoT applications.

CSS. Is a styling language used to design and visually enhance web pages. It controls layout, colors, fonts, and responsiveness, making web interfaces more attractive and user-friendly. In IoT systems, CSS ensures that dashboards and data displays are clear, organized, and accessible across different devices.

Bootstrap. Is a popular front-end framework that offers pre-built CSS and JavaScript components to help developers design responsive, mobile-friendly web interfaces quickly. It simplifies the development process by providing grids, buttons, forms, and navigation elements. Bootstrap is commonly used to create professional-looking dashboards for monitoring poultry farm conditions.

Roboflow. Is an online platform that helps manage, label, and preprocess image datasets for machine learning, especially in computer vision tasks. It simplifies preparing training data and deploying models for tasks like object detection or health monitoring in poultry farms by analyzing images of animals or their environment.

ARIMA. Refers to the process where a pre-trained machine learning model analyzes new data to make predictions or decisions. In smart poultry farming, inference enables the system to evaluate real-time environmental data such as temperature and air quality, providing actionable insights to optimize chicken growth and health based on learned patterns.

NumP. Is a Python library for numerical computing, providing efficient support for large arrays and matrices along with mathematical functions. It is essential for handling sensor data, performing calculations, and preparing inputs for machine learning models in IoT applications where fast and accurate data manipulation is required.

Firebase. Is a cloud-hosted NoSQL database that stores data in JSON format and synchronizes it in real time across connected devices. It is scalable, serverless, and supports offline access, making it suitable for applications such

as IoT monitoring, mobile apps, and poultry farm management systems. Firebase also provides built-in authentication, analytics, and cloud functions, enabling secure and automated data handling. Its real-time database capabilities allow farmers to receive instant updates on poultry conditions, improving response time to environmental changes.

ArduinoIDE. Is a programming environment used to write, compile, and upload code to Arduino microcontrollers. It supports coding in C/C++ and is widely used to develop control programs for sensors and actuators in IoT projects. For poultry farming, Arduino IDE helps program hardware devices to monitor environmental factors and automate tasks. It allows integration with various sensors such as temperature, humidity, and gas sensors to collect critical farm data. Additionally, Arduino IDE enables scheduling and automation of equipment like fans, heaters, and feeders, reducing manual labor.

Visual Studio Code. Is a powerful, lightweight source code editor that supports multiple programming languages and offers tools for debugging, version control, and extensions. It is widely used by developers working on both hardware programming and web development in IoT projects, providing an efficient environment for building integrated poultry monitoring systems. VS Code supports extensions that connect directly to IoT devices, allowing for live monitoring and debugging of code. It also facilitates collaboration among developers through integrated Git support, ensuring version control and streamlined project management.

Table 4. Hardware Specifications

Sensor	Measured Parameter	Specification	Detection / Coverage	Operating Voltage
DHT22	Temperature & Humidity	Temp: -40°C to 80°C, Humidity: 0–100% RH, Accuracy: ±0.5°C, ±2–5% RH	Recommended coverage: up to 20 meters line of sight. Can be deployed for larger areas using multiple sensors to ensure uniform environmental monitoring.	3.3–6 V
MQ-137	Ammonia Gas (NH_3)	Detectable Gas: 5–500 ppm, Sensitivity: 10–500 ppm	Recommended coverage: effective for up to 10–15 m^2 per sensor. Multiple sensors may be required for larger poultry houses to monitor gas levels uniformly.	5 V

Project Development

The system titled "Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth" developed using the Agile Scrum methodology. This iterative and flexible approach allows for continuous feedback and collaboration, ensuring the system meets the specific needs of poultry farmers in Nasugbu, Batangas.

The project will be organized into one- to two-week sprints, with regular stand-up meetings to track progress and resolve issues. Key components such as IoT-based environmental monitoring and machine learning for EQI analysis developed and integrated incrementally. This approach enables rapid, adaptive development focused on optimizing chicken growth through real-time environmental data.

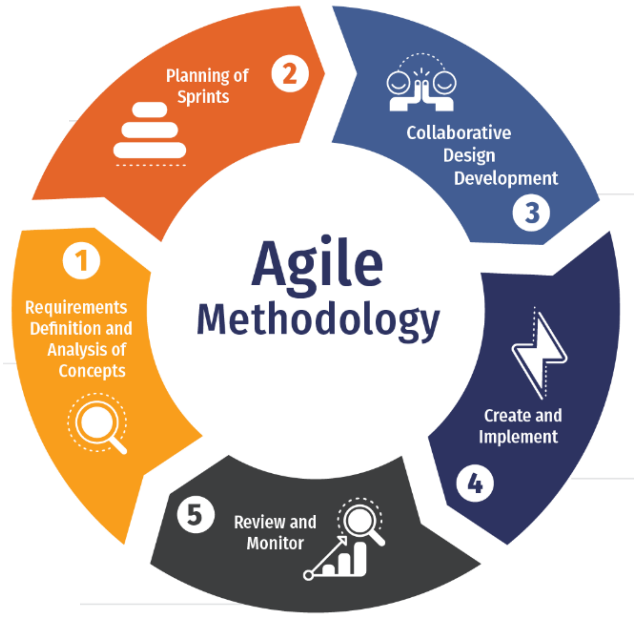


Figure 10. Agile Scrum Methodology

Scope

This research aims to develop and implement an IoT-based poultry monitoring system for poultry farmers in Nasugbu, Batangas. The main users of the system are farming administrators and technical staff. The system is designed to give breeders real-time access to important environmental data that directly affects the health and productivity of their poultry. Through the use of sensors, it can monitor temperature, humidity, and ammonia levels inside the poultry houses. Aside from monitoring, the system will also include a prediction feature that can help forecast possible risks, like sudden environmental changes, so that farmers can take preventive action ahead of time. ICT professionals will be responsible for maintaining the system, checking the accuracy of the data, handling the predictive functions, and providing technical support. Overall, the goal is to make farm operations more efficient, improve the living conditions of the poultry, and support sustainable poultry management by combining real-time monitoring with predictive insights.

Design

The system follows a modular, user-focused design to ensure streamlined functionality and ease of use. It integrates several key modules that work together to monitor and manage the environmental conditions of poultry farms. The Login Module provides secure authentication for different users, ensuring that only authorized personnel can access sensitive data. The Settings Module allows users to modify their credentials and customize the dashboard layout based on their preferences. Central to the system is the Dashboard Module, which consolidates and displays real-time analytics of temperature, humidity, ammonia levels collected from sensors in the poultry environment. The Notification Module sends automatic alerts based on the Environmental Quality Index (EQI), helping users respond promptly to unfavorable conditions. Additionally, there are individual data logging modules such as Temperature, Humidity, Ammonia each dedicated to storing and organizing environmental records for analysis and reporting. This modular approach not only ensures clarity and usability but also supports effective environmental monitoring to improve poultry health and farm efficiency.

Product Backlogs

The product backlog for admin or farmer users in the Smart Poultry Farming system focuses on providing critical tools for managing environmental conditions within poultry facilities. As the primary users of the IoT system, admins and farmers are responsible for monitoring real-time sensor data, setting thresholds for environmental quality indicators, and responding to alerts that may impact poultry health and growth. The backlog includes modules for

logging in securely, adjusting system settings, accessing the main dashboard for analytics, and receiving notifications based on Environmental Quality Index (EQI) statuses. Each backlog item is aimed at enhancing decision-making, automating data collection, and streamlining farm monitoring to support optimal chicken growth and farm productivity.

Additional features within the backlog include historical data visualization, which allows users to track trends in temperature, humidity, and ammonia levels over time. Users can also configure automated responses, such as activating ventilation systems or triggering alarms, when environmental thresholds are exceeded. The system emphasizes mobile accessibility, enabling farmers to monitor and manage the farm remotely through smartphones or tablets. Cloud integration ensures that all collected data are securely backed up and easily retrievable for reporting or auditing purposes. Real-time notifications can be customized according to different levels of urgency, assisting users in prioritizing appropriate actions. Predictive analytics modules provide recommendations based on historical data and environmental trends, supporting proactive farm management. Lastly, feedback and help modules allow users to report system issues, request enhancements, and access tutorials, ensuring continuous improvement and reliable system support.

Table 5. Product Backlog for Admin

Story User	Description	User/Role	Priority	Estimated Effort Days
Login Module	As a user Secure sign-in/sign-out with lockout: allow only 3 failed attempts, then enforce a 10-minute cooldown; send SMS on lockout; record events.	Admin, Farmer	High	1 day
Settings Module	As an admin, I want to customize threshold values for alerts so I can adapt to my farm's needs	Admin	Medium	1.5 days
Dashboard Module	As a farmer, I want to see real-time environmental data on my dashboard so I can quickly assess farm conditions	Farmer	High	1.5 days
Notification Module (SMS)	As a user, I want to receive automatic notifications when temperature, humidity, or gas levels are abnormal, and Automatic SMS.	Admin, Farmer	High	1 day
Temperature Monitoring	As a farmer, I want to view current and historical temperature data so I can ensure chickens are kept at optimal warmth	Farmer	High	1 day
Humidity Monitoring	As a farmer, I want to track humidity levels to maintain the right moisture conditions for chicken health	Farmer	High	1 day
Ammonia Monitoring	As a user, I want to get alerts if ammonia levels are high so I can ventilate the coop and avoid health risks	Admin, Farmer	High	1 day
EQI Computation Integration	As an admin, I want to view an overall EQI score calculated from environmental data so I can quickly assess farm health	Admin	High	2 days
Sensor Data Storage	As a system, I want to automatically store sensor data to a secure database so it can be used for analytics	System	High	2 days
Graph & Chart Visualization	As a user, I want to view trends in the form of graphs and charts so I can make better decisions for my chickens	Admin, Farmer	Medium	2 days
System Health Checker	As an admin, I want to be notified if any sensors go offline or report incorrect values so I can troubleshoot quickly	Admin	Medium	1.5 days
Manual Sensor Trigger	As a user, I want to manually trigger a sensor reading in case I suspect inaccurate values	Admin, Farmer	Low	1 day
System Logging	As an admin, I want the system to keep logs of all major actions (logins, alerts, setting changes) for security and auditing	Admin	Medium	1.5 days

The product backlog for the Smart Poultry Farming system includes essential features that support real-time environmental monitoring and chicken growth optimization using IoT and machine learning. Key components include

secure user authentication, real-time dashboard display, automated sensor data storage, alert notifications, and historical data analytics. A machine learning-based EQI calculation module evaluates environmental conditions, while customizable settings and user management enhance flexibility and control. Each feature is prioritized and estimated to guide development, ensuring a user-focused, efficient, and responsive farming system.

Table 6. Main Product Backlog (General Features)

Story User	Description	User/Role	Priority	Estimated Effort Days
User Authentication	Login/logout for both farmer and admin users	Admin, Farmer	High	2 days
Dashboard Display	Real-time visualization of temperature, humidity, gas levels, and EQI	Admin, Farmer	High	2 days
Notification System	Alert system for abnormal sensor values (e.g., too hot, high ammonia)	Admin, Farmer	High	1.5 days
Senso Data Storage	Automatically store sensor data in the web-based platform	System	High	2 days
Analytics Module	Historical data analysis and graph generation for chicken growth optimization	Admin, Farmer	Medium	3 days
EQI Calculation Module	Derive Environmental Quality Index using machine learning based on sensor inputs	System	High	4 days
Settings Module	Allow users to customize threshold values and notification preferences	Admin	Medium	1.5 days
User Management	Admin can add/remove farmer accounts and assign roles	Admin	Medium	1.5 days

Sprint Planning for Admin

This sprint plan outlines a four-phase development strategy for the admin interface of the Smart Poultry Farming system. Each sprint builds progressively from securing system access and setting up dashboard controls, to integrating real-time sensor data, alerts, and final testing. The goal is to equip the admin/farmer with full control over monitoring and responding to

environmental conditions, ensuring optimal chicken growth through the system's reliable and intuitive functionality. The initial phase focuses on secure admin access and user management, ensuring that only authorized personnel can operate and modify system settings. Subsequent development integrates data visualization and analytics tools, enabling admins to interpret real-time sensor data and make informed decisions based on the Environmental Quality Index (EQI). The final phase ensures overall system reliability, including performance monitoring, logging of admin activities, and the ability to manually trigger sensor readings when necessary.

Table 7. Sprint Planning for Admin Features

Sprint	Goal	Tasks	Estimated Hours/Day	Expected Outcome
1	Set up secure admin access and user control	Develop login/logout functionality- Implement role-based access control- Create user management (add/edit/delete farmers)	6–8 hours/day for 3 days	Admin can securely access the system and manage farmer accounts
2	Configure system thresholds and notifications	Create settings module for threshold customization- Set up notification triggers based on sensor thresholds- Test notification alerts (SMS/email/in-app)	6–8 hours/day for 3 days	Admin can set custom alert thresholds and receive timely notifications
3	View and analyze environmental data	Develop admin dashboard with real-time sensor data- Integrate EQI score display- Enable access to detailed charts and analytics	6–8 hours/day for 4 days	Admin can monitor live data and make informed decisions using the EQI and graphs
4	Ensure system reliability and logging	Implement system health checker for sensor status- Add system logging for admin activities- Set up manual sensor trigger function	6–8 hours/day for 3 days	Admin can monitor system performance, view logs, and manually trigger data readings if needed

Daily Scrum

The development team holds a daily 2 hour and 30-minute scrum meeting to ensure alignment toward achieving the sprint goals for the IoT-based environmental monitoring system in poultry farming. These sessions help team members identify and address obstacles, adapt the sprint or product backlog if necessary, and maintain progress toward enhancing poultry farm management through real-time EQI monitoring and data-driven decision-making.

Sprint Execution

During sprint execution, developers work closely with the product owner, including poultry farm stakeholders and system administrators, to define sprint deliverables and estimate workloads. This collaboration ensures the development focus remains on integrating real-time IoT sensor data and optimizing environmental conditions such as temperature, humidity, and gas levels to support healthy poultry growth.

Sprint Review Meeting

the review, the development team presents the system functions, such as EQI analytics, environmental sensor logging, and the prediction feature that helps forecast risks affecting poultry health. This process makes it easier to evaluate if the system's components, including real-time monitoring and predictive insights, are truly meeting the operational needs of poultry farmers in maintaining optimal living conditions for their chickens.

Sprint Retrospective

At the end of each sprint, the team conducts a retrospective to evaluate their work and collaboration. Key areas of discussion include communication, tool efficiency, and development workflow. With the project focused on enhancing poultry farming through IoT-based EQI monitoring, the team reflects on completed milestones and documents areas for technical improvement. For instance, initial retrospectives might highlight that core sensor modules like temperature and humidity were successfully deployed during the first month.

Project Testing and Evaluation

The system goes through testing to make sure all its parts are reliable and accurate. Each module, such as temperature, humidity, and ammonia sensing, is tested one by one to check if the data is correct and if it works well with the dashboard. The prediction feature is also tested to see if it can give accurate forecasts of possible risks, like sudden changes in the environment that could affect poultry health. This whole testing process makes sure the IoT-based monitoring system is stable, easy to use, and ready to be applied in actual poultry farms. In the end, the goal is to help improve chicken growth by combining environmental control with predictive insights.

Software Testing

The figure below illustrates the comprehensive software testing process implemented during the development of the Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring System. This testing ensured that each software module performed reliably and accurately in monitoring poultry farm environmental conditions. Functional testing was

carried out through unit testing for core features such as sensor data collection, EQI computation, and user authentication. Integration testing validated the interaction between modules including temperature, humidity, ammonia, and carbon dioxide monitoring. System testing was conducted to ensure that the entire platform functioned as a unified solution, and acceptance testing was performed to confirm that the system met the needs and expectations of poultry farm operators. In addition to functional testing, non-functional testing addressed essential system qualities such as performance under sensor load, secure handling of environmental and system data, ease of use for farm staff with varying levels of technical knowledge, and compatibility across various devices used in agricultural settings. This robust testing framework guaranteed that the EQI-based monitoring system was stable, secure, and intuitive, making it fully prepared for deployment in real-world poultry farming to support optimal chicken growth through continuous environmental control.

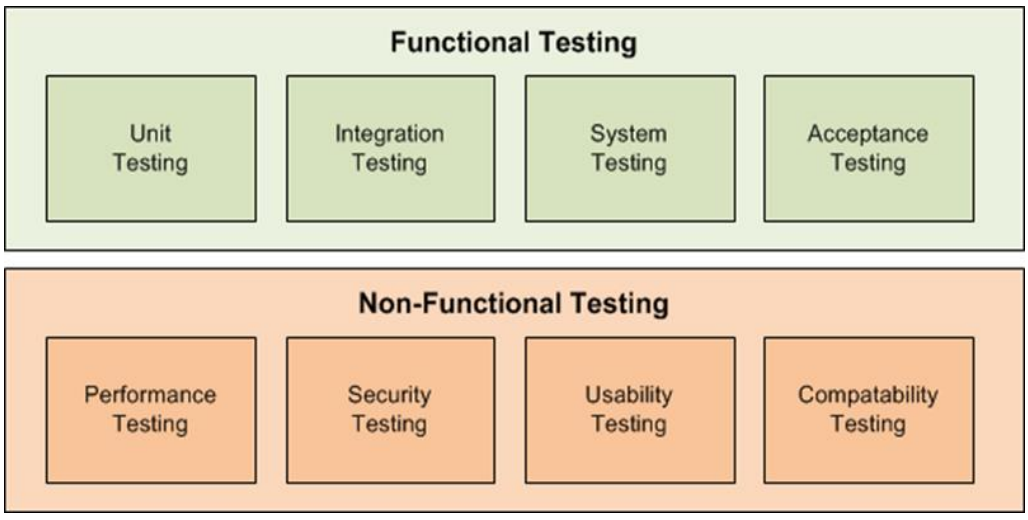


Figure 11. Levels of Testing

Unit Testing

Unit testing was conducted to verify that each individual component of the system including sensor data collection, environmental condition monitoring, and EQI computation functioned correctly in isolation. By testing these modules independently, the development team was able to detect and fix errors at an early stage, ensuring each feature worked as expected before integrating them into the full smart poultry monitoring system. This process contributed significantly to the overall reliability and stability of the EQI-based solution aimed at enhancing chicken growth through precise environmental control. Test cases were designed to cover a variety of scenarios, including normal, boundary, and unexpected input values, to ensure robustness. Automated scripts were used where possible to streamline repeated testing and maintain consistency in results. Additionally, the outcomes of unit tests were documented carefully to provide a reference for future maintenance and system upgrades.

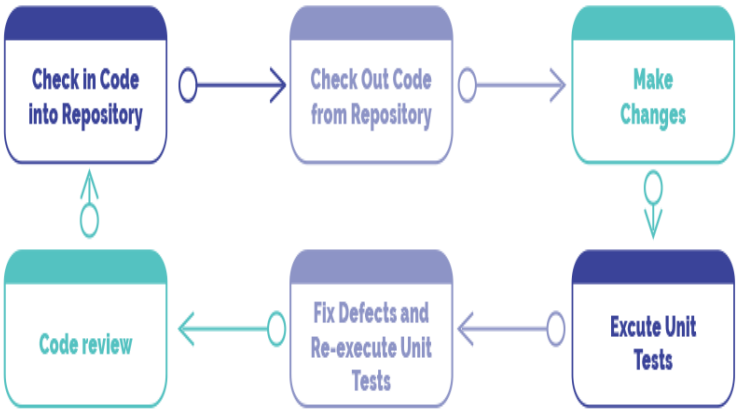


Figure 12. Unit Testing

Integration Testing

Integration testing ensured that all core modules such as sensor input processing, environmental monitoring, and EQI data analytics worked together seamlessly. This phase validated the proper communication and interaction between the system's components, helping identify any issues in data transfer, synchronization, or functional logic between modules. By doing so, the researchers confirmed that the integrated system reliably delivered accurate and real-time environmental readings necessary for poultry growth optimization.

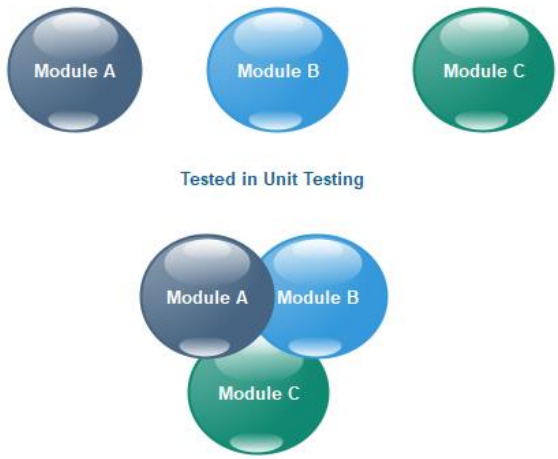


Figure 13. Integration Testing

System Testing

System testing was conducted to evaluate the Smart Poultry Monitoring System as a fully integrated solution. This phase examined whether the sensors, IoT gateway, cloud database, and web dashboard functioned correctly as one complete workflow. End-to-end scenarios were executed to ensure accurate data collection, transmission, processing, and visualization

under real operating conditions. Performance, reliability, and compatibility were also assessed to verify that the system could handle continuous sensor updates and user access without failure. The results confirmed that the entire system met its functional requirements and was stable enough for acceptance testing and deployment.

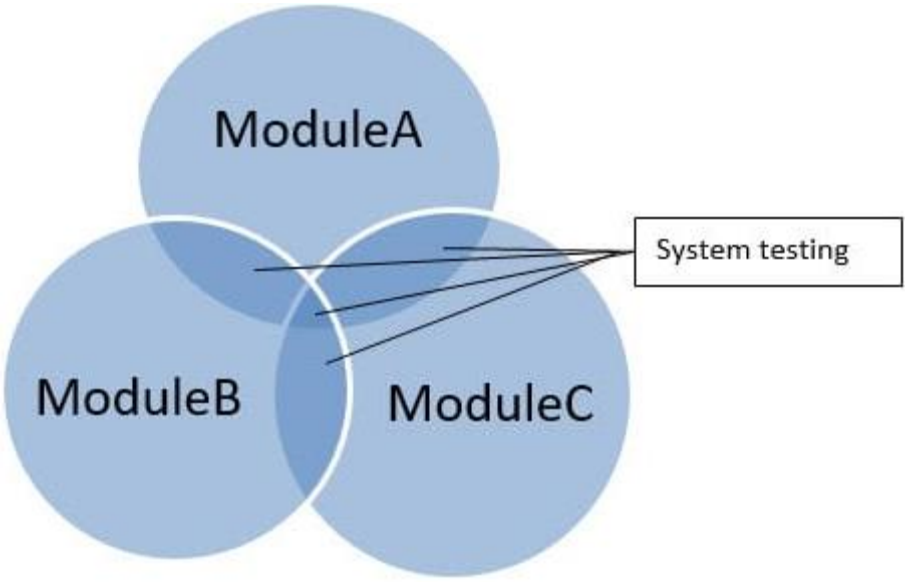


Figure 14. System Testing

Acceptance Testing

Acceptance testing was carried out to evaluate whether the IoT-based poultry monitoring system met the specific needs of poultry farmers in Nasugbu, Batangas. This involved confirming the correct implementation of features like real-time environmental tracking, sensor-based alerts, and EQI monitoring. Final approval was granted after the system successfully aligned with the operational goals and usability expectations defined by stakeholders, ensuring it was ready for deployment in an actual poultry farming environment.

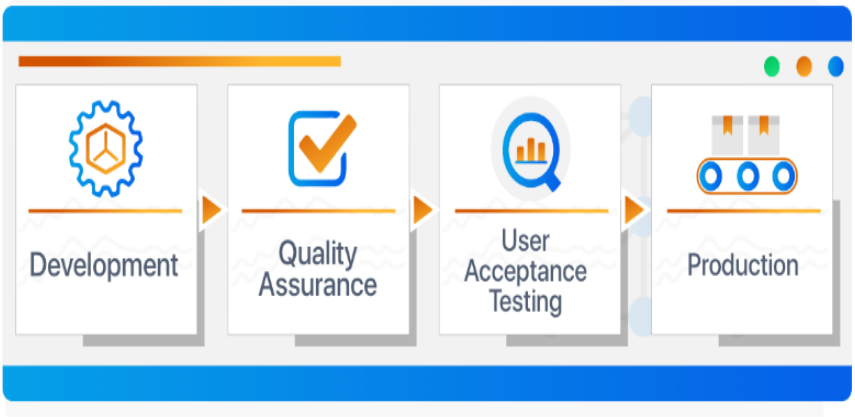


Figure 15. User Acceptance Testing

DHT22 Temperature and Humidity Sensor Calibration

The calibration of the DHT22 sensor is essential to ensure accurate monitoring of temperature and humidity within the poultry environment. This process begins with verifying proper connections, power supply, and sensor initialization. Calibration involves measuring sensor output at selected reference conditions and comparing these with values from a trusted thermometer and hygrometer. The resulting discrepancies are recorded as offsets, which are then applied to adjust the sensor readings in the system. Accurate calibration ensures reliable Environmental Quality Index (EQI) calculations, which are crucial for the machine learning model used to optimize poultry growth conditions.

Ammonia Sensors Calibrations

Calibrating ammonia sensors are essential for accurate gas level monitoring in the poultry environment. The process involves sensor preheating, exposing it to known gas concentrations, and comparing the readings to a reliable reference. Any differences are used to apply correction offsets. This ensures accurate Environmental Quality Index (EQI) values, supporting better air quality control and optimized chicken growth through machine learning.

Environmental Quality Requirements

Table 8 outlines the general acceptable environmental conditions necessary for healthy poultry growth. It specifies ideal ranges for temperature, humidity, ammonia (NH_3) levels. For temperature, chicks in the brooding stage (0–2 weeks) require a warmer environment between 32–35°C, while older chickens in the grow-out stage (3–6 weeks) thrive at cooler temperatures between 21–26°C. Humidity should be maintained between 50% and 70% relative humidity (RH) to support respiratory health and comfort. The acceptable limit for ammonia is ≤ 25 ppm, ensuring proper air quality to avoid stress or health issues in poultry.

In addition to temperature, proper humidity control is essential for maintaining respiratory health and minimizing the risk of dehydration or excessive moisture in litter materials. A relative humidity (RH) level between 50% and 70% provides a balanced environment that enhances comfort while reducing the likelihood of respiratory infections and pathogen growth.

Table 8. Required Environmental Conditions for Healthy Poultry Growth

Environmental Parameter	Required Levels
Temperature	Brooding (0–2 weeks): 32–35°C Grow-out (3–6 weeks): 21–26°C
Humidity	50%–70% Relative Humidity (RH)
Ammonia (NH_3) Level	≤ 25 ppm (parts per million)

Target Quality Requirements

Table 9 provides a more detailed breakdown of optimal environmental parameters based on the chickens' age. It categorizes the requirements week

by week, from 0 to 6 weeks and above. For instance, during the first week, chicks need temperatures between 32–35°C, humidity levels of 60–70%, ammonia levels of ≤ 10 ppm. As the birds mature, the ideal temperature and humidity decrease gradually, and slightly higher concentrations of ammonia become acceptable. This table acts as a practical reference for adjusting environmental settings over time, helping to maintain ideal growth conditions and contributing to the accuracy of the Environmental Quality Index (EQI) used in the project. This table serves as a practical reference for dynamically adjusting environmental control systems over time, supporting consistent growth conditions and enhancing the reliability and accuracy of the Environmental Quality Index (EQI) applied in the study.

Table 9. Threshold Levels Based on Chicken Age

Age (Weeks)	Temperature (°C)	Humidity (%)	Ammonia (NH_3, ppm)
0 – 1	32 – 35	60 – 70	≤ 10
2 – 3	29 – 32	55 – 70	≤ 15
4 – 5	24 – 28	50 – 65	≤ 20
6 and above	21 – 26	50 – 60	≤ 25

Table 10 presents the threshold levels for the Environmental Quality Index (EQI), categorizing poultry house conditions into three ranges based on environmental data. An EQI score of 80–100 indicates excellent and optimal conditions, 60–79 reflects moderate quality with minor environmental concerns, while 0–59 represents poor conditions, signaling possible stress or an unhealthy environment for the chickens. This classification helps in evaluating the overall quality of the poultry environment and guides necessary adjustments for maintaining ideal conditions.

Table 10. EQI Threshold Levels and Environmental Condition Status

EQI Range	Status	Description
80-100	Excellent	Optimal conditions for poultry
60-79	Moderate	Acceptable but with minor environmental concerns
0-59	Poor	Unhealthy or unsafe environment

System Flowchart

The flowchart outlines the architecture of a smart poultry monitoring system, structured into four distinct stages ranging from on-site hardware to user interaction. It begins in the Physical Farm layer, where an ESP8266 microcontroller collects environmental data from temperature, humidity (DHT22), and ammonia (MQ137) sensors, automatically triggering relays for exhaust fans or lights if local safety thresholds are breached. This sensor data is formatted as JSON and transmitted via Wi-Fi through the Edge & Transmission layer to the Cloud & Processing layer, where a Firebase Database serves as the central storage hub. Inside the cloud layer, the system performs two key parallel tasks: running an ARIMA model to generate future forecasts and checking data against settings to trigger SMS alerts if conditions are unsafe. Finally, the User Interface allows authenticated farmers to log in to a web app to view real-time dashboards and forecasts, input chicken inventory data, and modify system configurations.

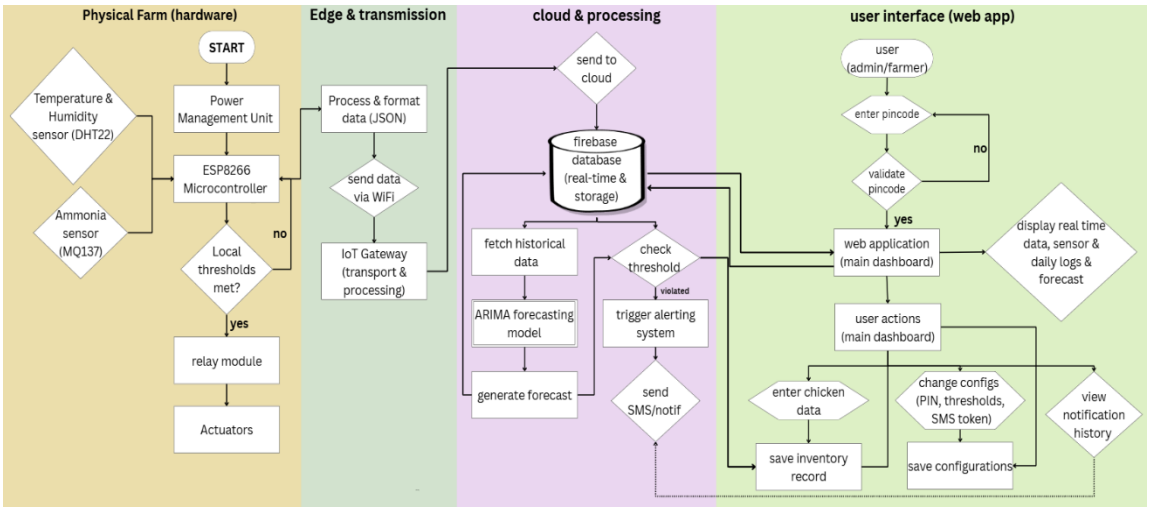


Figure 16. System Flowchart of Smart Farming System

The flowchart illustrates the end-to-end data workflow of an IoT monitoring system, segmented into three distinct layers: Local Hardware (Edge), Cloud & Processing, and User Interface. The process begins at the Local Hardware layer, where an Arduino microcontroller initializes, connects to Wi-Fi, and continuously cycles through reading environmental sensors (DHT22 for temperature/humidity and MQ137 for ammonia) before packaging the data into JSON and transmitting it via HTTPS. The workflow then moves to the Cloud & Processing layer, where this raw data is stored in a Firebase Realtime Database and subsequently processed by a server-side Python script using an ARIMA model to generate predictive forecasts. Finally, the system outputs to the User Interface, where a web or mobile dashboard pulls both the real-time

sensor readings and the forecasted data to display comprehensive environmental insights to the user.

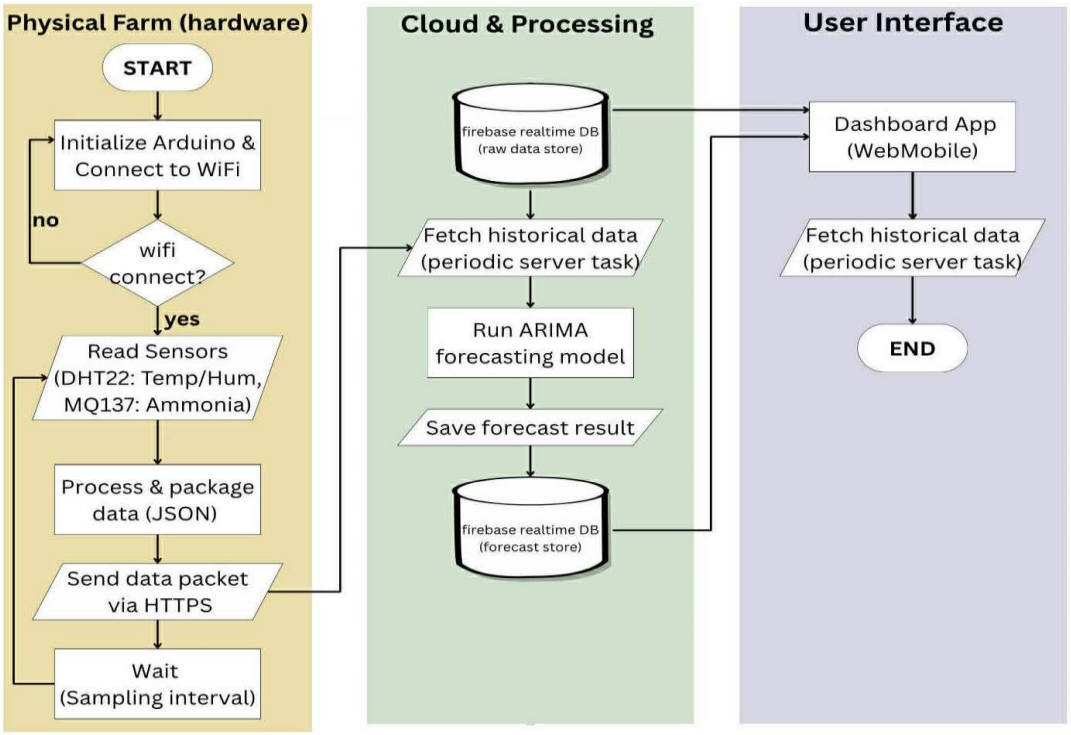


Figure 17. System Flowchart of Hardware

Use Case Diagram

Figure 17 illustrates the interactions between a user role, defined as admin/farmers, and the various functionalities of a system. Users who have direct access to six distinct use cases displayed as ovals: pincode, dashboard, monitoring, inventory, notification, and settings. Lines connect the actor to each of these use cases, indicating that the admin/farmers can engage with all these features. The use cases are also connected vertically in a sequence from pincode at the top to settings at the bottom, which may suggest a navigational flow, a hierarchical menu structure, or a sequence of dependencies, such as requiring a pincode entry before accessing the dashboard and subsequent functions.

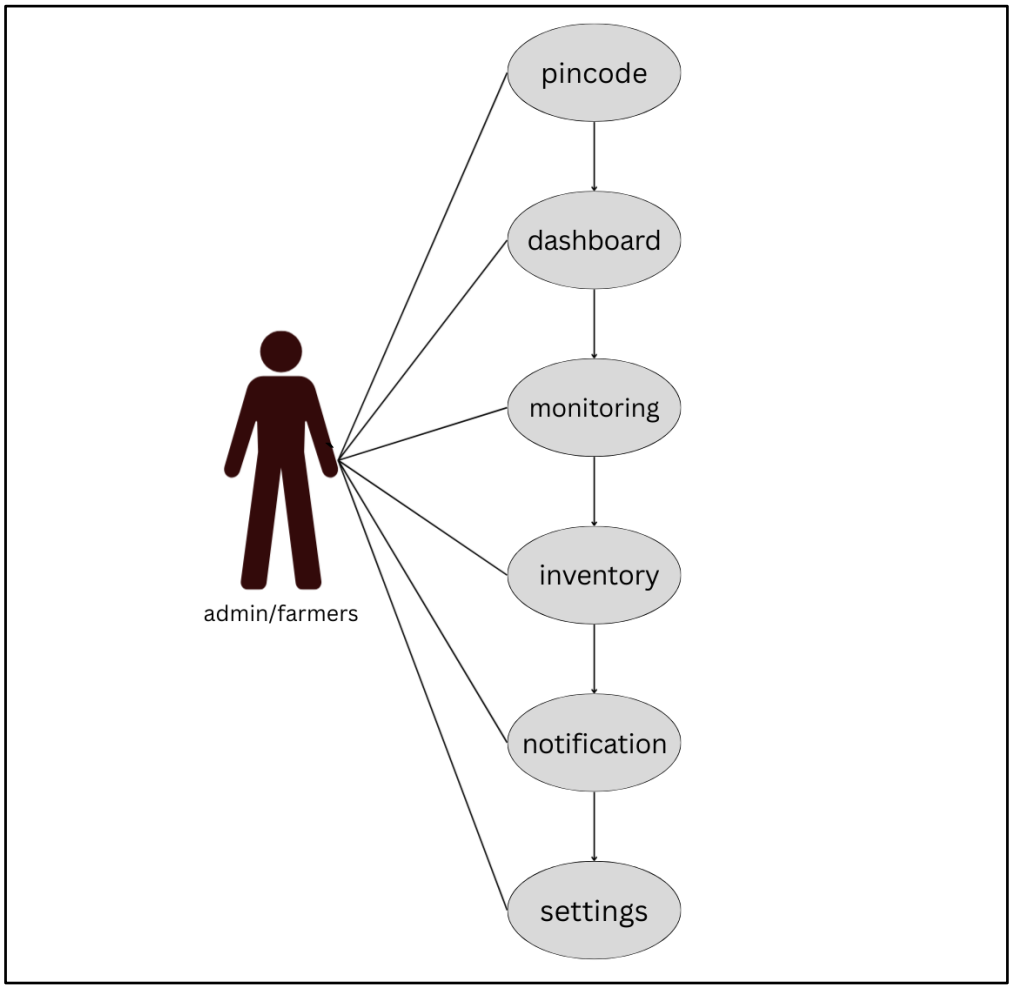


Figure 18. Used Case Diagram

Evaluation Procedure

For the *Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth* project, beta testing was conducted to assess the system's overall quality and performance. Feedback was collected from poultry farmers, veterinarian and IT experts using a Likert scale, focusing on key aspects such as usability, efficiency, and reliability. The evaluation process was guided by the ISO 25010 quality standard, a widely recognized framework for measuring software quality.

The assessment emphasized critical quality characteristics including functionality, security, maintainability, and portability to ensure the system meets industry standards for poultry farm environmental monitoring. Insights gained from this evaluation helped identify areas requiring enhancement, which were addressed to improve the system's effectiveness and user experience. Adhering to the ISO 25010 model ensured that the IoT-based solution reliably supports poultry farm operations while maintaining high standards of software quality and performance.

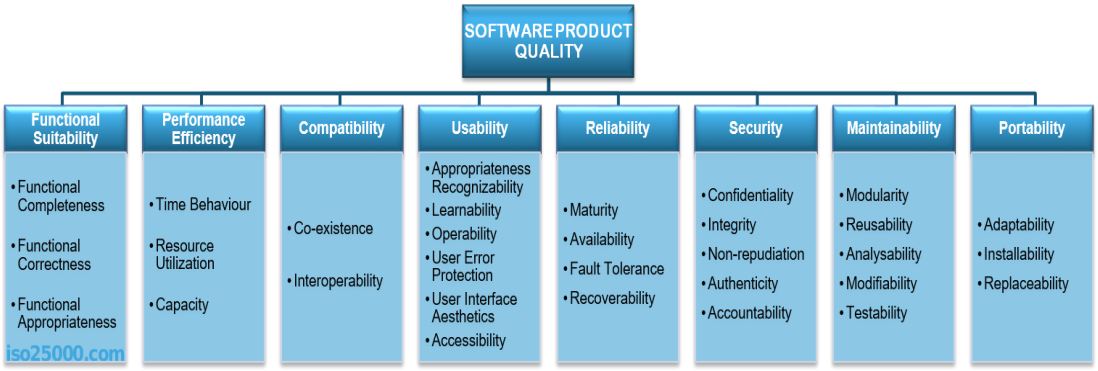


Figure 19. Quality Characteristics of ISO 25010 Standard

Population of the Study

This study on *Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth* involves three key groups: IT experts, Veterinary doctors, and poultry farmers. These groups provide essential perspectives needed to develop an effective and user-friendly monitoring system tailored to the needs of local poultry operations.

Poultry farmers offer practical insights on how the IoT system helps in daily environmental monitoring, supporting healthier chicken growth by

managing temperature, humidity, and ammonia on the farm. Their feedback ensures that the system meets real-world farming challenges and improves operational efficiency.

Veterinary doctors play a key role in monitoring and maintaining the health and welfare of the chickens. Their expertise ensures that the system's data on temperature, humidity, and air quality accurately reflects the birds' living conditions. Their feedback is essential in improving the system's functionality, particularly in aligning environmental monitoring with animal health standards and optimizing chicken growth and well-being.

IT experts focus on evaluating the technical framework, including system security, scalability, and the integration of IoT sensors with machine learning analytics. Their expertise guarantees that the system is reliable, adaptable, and capable of handling complex data for precise environmental quality monitoring.

By combining insights from IT experts, admins, and farmers, the study aims to build a robust, user-centered IoT-based environmental monitoring solution that supports sustainable and optimized poultry farming in Nasugbu, Batangas.

Table 11. Respondents of the Research Project

CATEGORY OF RESPONDENTS	SAMPLE
Farmers	30
IT Expert	5
Veterinarian	5
Total	**40**

Data Collection Instrument

To evaluate the accuracy and reliability of the IoT-based environmental monitoring system, the researcher utilized various assessment tools. The primary focus was to determine how well the system addressed the needs of poultry farmers by collecting feedback on its performance, ease of use, and overall acceptance. This feedback was crucial for identifying areas requiring improvement and for guiding enhancements to the system's functionality.

The researchers conducted structured surveys to measure the system's effectiveness in supporting poultry farmers with environmental monitoring. A 4-point Likert scale was employed to gauge the degree of agreement, eliminating neutral responses to encourage decisive feedback. The survey was handed out as printed questionnaires to collect responses directly from participants, making sure their answers were accurate and reliable.

For analyzing the data, the responses were scored as follows: 1 for Strongly Disagree, 2 for Disagree, 3 for Agree, and 4 for Strongly Agree. The mean score was calculated using the formula:

$$(\bar{x}) = \frac{\left(\Sigma\ (f\ \times\ x)\right)}{N}$$

Where f represents the frequency of each response, x is the numerical value assigned to that response, and N is the total number of respondents. This method provided a clear measure of user opinions regarding the system's effectiveness.

The standard deviation was also calculated to measure the dispersion of responses around the mean, using the formula:

$$s = \sqrt{\frac{\Sigma(\bar{x} - x)^2}{(n-1)}}$$

Where x represents each individual response, x is the mean score, and N is the total number of respondents. A lower standard deviation indicates that the responses were closely clustered around the mean, while a higher standard deviation shows more variation in user opinions.

Data encoding and analysis were performed using Microsoft Excel to ensure organized and accurate processing of survey results.

Parameter Evaluation and EQI Formula

The system utilized a quantitative scoring model to analyze the poultry environment based on three essential parameters: temperature, humidity, and ammonia concentration. These variables were evaluated relative to the threshold values assigned for every chicken age group. Each sensor reading was processed and converted into an equivalent environmental score, which was later aggregated to form the Environmental Quality Index (EQI). A value within the prescribed range automatically yielded a perfect score of 100, indicating an optimal poultry condition.

Temperature Scoring

Temperature was prioritized as the parameter with the strongest impact on poultry growth. Therefore, readings outside the acceptable thermal range

incurred a high penalty. The deviation was multiplied by a penalty factor to reduce the score proportionally.

$$Penalty = (difference \times 20)$$

$$TemperatureScore = 100 - Penalty$$

where *difference* refers to the magnitude of deviation below or above the ideal temperature tolerance.

Humidity Scoring

Humidity influenced bird comfort and respiratory regulation. It was assigned a moderate penalty weight compared to temperature. Scores decreased gradually as humidity deviated from the recommended range.

$$Penalty = (difference \times 2)$$

$$HumidityScore = 100 - Penalty$$

Ammonia Scoring

Ammonia concentration was evaluated relative to the maximum safe limit. Higher levels implied greater risk, therefore the scoring used proportional deduction.

$$AmmoniaScore = 100 - (\frac{Ammonia}{MaxAmmonia} \times 100)$$

Upon computing the individual parameters, the system generated the final Environmental Quality Index (EQI) through averaging. This value represents the overall condition of the poultry house. The EQI enables farmers and admins to quickly assess environmental status without analyzing each parameter separately. Regular monitoring of EQI trends helps identify recurring issues in temperature, humidity, or ammonia levels, and alerts can be generated when the EQI falls below a predefined threshold, prompting timely interventions to protect poultry health. By tracking the EQI over time, farm managers can evaluate the effectiveness of implemented interventions and adjust management strategies accordingly. Additionally, the EQI facilitates comparison between different poultry houses or production cycles, providing insights into overall farm performance.

$$EQI = \frac{TemperatureScore + HumidityScore + AmmoniaScore}{3}$$

A higher EQI signifies a safer and more suitable environment for chicken development, while lower values indicate threshold violations and potential risk factors. This standardized index also assists in prioritizing corrective actions, such as adjusting ventilation or humidity levels, and supports more informed management decisions across different poultry houses or farm locations.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presented the results and the discussion supporting the methodology in achieving the objectives.

Actual Arduino-based Smart Farming sensor setup displaying temperature, humidity, and ammonia readings.

This figure shows the prototype of the Arduino-based sensor module that we developed for the Smart Poultry Farming System. It has an LCD screen that displays real-time readings, a DHT22 sensor for measuring temperature and humidity, and an MQ137 sensor for detecting ammonia levels inside the poultry house. We also added manual control buttons so users can switch between modes when needed. These components work together to continuously monitor the environment, which is important for keeping the chickens healthy and growing well. The sensor module collects the data and sends it to the web-based dashboard, where it's processed and analyzed to compute the Environmental Quality Index (EQI).

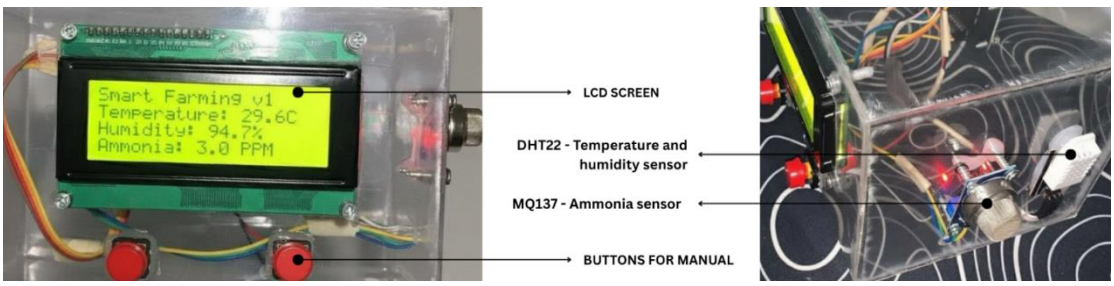


Figure 20. Smart Farming Sensor

Connection of the ESP8266 microcontroller and relay module used to automate environmental controls such as fans and lights based on sensor readings.

Figure 20 shows the ESP8266 microcontroller connected to a relay module, which serves as the automation part of the Smart Poultry Farming System. The ESP8266 acts as the main controller that receives data from sensors like the DHT22 and MQ137, processes the readings, and sends commands to control different devices. The relay module works like a switch that turns equipment such as fans and lights on or off depending on the environmental conditions detected by the sensors. With this setup, the system can automatically adjust the temperature, humidity, and air quality inside the poultry house to maintain a comfortable and healthy environment for the chickens. Overall, this setup highlights how the system combines IoT-based monitoring with automatic control to make poultry farm management more efficient.

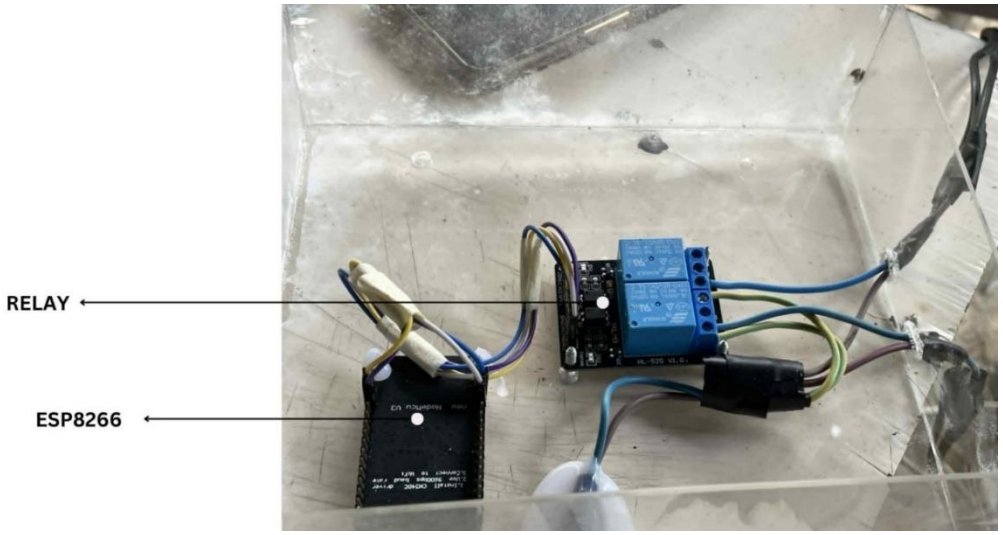


Figure 21. ESP8266 Microcontroller Connected to Relay

Chicken Pen used for system implementation.

Figure 22 shows the actual chicken pen where the Smart Poultry Farming System was tested and evaluated. As seen in figure 22 (*a*), the enclosure was made using simple, locally available materials that provided enough ventilation and protection for the chickens. To help control the temperature, four exhaust fans were installed, two on each side of the pen as shown in figure 22 (*b*). Inside the enclosure of figure 22, (*c*) shows the placement of the IoT-based sensors used to track temperature, humidity, and ammonia levels, making sure the data collected was accurate and consistent. Lastly, in figure 22 (*d*) shows how the system could be monitored and controlled remotely through a mobile interface. This setup helped the researchers observe how well the system could automatically maintain the right living conditions for the chickens and support their healthy growth and well-being.

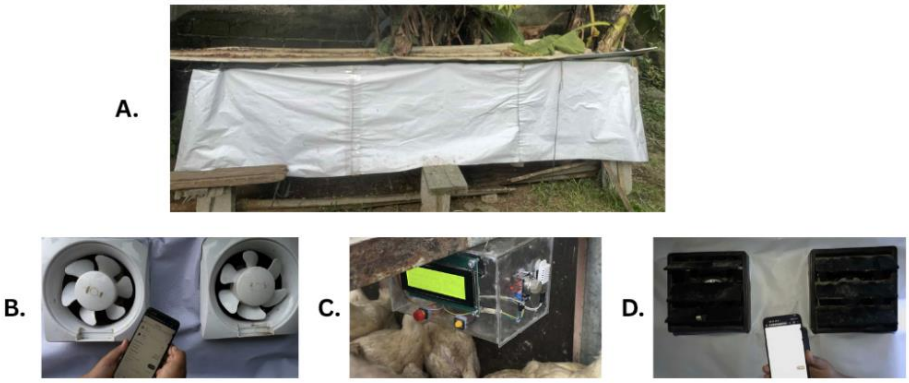
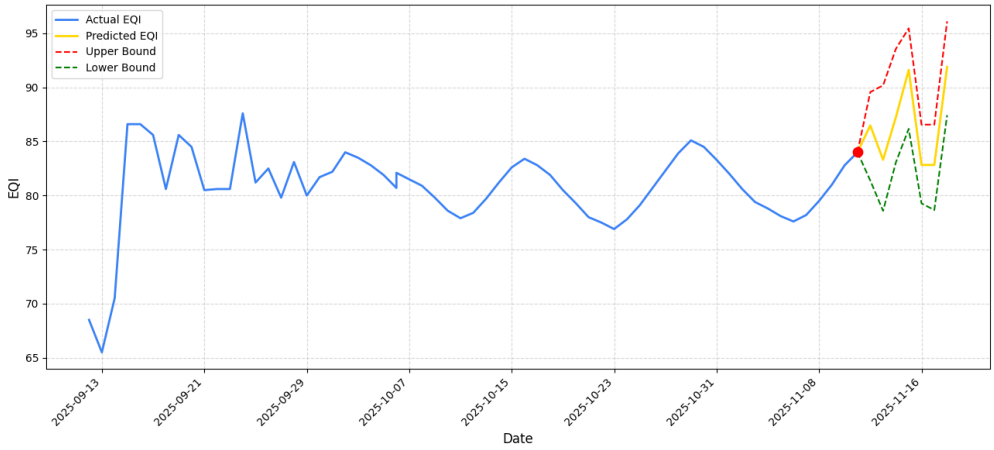


Figure 22. Chicken Pen used in the implementation

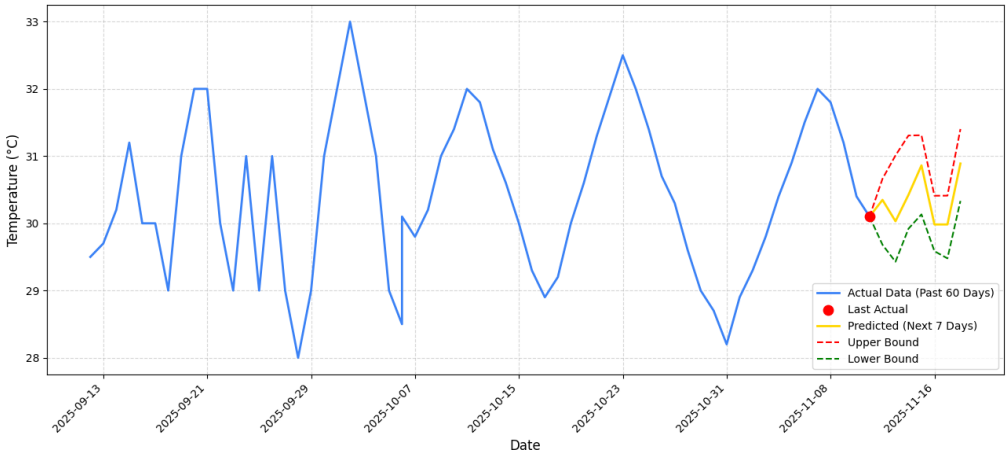
Predicting Optimal Environmental Conditions for Chicken Growth Using Pre-trained Machine Learning Models.

The figures forecasted environmental metrics for the next seven days, based on the preceding 60 days of actual data, where the red dot marks the last recorded actual value. Figure 23 (*a*) shows EQI (Environmental Quality Index), which has been generally decreasing over the past 60 days, settling around 67% recently, and is predicted to remain relatively stable in the 66% to 68% range for the next week. As shown in figure 23 (*b*) illustrates the Temperature, which exhibits significant daily fluctuations, recently measured at about 30.1 °C, the forecast predicts continued, though slightly muted, oscillations in the 30 °C to 31°C range, with a possible peak near 31°C. In figure 23 (*c*) presents the Humidity, which has been in the 75 to 85 range recently, peaking at 84 at the last actual reading; the prediction shows an expected increase, fluctuating between approximately 83 and 91 over the next seven days. Finally, in figure 23 (*d*) tracks ammonia, which has consistently fluctuated around 2.5 ppm, the prediction shows a sharp, yet uncertain, increase for the next week, with the forecast median rising to a peak around 3.6 ppm and a wide prediction band ranging from 2.1ppm to 4.0 ppm.

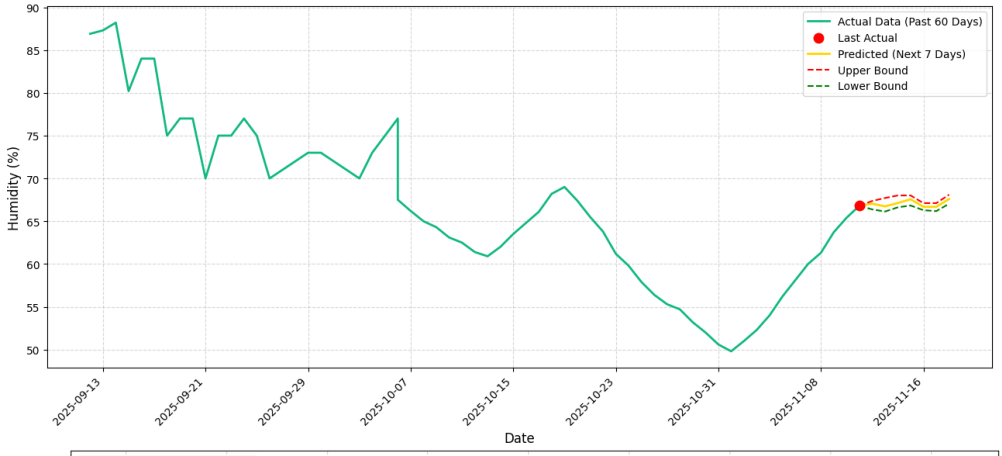
A.



B.



C.



D.

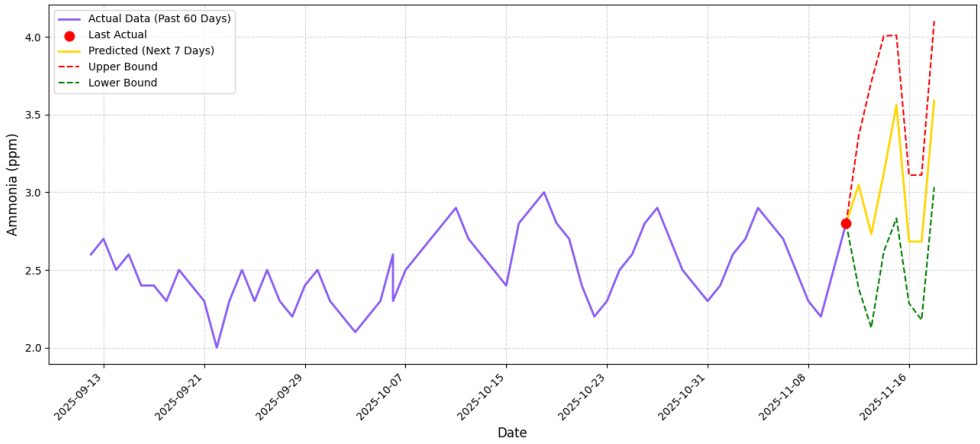


Figure 23. Prediction Trends

Developed a user-friendly web-based system that displays real-time farm data, EQI trends, and uses machine learning to provide smart insights for better farm management.

Figures show a clean, modern design with an emphasis on data visualization and accessibility. Both views feature a left-hand navigation sidebar with an orange and white color scheme, containing icons for Dashboards, Monitoring, Notification, and Settings, along with a prominent orange circle with a plus sign for quick actions. Shown in figure 24 (*a*), which appears to be the main dashboard/home screen, provides an overview of real-time status and a large central EQI Prediction for the Next 7 Days graph. Figure 24 (*b*), likely the dedicated "Monitoring" section, is a detailed view that organizes information into four cards: Temperature Data, Humidity Data, Ammonia Data, and Chicken Count, each providing a separate chart (historical data plus 7-day forecast) or a summary visualization (the donut chart), along with buttons to Download Data and Download Chart, maintaining a clear, modular layout focused on comprehensive data analysis.

A.



B.



Figure 24. System Dashboard & Prediction

Test and Evaluate the Software using ISO 25010

Table 12 presents the IT Specialist's evaluation of the IoT system's Functional Suitability, resulting in an overall mean score 3.47 (SD = 0.48), which is interpreted as Strongly Agree. This indicates a high level of confidence in the system's core capabilities from a technical expert. The strongest agreement was recorded for Functional Completeness (3.80, SD = 0.45), confirming the system covers all specified tasks like sensor data collection, computing EQI,

and generating growth predictions. The specialist also Strongly Agreed on Functional Correctness (3.40, SD = 0.55), suggesting the system provides accurate results with the necessary precision. However, the item Functional Appropriateness received a lower mean score of 3.20 (SD= 0.45), placing it in the less emphatic Agree category (2.51-3.25), which suggests a slight reservation about the degree to which the system helps accomplish the ultimate goal of monitoring poultry house conditions and supporting farm decisions.

Table 12. Functional Suitability based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Functional Completeness – The system covers all the specified tasks such as collecting sensor data, computing EQI, and generating growth predictions.	3.80	0.45	Strongly Agree
2. Functional Correctness – The system provides accurate sensor readings and EQI results with the needed degree of precision.	3.40	0.55	Strongly Agree
3. Functional Appropriateness – The system helps accomplish the goal of monitoring poultry house conditions and supporting farm decisions.	3.20	0.45	Agree
Composite Mean	**3.47**	**0.48**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 13 summarizes the IT Specialist's assessment of the IoT system's Performance Efficiency, which is highly positive with an overall mean score of 3.47 (SD = 0.52), interpreted as Strongly Agree. This strong result confirms the system's technical effectiveness in handling its workload. The highest agreement was recorded for Capacity (Mean = 3.60, SD = 0.52), indicating a strong belief that the system can manage multiple data inputs from sensors

without experiencing performance degradation. The specialist also Strongly Agreed on the system's efficient Time Behavior (3.40, SD = 0.52), affirming that real-time sensor data is processed and displayed quickly without delays, and its effective Resource Utilization (3.40, SD = 0.52), ensuring the system runs without slowing down or overloading the web server or connected devices. Collectively, these findings validate the system's stability and speed under operational conditions from a technical perspective.

Table 13. Performance Efficiency based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Time Behavior – The system processes and displays real-time sensor data quickly and without delays.	3.40	0.52	Strongly Agree
2. Resource Utilization – The system runs efficiently without slowing down or overloading the web server or connected devices.	3.40	0.52	Strongly Agree
3. Capacity – The system can handle multiple data inputs from sensors without performance issues.	3.60	0.52	Strongly Agree
Composite Mean	**3.47**	**0.52**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 14 summarizes the IT Specialist's assessment of the IoT system's Compatibility, resulting in a highly favorable overall mean score of 3.50 (SD = 0.55), which is interpreted as Strongly agree. This strong agreement confirms the system's technical soundness in integrating with existing environments. The specialist Strongly Agreed most strongly with the concept of Co-existence (Mean = 3.60, SD = 0.55), indicating high confidence that the system runs properly while effectively sharing resources such as the internet, local network,

and databases with other applications. High agreement was also found for Interoperability (Mean = 3.40, SD = 0.55), affirming that the system can effectively exchange data among its internal components, including the sensors, database, and the web dashboard. These findings collectively validate that the IoT system is technically equipped to function reliably within a typical farm's IT infrastructure. Additionally, the table highlights individual dimension scores, allowing readers to identify specific strengths in the system's technical integration. The standard deviation values indicate consistency in the specialist's evaluation across all assessed parameters. Overall, Table 20 provides a clear quantitative representation of the system's compatibility performance, supporting its implementation within real-world poultry farm operations.

Table 14. Compatibility based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Co-existence – The system runs properly while sharing resources (internet, local network, databases) with other applications.	3.60	0.55	Strongly Agree
2. Interoperability – The system can exchange data effectively between the sensors, database, and web dashboard.	3.40	0.55	Strongly Agree
Composite Mean	**3.50**	**0.55**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 15 presents the IT Specialist's assessment of the IoT system's Interaction Capability, resulting in an overall mean score of 3.28 (SD = 0.27), which is interpreted as Strongly Agree. This score indicates a high degree of

confidence in the system's overall usability. Strong agreement (Mean = 3.40) was found across several key usability aspects, including Appropriateness Recognizability, Learnability, User Error Protection, Inclusivity, and User Assistance. However, three components were rated as Agree (2.51-3.25): Operability (3.20, SD = 0.45), User Engagement (3.00, SD = 0.00), and Self-Descriptiveness (3.00, SD = 0.00). The lower scores for Operability, Engagement, and Self-Descriptiveness suggest that while the system is highly learnable and protective, the IT specialist perceives slight room for improvement in making the daily operation more immediately intuitive, engaging for continuous use, and clearer in explaining its own functions without extra assistance.

Table 15. Interaction Capability based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Appropriateness Recognizability – Users can easily recognize that the system is suitable for monitoring poultry farm conditions.	3.40	0.55	Strongly Agree
2. Learnability – Users can quickly learn how to use the system to monitor farm conditions.	3.40	0.55	Strongly Agree
3. Operability – The system is easy to operate and control for daily farm monitoring tasks.	3.20	0.45	Agree
4. User Error Protection – The system minimizes user errors and provides clear feedback when errors occur.	3.40	0.55	Strongly Agree
5. User Engagement – The system keeps users actively engaged and motivated while monitoring poultry farm conditions.	3.00	0.00	Agree
6. Inclusivity – The system can be easily used by people with different technical skills or experience levels.	3.40	0.55	Strongly Agree
7. User Assistance – The system provides helpful instructions, guides, or tooltips when users need support.	3.40	0.55	Strongly Agree
8. Self-Descriptiveness – The system clearly describes its functions and actions, so users can understand what to do without extra help.	3.00	0.00	Agree
Composite Mean	**3.28**	**0.27**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 16 presents the IT Specialist's evaluation of the IoT system's Reliability, resulting in a strong overall mean score of 3.45 (SD = 0.55), which is interpreted as Strongly Agree. This score indicates a high degree of confidence from a technical expert in the system's ability to operate dependably. The highest agreement was recorded for Faultlessness (Mean = 3.60, SD = 0.55), affirming that the system performs its functions accurately and consistently without generating incorrect data or readings. Furthermore,

the specialist Strongly Agreed that the system maintains high Availability (Mean = 3.40), is capable of Fault Tolerance (Mean = 3.40) by operating correctly even if sensors temporarily fail, and demonstrates strong Recoverability (Mean = 3.40), allowing it to resume normal operation after interruptions like power loss or errors. These results collectively validate the system's technical robustness and resilience under various operational scenarios. In addition, the table provides a clear overview of individual reliability dimensions, highlighting specific areas where the system excels. The low standard deviation values indicate consistent expert agreement across all assessed attributes. Overall, Table 22 offers a quantitative and comprehensive representation of the IoT system's reliability, supporting its suitability for continuous use in smart poultry farm operations.

Table 16. Reliability based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Faultlessness – The system performs its functions accurately and consistently without producing wrong EQI data or readings.	3.60	0.55	Strongly Agree
2. Availability – The system is always accessible and ready for use whenever the poultry environment needs to be monitored.	3.40	0.55	Strongly Agree
3. Fault Tolerance – The system continues to operate correctly even when some IoT sensors temporarily fail or disconnect.	3.40	0.55	Strongly Agree
4. Recoverability – The system can recover data and resume normal operation automatically after power loss, internet interruption, or system error.	3.40	0.55	Strongly Agree
Composite Mean	**3.45**	**0.55**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 17 presents the IT Specialist's assessment of the IoT system's Security, resulting in an overall mean score of 3.27 (SD = 0.41), which is interpreted as Strongly agree. This indicates a positive view of the system's security features, though closer examination reveals nuance. The specialist Strongly Agreed on two critical areas: Integrity (3.60, SD = 0.55), confirming that the system highly protects data from unauthorized modification, and Accountability (3.40, SD = 0.55), noting the presence of detailed user action logs. However, four components, Confidentiality, Non-repudiation, and Authenticity (all at 3.20), and Resistance (3.00) all fell within the less emphatic Agree range (2.51-3.25). This suggests that while core data integrity and user accountability are strong, the specialist sees potential for improvement in areas such as restricting access to authorized users, verifying user identity, providing non-repudiation of actions, and, particularly, bolstering the system's ability to withstand external cyber threats.

Table 17. Security based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Confidentiality – The system ensures that poultry farm data and EQI records are only accessible to authorized users.	3.20	0.45	Agree
2. Integrity – The system protects data from being modified, deleted, or corrupted by errors or unauthorized actions.	3.60	0.55	Strongly Agree
3. Non-repudiation – The system records all user activities, ensuring that any action taken within the system can be verified later.	3.20	0.45	Agree
4. Accountability – The system keeps detailed logs that can trace every user's action during system operation.	3.40	0.55	Strongly Agree
5. Authenticity – The system verifies the identity of users to ensure that only legitimate accounts can access and manage data.	3.20	0.45	Agree
6. Resistance – The system can withstand potential cyber threats, hacking attempts, or unauthorized access without compromising data security.	3.00	0.00	Agree
Composite Mean	**3.27**	**0.41**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 18 summarizes the IT Specialist's assessment of the IoT system's Maintainability, resulting in an overall mean score of 3.24 (SD = 0.31), which is interpreted as Agree. This indicates a moderate level of confidence in the system's ease of upkeep. The strongest agreement was found for Modifiability (3.60, SD = 0.55), confirming that the system can be updated or improved without causing new errors or performance problems. The specialist also Strongly Agreed that the system is built with Modularity (3.40) composed of separate parts to prevent localized changes from affecting the whole. However, the components of Reusability (3.20), Analyzability (3.00), and Testability (3.00)

all fall within the less emphatic Agree range. This suggests that the specialist perceives room for improvement in making system components readily reusable for other applications, providing clearer logs for easier error analysis, and facilitating the testing of new updates to ensure they work as intended.

Table 18. Maintability based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Modularity – Is the system composed of separate parts (IoT sensors, database, dashboard, ML model) so that changes in one part do not affect the others?	3.40	0.55	Strongly Agree
2. Reusability – Can the system's modules, such as the sensor setup or data-processing features, be reused for other smart farming systems or applications?	3.20	0.45	Agree
3. Analysability – Does the system provide clear logs or messages that help developers identify and analyze errors or failures easily?	3.00	0.00	Agree
4. Modifiability – Can the system be modified, updated, or improved without causing new errors or performance problems?	3.60	0.55	Strongly Agree
5. Testability – Can the system be tested easily to ensure that each new update or feature works as intended?	3.00	0.00	Agree
Composite Mean	**3.24**	**0.31**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 19 presents the IT Specialist's assessment of the IoT system's Flexibility, which resulted in a moderate overall mean score of 3.10 (SD = 0.22), interpreted as Agree. This score suggests that while the system is considered adequately flexible, it is not emphatically endorsed by the technical expert in this area. The highest mean scores 3.20 were recorded for both Adaptability and Scalability, indicating a slightly higher, though still moderate, agreement

that the system can function properly on different devices and handle additional users or data inputs without affecting performance. Conversely, the components of Install ability and Replaceability both received lower mean scores of 3.00. This suggests that the specialist perceives minor limitations regarding the ease of initial installation configuration and the ability of the system to be upgraded or replaced with a new version without disrupting ongoing operations or causing data loss.

Table 19. Flexibility based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Adaptability – Can the system adapt and function properly on different devices (such as desktop, tablet, or mobile) and browsers without issues?	3.20	0.45	Agree
2. Scalability – Can the system handle additional users, sensors, or data inputs without slowing down or affecting its performance?	3.20	0.45	Agree
3. Installability – Can the system be easily installed, accessed, or configured using a web browser without requiring complex setup?	3.00	0.00	Agree
4. Replaceability – Can the system be upgraded or replaced with a new version without disrupting ongoing operations or losing data?	3.00	0.00	Agree
Composite Mean	**3.10**	**0.22**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 20 presents the IT Specialist's assessment of the IoT system's Safety, resulting in an overall mean score of 3.12 (SD = 0.20), which is interpreted as Agree. This suggests a moderate but not emphatic endorsement of the system's safety features from the technical expert. The only item to receive a Strongly Agree interpretation was Fail Safe (3.40, SD = 0.55),

confirming a strong belief in the system's ability to automatically activate a safety response when errors occur. Conversely, four safety components, Operational Constraint (3.20), Risk Identification (3.00), Hazard Warning (3.00), and Safe Integration (3.00) all fell within the less emphatic Agree range. This indicates that while the system's automatic recovery is reliable, the IT specialist perceives potential for improvement in areas such as preventing unsafe actions, detecting potential risks, alerting the user to problems, and ensuring secure integration with external devices without causing safety risks.

Table 20. Safety based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Operational Constraint – Does the system operate safely under normal conditions and prevent unsafe actions or data loss during monitoring?	3.20	0.45	Agree
2. Risk Identification – Can the system detect and identify possible risks, such as inaccurate sensor readings or device malfunctions?	3.00	0.00	Agree
3. Fail Safe – Does the system automatically activate a safety response (like data recovery or system reset) when an error or failure occurs?	3.40	0.55	Strongly Agree
4. Hazard Warning – Does the system alert or notify the user immediately when there are abnormal readings or system problems that may affect safety?	3.00	0.00	Agree
5. Safe Integration – Can the system connect and integrate with IoT devices or external platforms securely without causing harm or security risks?	3.00	0.00	Agree
Composite Mean	**3.12**	**0.20**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

The results of the table 21 shows that IT specialists strongly agree that the system meets most core software quality standards, with high ratings in

Functional Suitability (3.73), Performance Efficiency (3.47), Compatibility (3.50), Reliability (3.45), Security (3.27), Maintainability (3.30), and Interaction Capability (3.28), indicating that the system performs its intended functions well, operates efficiently, remains stable, protects data, and is easy to maintain and use. Flexibility (3.18) and Safety (3.12) received “Agree,” suggesting these areas are acceptable but still have room for improvement. Overall, the system obtained an Overall Mean of 3.36, interpreted as “Strongly Agree,” demonstrating that IT specialists view the system as high-quality and aligned with ISO 25010 software standards.

Table 21. Result of ISO 25010 based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
Functional Suitability	3.73	0.48	Strongly Agree
Performance Efficiency	3.47	0.48	Strongly Agree
Compatibility	3.50	0.55	Strongly Agree
Interaction Capability	3.28	0.53	Strongly Agree
Reliability	3.45	0.55	Strongly Agree
Security	3.27	0.41	Strongly Agree
Maintainability	3.30	0.31	Strongly Agree
Flexibility	3.18	0.34	Agree
Safety	3.12	0.20	Agree
Overall Mean	**3.36**	**0.43**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 22, which presents the veterinarian's perception of the IoT system's Functional Suitability (a component of Perceived Usefulness),

demonstrates an overwhelmingly positive assessment, with an overall mean score of 3.73 and a standard deviation of 0.44, which is interpreted as Strongly Agree. The veterinarian provided the highest possible rating of 4.00 (SD = 0.32) for Functional Appropriateness, strongly agreeing that the system is highly effective in achieving the core goal of monitoring poultry house conditions and supporting farm decisions. Similarly high scores were given to Functional Completeness (3.60, SD = 0.48), confirming the system covers all necessary tasks like collecting sensor data and generating predictions, and Functional Correctness (3.60, SD = 0.52), indicating the system provides accurate sensor readings and results with the required precision. These results affirm that, from an expert perspective, the IoT system is functionally robust, reliable, and highly suitable for its intended application in poultry management.

Table 22. Functional Suitability based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Functional Completeness – The system covers all the specified tasks such as collecting sensor data, computing EQI, and generating growth predictions.	3.60	0.48	Strongly Agree
2. Functional Correctness – The system provides accurate sensor readings and EQI results with the needed degree of precision.	3.60	0.52	Strongly Agree
3. Functional Appropriateness – The system helps accomplish the goal of monitoring poultry house conditions and supporting farm decisions.	4.00	0.32	Strongly Agree
Composite Mean	**3.73**	**0.44**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 23, which reports the veterinarian's assessment of the IoT system's Performance Efficiency, shows a highly favorable evaluation, resulting in an overall mean score of 3.47 (SD = 0.52), which is interpreted as Strongly Agree. This strong agreement indicates that the system is deemed robust and reliable in its operation. Specifically, the veterinarian strongly agreed that the system effectively handles Capacity, being able to manage multiple data inputs from sensors without performance issues (Mean = 3.60, SD = 0.52). High agreement was also found for Time behavior (3.40, SD = 0.52), confirming that the system processes and displays real-time sensor data quickly and without significant delays, and Resource Utilization (3.40, SD = 0.52), indicating the system runs efficiently without slowing down or overloading its web server or connected devices. Collectively, these results attest that, from an expert perspective, the system performs optimally under typical operational demands.

Table 23. Performance Efficiency based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Time Behaviour – The system processes and displays real-time sensor data quickly and without delays.	3.40	0.52	Strongly Agree
2. Resource Utilization – The system runs efficiently without slowing down or overloading the web server or connected devices.	3.40	0.52	Strongly Agree
3. Capacity – The system can handle multiple data inputs from sensors without performance issues.	3.60	0.52	Strongly Agree
Composite Mean	**3.47**	**0.52**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 24, which presents the veterinarian's evaluation of the IoT system's Compatibility, yields a highly positive assessment, with an overall mean score of 3.50 (SD = 0.52), interpreted as Strongly Agree. This strong result confirms the system's ability to integrate seamlessly with existing technological environments. Specifically, the veterinarian Strongly Agreed that the system is effective in Co-existence (Mean = 3.60, SD = 0.52), meaning it runs properly and shares resources with other applications like the internet, local network, and databases without conflicts. They also Strongly Agreed on the system's high Interoperability (Mean = 3.40, SD = 0.52), indicating it can effectively exchange data among all its components, including the sensors, database, and the web dashboard. These findings affirm that the IoT system is technologically sound and ready for real-world deployment alongside existing farm technology infrastructure.

Table 24. Compatibility based on Veterinarian’s Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Co-existence – The system runs properly while sharing resources (internet, local network, databases) with other applications.	3.60	0.52	Strongly Agree
2. Interoperability – The system can exchange data effectively between the sensors, database, and web dashboard.	3.40	0.52	Strongly Agree
Composite Mean	**3.50**	**0.52**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 25, which assesses the veterinarian’s perception of the IoT system’s Interaction Capability, shows a very high level of agreement regarding

its usability, yielding an overall mean score of 3.43 (SD = 0.45), interpreted as Strongly Agree. The highest rated item is Operability (Mean = 3.80, SD = 0.53), indicating strong agreement that the system is easy to operate and control for daily monitoring tasks. Similarly high scores were given to Self-Descriptiveness (3.60), and strong agreement was found for factors like Appropriateness Recognizability, Learnability, User Error Protection, Inclusivity, and User Assistance (all with a mean of 3.40). However, the single point of deviation is User Engagement, which received a mean of 3.00 (SD = 0.00), falling into the less emphatic Agree category, suggesting that while the system is highly usable and functional, there may be slight room for improvement in features that actively keep users engaged and motivated during continuous monitoring.

Table 25. Interaction Capability based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Appropriateness Recognizability – Users can easily recognize that the system is suitable for monitoring poultry farm conditions.	3.40	0.52	Strongly Agree
2. Learnability – Users can quickly learn how to use the system to monitor farm conditions.	3.40	0.52	Strongly Agree
3. Operability – The system is easy to operate and control for daily farm monitoring tasks.	3.80	0.53	Strongly Agree
4. User Error Protection – The system minimizes user errors and provides clear feedback when errors occur.	3.40	0.52	Strongly Agree
5. User Engagement – The system keeps users actively engaged and motivated while monitoring poultry farm conditions.	3.00	0.00	Agree
6. Inclusivity – The system can be easily used by people with different technical skills or experience levels.	3.40	0.52	Strongly Agree
7. User Assistance – The system provides helpful instructions, guides, or tooltips when users need support.	3.40	0.52	Strongly Agree
8. Self-Descriptiveness – The system clearly describes its functions and actions, so users can understand what to do without extra help.	3.60	0.48	Strongly Agree
Composite Mean	**3.43**	**0.45**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 26, which presents the veterinarian's evaluation of the IoT system's Reliability, indicates a high degree of confidence in the system's dependable operation, resulting in an overall mean score of 3.60 (SD = 0.51), which is interpreted as Strongly Agree. The highest mean scores (3.80) were recorded for both Faultlessness and Recoverability. This signifies a strong agreement that the system performs functions accurately and consistently without producing erroneous data, and that it can efficiently recover and resume

operation following interruptions like power loss or system errors. Furthermore, the veterinarian Strongly Agreed that the system maintains high Availability (Mean = 3.40) for monitoring, and demonstrates strong Fault Tolerance (Mean = 3.40), allowing it to continue operating correctly even if some IoT sensors temporarily fail or disconnect. These findings confirm the system's robustness, accuracy, and resilience under various operating conditions, which are critical traits for reliable farm management technology.

Table 26. Reliability based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Faultlessness – The system performs its functions accurately and consistently without producing wrong EQI data or readings.	3.80	0.48	Strongly Agree
2. Availability – The system is always accessible and ready for use whenever the poultry environment needs to be monitored.	3.40	0.52	Strongly Agree
3. Fault Tolerance – The system continues to operate correctly even when some IoT sensors temporarily fail or disconnect.	3.40	0.52	Strongly Agree
4. Recoverability – The system can recover data and resume normal operation automatically after power loss, internet interruption, or system error.	3.80	0.52	Strongly Agree
Composite Mean	**3.60**	**0.51**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 27 presents the veterinarian's evaluation of the IoT system's Security, yielding an overall mean score of3.30 (SD = 0.44), which is interpreted as Strongly Agree (based on the 3.26-4.0 range). This result suggests a generally high confidence in the system's security posture. The highest mean scores were recorded for Integrity (3.60, SD = 0.52) and Confidentiality (3.40,

SD = 0.48), indicating strong agreement that the system effectively protects data from modification or unauthorized corruption, and ensures data is only accessible to authorized users. However, four security components non-repudiation, Authenticity, Accountability, and Resistance each received a mean score of 3.20, which falls into the less definitive Agree category (2.51-3.25). This suggests that while the system's core data integrity and access control are strongly affirmed, there is a perceived margin for improvement in areas like detailed user activity logging, verification of user identity, and resistance to external cyber threats.

Table 27. Security based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Confidentiality – The system ensures that poultry farm data and EQI records are only accessible to authorized users.	3.40	0.48	Strongly Agree
2. Integrity – The system protects data from being modified, deleted, or corrupted by errors or unauthorized actions.	3.60	0.52	Strongly Agree
3. Non-repudiation – The system records all user activities, ensuring that any action taken within the system can be verified later.	3.20	0.42	Agree
4. Accountability – The system keeps detailed logs that can trace every user's action during system operation.	3.20	0.48	Agree
5. Authenticity – The system verifies the identity of users to ensure that only legitimate accounts can access and manage data.	3.20	0.42	Agree
6. Resistance – The system can withstand potential cyber threats, hacking attempts, or unauthorized access without compromising data security.	3.20	0.32	Agree
Composite Mean	**3.30**	**0.44**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 28 presents the veterinarian's assessment of the IoT system's Maintainability, resulting in an overall mean score of 3.08 (SD = 0.31), which is interpreted as Agree. This suggests that while the system is considered maintainable, the endorsement is less emphatic than for other quality factors. The components of Modifiability and Testability received the highest means (3.20), indicating that the system can be modified, updated, and tested without significant performance issues. However, the factors of Modularity, Reusability, and Analyzability all received a mean score of 3.00, placing them comfortably in the Agree category. This implies that while the system is structured with separable parts and provides logs for error analysis, and its modules can be potentially reused, there is perceived room for improvement in these areas to optimize long-term maintenance, especially regarding the system's internal structure and ease of fault identification by developers.

Table 28. Maintability based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Modularity – Is the system composed of separate parts (IoT sensors, database, dashboard, ML model) so that changes in one part do not affect the others?	3.00	0.42	Agree
2. Reusability – Can the system's modules, such as the sensor setup or data-processing features, be reused for other smart farming systems or applications?	3.00	0.32	Agree
3. Analysability – Does the system provide clear logs or messages that help developers identify and analyze errors or failures easily?	3.00	0.00	Agree
4. Modifiability – Can the system be modified, updated, or improved without causing new errors or performance problems?	3.20	0.52	Agree
5. Testability – Can the system be tested easily to ensure that each new update or feature works as intended?	3.20	0.32	Agree
Composite Mean	**3.08**	**0.31**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 29 presents the veterinarian's assessment of the IoT system's Flexibility, yielding an overall mean score of 3.05 (SD = 0.24), which is interpreted as Agree. This suggests that while the system is generally considered flexible, the endorsement is moderate compared to the system's core functional and performance qualities. The highest mean was recorded for Install ability (3.20, SD = 0.32), indicating a high degree of agreement that the system can be easily installed, accessed, or configured using a web browser without requiring a complex setup. However, the components of Adaptability (functioning on different devices), Scalability (handling additional users or data inputs), and Replaceability (upgrading to a new version) all received a mean

score of 3.00. These results imply that while the system is easy to deploy, the veterinarian perceives potential, albeit minor, limitations regarding the system's long-term ability to adapt to diverse devices, scale up its operation with growing demand, or undergo seamless future upgrades.

Table 29. Flexibility based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Adaptability – Can the system adapt and function properly on different devices (such as desktop, tablet, or mobile) and browsers without issues?	3.00	0.32	Agree
2. Scalability – Can the system handle additional users, sensors, or data inputs without slowing down or affecting its performance?	3.00	0.32	Agree
3. Installability – Can the system be easily installed, accessed, or configured using a web browser without requiring complex setup?	3.20	0.32	Agree
4. Replaceability – Can the system be upgraded or replaced with a new version without disrupting ongoing operations or losing data?	3.00	0.00	Agree
Composite Mean	**3.05**	**0.24**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 30 presents the veterinarian's assessment of the IoT system's Safety features, yielding an overall mean score of 3.48 (SD = 0.47), which is interpreted as Strongly Agree. This strong result confirms the system is viewed as safe and reliable in preventing harm and mitigating risks during operation. The highest mean score (3.80, SD = 0.53) was attributed to Operational Constraint, indicating a high level of agreement that the system operates safely and prevents unsafe actions or data loss under normal conditions. Strong

agreement was also recorded for Safe Integration (3.60), Risk Identification (3.40), and Hazard Warning (3.40). However, the item Fail Safe received a mean score of 3.20 (SD = 0.48), placing it in the less emphatic Agree category, suggesting that while the system has high overall safety, the automatic activation of safety responses (like data recovery or system reset) when errors occur is viewed as slightly less assured than the other safety components.

Table 30. Safety based on Veterinarian's Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Operational Constraint – Does the system operate safely under normal conditions and prevent unsafe actions or data loss during monitoring?	3.80	0.53	Strongly Agree
2. Risk Identification – Can the system detect and identify possible risks, such as inaccurate sensor readings or device malfunctions?	3.40	0.42	Strongly Agree
3. Fail Safe – Does the system automatically activate a safety response (like data recovery or system reset) when an error or failure occurs?	3.20	0.48	Agree
4. Hazard Warning – Does the system alert or notify the user immediately when there are abnormal readings or system problems that may affect safety?	3.40	0.42	Strongly Agree
5. Safe Integration – Can the system connect and integrate with IoT devices or external platforms securely without causing harm or security risks?	3.60	0.48	Strongly Agree
Composite Mean	**3.48**	**0.47**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 31 shows a strong positive perception of the system's overall quality, with high ratings in Functional Suitability (3.73), Performance Efficiency (3.47), Compatibility (3.50), Interaction Capability (3.43), Reliability (3.60), Security (3.30), and Safety (3.48), all interpreted as "Strongly Agree," indicating

that the system effectively performs its intended functions, operates efficiently, integrates well with different platforms, supports smooth user interaction, maintains stability, safeguards data, and ensures safe operation. Meanwhile, Maintainability (3.08) and Flexibility (3.10) received an “Agree,” suggesting these areas are acceptable but could benefit from further enhancement to improve ease of updates and adaptability. With an Overall Mean of 3.67, interpreted as “Strongly Agree,” the results confirm that veterinarians view the system as highly reliable, functional, and aligned with ISO 25010 quality standards.

Table 31. Result of ISO 25010 based on Veterinarian’s Response

INDICATORS	MEAN	SD	INTERPRETATION
Functional Suitability	3.73	0.37	Strongly Agree
Performance Efficiency	3.47	0.55	Strongly Agree
Compatibility	3.50	0.55	Strongly Agree
Interaction Capability	3.43	0.47	Strongly Agree
Reliability	3.60	0.50	Strongly Agree
Security	3.30	0.48	Strongly Agree
Maintainability	3.08	0.18	Agree
Flexibility	3.10	0.23	Agree
Safety	3.48	0.51	Strongly Agree
Overall Mean	**3.67**	**0.52**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Results of Survey Answered by the Respondents

Table 32 presents the respondents' assessment of the Perceived Usefulness of the IoT-based system in poultry farming. The results show an overall mean of 3.55 with a standard deviation of 0.54, interpreted as "Strongly Agree." This indicates that respondents highly recognize the system's usefulness in improving poultry operations. Specifically, they agree that IoT devices effectively monitor poultry health (mean = 3.74), provide real-time environmental updates (mean = 3.40), help maintain ideal conditions (mean = 3.54), and contribute to better chicken growth performance (mean = 3.51). Overall, the findings suggest that the IoT-based system is perceived as a valuable tool in enhancing productivity and management efficiency in poultry farming.

Table 32. Perceived Usefulness based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. IoT devices in poultry farming help monitor the health and well-being of poultry.	3.74	0.44	Strongly Agree
2. IoT devices in poultry farming provide real-time monitoring and notifications about environmental conditions (temperature, humidity, and ammonia).	3.40	0.60	Strongly Agree
3. The IoT-based system helps maintain ideal environmental conditions inside the poultry house.	3.54	0.61	Strongly Agree
4. The system contributes to improving the growth performance of chickens.	3.51	0.51	Strongly Agree
Composite Mean	**3.55**	**0.54**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 33 presents the respondents' evaluation of the Perceived Ease of Use of the IoT-based system in poultry farming. The results show an overall mean of 3.39 with a standard deviation of 0.57**,** interpreted as "Strongly Agree." This indicates that the respondents found the system easy to use and operate. They strongly agreed that the IoT devices are simple to manage for daily farm monitoring (mean = 3.46), the system is user-friendly for overseeing poultry health and environment (mean = 3.51), and alerts are easy to understand (mean = 3.43). Although the web dashboard was rated slightly lower (mean = 3.14, "Agree"), the overall findings suggest that the system is generally accessible, intuitive, and convenient for farmers to operate effectively.

Table 33. Perceived Ease of Used based on Respondents' Response

	INDICATORS	MEAN	SD	INTERPRETATION
1.	It is easy to operate the IoT devices for poultry farm monitoring on a daily basis.	3.46	0.56	Strongly Agree
2.	The system is user-friendly for farmers managing poultry health and environmental conditions.	3.51	0.61	Strongly Agree
3.	The web dashboard/interface for viewing sensor data is easy to navigate.	3.14	0.60	Agree
4.	Receiving and understanding real-time alerts from the system is straightforward.	3.43	0.50	Strongly Agree
Composite Mean		**3.39**	**0.57**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 34, which evaluates the level of Personal Innovation among the study's respondents, strongly suggests a high degree of technological readiness and openness to change within the group. The overall mean score for Personal Innovation is (3.46) with a standard deviation of (0.55) is

categorized as Strongly Agree. This result is consistently supported by the individual items, all of which fall within the "Strongly Agree" range. The highest mean was recorded for feeling excited about advanced technologies in farm operations (3.57), while the lowest was for being open to new IoT systems (3.37). Collectively, the findings indicate that the respondents are highly inclined to seek, experiment with, and integrate novel technologies and innovative systems into their poultry farming practices, reflecting a proactive and progressive mindset essential for modern agricultural adoption.

Table 34. Personal Innovation based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I enjoy trying out new technologies related to poultry farming.	3.49	0.61	Strongly Agree
2. I like experimenting with innovative systems that can improve poultry production.	3.43	0.50	Strongly Agree
3. I consider myself open to using new IoT systems in my work.	3.37	0.55	Strongly Agree
4. I feel excited to use advanced technologies in my farm operations.	3.57	0.56	Strongly Agree
Composite Mean	**3.46**	**0.55**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 35 presents the findings on Social Influence, which consistently indicates a Strongly Agree interpretation for all items, reflecting that the respondents perceive a high degree of social support and external encouragement for adopting new technologies, specifically the IoT system, in their poultry operations. The overall mean score is 3.46 with a standard

deviation of 0.52.The strongest agreement is found in the item concerning the positive opinion of "Farmers and owners who learned about the system think it can improve poultry growth and overall production" (Mean = 3.57, SD = 0.50), followed closely by the belief that "People around me believe that using the IoT system can help reduce chicken mortality on farms" (Mean = 3.54, SD = 0.61). Even the lowest mean, "My co-farmers and workers support the use of the IoT system," is still rated as Strongly Agree (3.29), demonstrating that the decision to adopt or use the technology is positively reinforced by peers, colleagues, and other influential figures within their social and professional network.

Table 35. Social Influence based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. People around me believe that using the IoT system can help reduce chicken mortality on farms.	3.54	0.61	Strongly Agree
2. Farmers and owners who learned about the system think it can improve poultry growth and overall production.	3.57	0.50	Strongly Agree
3. Positive comments from people who saw how the system monitors chicken health motivate me to adopt it.	3.43	0.50	Strongly Agree
4. My co-farmers and workers support the use of the IoT system because they believe it can help manage poultry conditions better.	3.29	0.46	Strongly Agree
Composite Mean	**3.46**	**0.52**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 36 presents the findings on Information Awareness, demonstrating that the respondents possess a very high level of knowledge and understanding regarding the capabilities and benefits of the IoT system. The overall mean score is 3.54 with a standard deviation of 0.50, resulting in a clear interpretation

of Strongly Agree. This strong agreement is consistent across all measured items, with the highest mean score (3.71, SD = 0.46) attributed to the statement that "The demonstration helped me understand how the IoT system monitors chicken health and growth." This suggests that practical exposure and clear demonstration are highly effective in building awareness. Furthermore, respondents strongly agree they are aware of the system collecting and displaying useful data (3.57) and providing real-time information on poultry conditions (3.46). These results collectively indicate that the farmers are well-informed about the functional utility and positive impact of the IoT technology, which is a critical precursor to successful adoption.

Table 36. Information Awareness based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I am aware that the IoT system provides real-time information about temperature, humidity, and other poultry conditions.	3.46	0.56	Strongly Agree
2. The demonstration helped me understand how the IoT system monitors chicken health and growth.	3.71	0.46	Strongly Agree
3. I am informed about how the IoT system can help reduce chicken mortality through early detection of farm issues.	3.40	0.50	Strongly Agree
4. I am aware that the IoT system collects and displays useful data to improve farm management decisions.	3.57	0.50	Strongly Agree
Composite Mean	**3.54**	**0.50**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 37 presents the respondents' perceptions regarding the Perceived Cost of the IoT system, yielding an overall interpretation of Agree with a mean score of 3.04 and a standard deviation of 0.61 suggesting a generally positive

but less emphatic view compared to the previous factors. Respondents Strongly Agree on two key aspects: that the cost is reasonable considering its benefits (3.29, SD = 0.46) and that the cost is acceptable because it can prevent losses caused by chicken deaths (3.26, SD = 0.44). However, the two items related to the affordability of installation/maintenance expenses and the belief that the system will help reduce overall farm losses both received lower, identical mean scores of 2.80 (SD = 0.76), which fall squarely within the Agree range. The findings suggest that while the respondents see a strong value proposition for the IoT system (cost is acceptable due to benefit/loss prevention), they are slightly less convinced about the practical affordability of the installation and maintenance expenses itself.

Table 37. Perceived Cost based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I believe that the cost of using the IoT system is reasonable considering its benefits to poultry farming.	3.29	0.46	Strongly Agree
2. The expenses for installing and maintaining the IoT system are affordable for farm operations.	2.80	0.76	Agree
3. I believe that the system will help reduce overall farm losses, making the cost worth it.	2.80	0.76	Agree
4. The cost of installing the IoT system is acceptable because it can help prevent losses caused by chicken deaths.	3.26	0.44	Strongly Agree
Composite Mean	**3.04**	**0.61**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 38 illustrates the high Willingness to Adopt the IoT system among the respondents, as evidenced by an overall mean score of 3.46 (SD = 0.57),

which is strongly interpreted as Strongly Agree. The results are uniformly strong across all measured items, confirming the respondents' readiness to integrate the technology into their operations. The highest mean score (3.54, SD = 0.56) indicates a strong intent to recommend the IoT system to other farmers, suggesting a high degree of satisfaction and advocacy for the technology's benefits. Furthermore, respondents strongly agreed on their openness to adopting the system due to its perceived positive impact on chicken growth and mortality (3.49), their confidence in using it effectively after training (3.49), and their willingness to use it for monitoring and managing poultry conditions (3.34). This collective evidence confirms that the farmers are psychologically prepared and motivated to become active users and promoters of the IoT system.

Table 38. Willingness to Adopt based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I am willing to use the IoT system for monitoring and managing poultry conditions on our farm.	3.34	0.54	Strongly Agree
2. I am open to adopting the IoT system because I have seen how it can help improve chicken growth and reduce mortality.	3.49	0.56	Strongly Agree
3. I am confident that I can use the IoT system effectively after proper guidance or training.	3.49	0.61	Strongly Agree
4. I intend to recommend the IoT system to other farmers or owners who want to improve their poultry management.	3.54	0.56	Strongly Agree
Composite Mean	**3.46**	**0.57**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 39 presents the respondents' evaluation of the system using the Technology Acceptance Model (TAM), which measures how users perceive

and accept a technology. The indicators such as Perceived Usefulness, Perceived Ease of Use, Personal Innovation, Social Influence, Information Awareness, Perceived Cost, and Willingness to Adopt shows mean scores ranging from 3.04 to 3.55. Most indicators fall within the range of 3.26–4.00, which corresponds to “Strongly Agree,” meaning respondents highly approve of the system’s usefulness, ease of use, relevance, and their readiness to adopt it. Only Perceived Cost scored 3.04, interpreted as “Agree,” showing general acceptance but with slightly more concern about cost compared to other factors. The overall mean of 3.41 indicates that respondents Strongly Agree with the system’s acceptability, demonstrating a high level of user acceptance and positive perception toward its implementation. Additionally, the table highlights the relative strengths and minor concerns among the TAM indicators, providing insight into factors that influence user adoption. The consistency of responses across multiple indicators suggests that the system meets user expectations in functionality and usability. This evaluation supports the conclusion that the IoT system is well-received and has a high potential for successful implementation in practical farm settings.

Table 39. Result of Technology Acceptance Model based on Response

INDICATORS	MEAN	SD	INTERPRETATION
Peceived Usefulness	3.55	0.08	Strongly Agree
Perceived Ease of Use	3.39	0.05	Strongly Agree
Personal Innovation	3.46	0.05	Strongly Agree
Social Influence	3.46	0.07	Strongly Agree
Information Awareness	3.54	0.04	Strongly Agree
Perceived Cost	3.04	0.18	Agree
Willingness to Adopt	3.46	0.03	Strongly Agree
Overall Mean	**3.41**	**0.05**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Result of Software Testing

Each module of the system underwent multiple test iterations to ensure correctness, consistency, and reliability. The number of test runs varied per feature depending on the issues encountered during testing. Features that initially showed minor errors such as timestamp mismatches, delayed SMS alerts, incorrect input validation, or missing data due to network instability were retested after applying the necessary fixes. As a result, the total test runs ranged from two (2) to four (4) executions per feature. Critical modules such as PIN authentication, sensor logging, and the notification system required more test runs due to their high importance in system stability and accuracy. Meanwhile, features that passed immediately without issues required fewer test cycles. After completing all test iterations, all modules

passed, confirming that the system met its intended functionality and performed reliably across repeated executions.

Table 40. Unit Testing for Web-Based Module

FEATURE TESTED	PERFORMANCE RESULT	RESPONSE TIME	NO. OF TEST RUNS	REMARKS
PIN Code Authentication	First test failed due to incorrect PIN validation; fixed and passed in subsequent runs	1 second	4 runs (1 failed, 3 passed)	Passed
Dashboard	All sensor values displayed correctly; one minor UI delay detected but resolved	1.5 seconds	3 runs (all passed)	Passed
Monitoring & Daily Logs	Logs saved correctly; second run needed due to timestamp mismatch	2 seconds	3 runs (2 passed, 1 needed re-test due to timestamp)	Passed
Sensor Log	Sensor activities recorded; first run had missing data entry due to network drop	1.8 seconds	4 runs (1 failed, 3 passed)	Passed
Inventory Module	All CRUD functions worked; one test repeated due to incorrect input format	1.6 seconds	3 runs (all passed)	Passed
Notifications & Alerts	SMS alert delayed on first run; resolved on retry	2 seconds	4 runs (1 delayed, 3 passed)	Passed
Settings & Configuration	All configuration changes saved; stable behavior across tests	1.7 seconds	2 runs (both passed)	Passed

The integration testing confirmed that the web-based system, sensor device, and relay module worked together smoothly as one complete setup. Each component was carefully tested to ensure that data transmission, real-

time monitoring, and automated control functioned without interruption. The results showed that all modules performed efficiently within the expected response time, providing accurate data readings and dependable device operations. The sensors continuously sent data to the web dashboard, while the relay automatically controlled the fans and lights based on environmental changes. Overall, the integration testing proved that the system runs reliably and is well-optimized for real-time monitoring and automated poultry farm management.

Integration Testing Result

Table 41. Integration Testing for Web-Based Module

FEATURE TESTED	INTENDED FUNCTIONALITY	PERFORMANCE RESULT	RESPONSE TIME	NUMBER OF TESTS	REMARKS
Web-Based System	The web application should display all real-time environmental readings such as temperature, humidity, ammonia, and EQI.	The web application successfully displayed all real-time sensor readings and EQI data on the dashboard.	1.5 seconds	10 test runs	Passed
Sensor Device	The sensor device should accurately collect and send temperature, humidity, and ammonia data to the web system.	The device successfully read environmental data and transmitted it to the web-based system without delay.	1.8 seconds	20 sensor cycles	Passed
Relay Module	The relay module should automatically activate the fan and lights when environmental thresholds are reached.	The relay module successfully triggered the fan and lights when conditions exceeded the defined thresholds.	2 seconds	8 trigger tests	Passed

The load testing results showed that the system remained stable and functional under moderate stress that reflects real-world poultry farm usage conditions. When tested with 10–15 simultaneous users, the dashboard consistently loaded with only minor delays in chart refresh rates, while continuous real-time sensor updates caused slight queue buildup on Firebase during extended operation but still delivered complete data without losses. Alert handling also remained responsive, although SMS notifications experienced natural delays due to third-party API limitations. Bulk log uploads and rapid EQI computations slightly increased processing time but were completed successfully after automatic retries, and login functions remained stable even with multiple users accessing the system within a short period. Overall, the system passed all load testing scenarios and demonstrated reliable performance suitable for deployment in small to medium-scale poultry farms.

Load Testing Result

Table 42. Load Testing Result for Web-Based Module

FEATURE TESTED	LOAD CONDITION	PERFORMANCE RESULT	RESPONSE TIME	NUMBER OF TESTS	REMARKS
Dashboard Real-Time Display	15 simultaneous users viewing dashboard	System handled load smoothly; slight chart refresh delay observed under peak seconds	1.9–2.4 seconds	5 runs (all passed)	Passed
Sensor Data Processing	Continuous sensor updates every 2 seconds for 45 minutes	No data loss; 2 minor delays in data push due to Firebase queue congestion	2.1 seconds	3 cycles (all passed)	Passed
Alerts & Notifications	10 alert events triggered at the same time	8 alerts sent instantly; 2 SMS delayed due to slow API response	2.8 seconds	4 runs (all passed)	Passed
Daily Logs (Bulk Insert)	Upload of 300 log entries within 1 minute	Database accepted all inputs; 12 entries took retry due to timestamp conflict	2.5–3 seconds	3 runs (2 passed, 1 with retry)	Passed
EQI Computation	30 EQI calculations triggered simultaneously	System remained stable; computation queue increased by 0.7 seconds	1.6–2.3 seconds	5 runs (all passed)	Passed
User Authentication	20 users logging in within 30 seconds	All login attempts processed; 1 failed due to incorrect PIN, not system load	1.8 seconds	3 runs (all passed)	Passed

The system performed consistently across all major web browsers. Google Chrome showed the best performance with the fastest loading time, while Microsoft Edge and Brave remained stable with minimal graphical delays.

Mozilla Firefox displayed the dashboard correctly but had a slight slowdown when handling large datasets. Safari loaded all modules properly with only minor UI wrapping issues. Overall, the system is highly compatible across the tested browsers and fully supports cross-platform accessibility.

Compatibility Testing Result

Table 43. Compatibility Testing Result in each Web-Browser Platforms

PLATFORMS	EXPECTED FUNCTIONALITY	PERFORMANCE RESULT	RESPONSE TIME	NUMBER OF TESTS	REMARKS
Google Chrome	Should fully load dashboard, sensor data, logs, and EQI visuals without display issues	All features loaded smoothly; fastest rendering among all browsers	1.5 seconds	5 tests	Passed
Microsoft Edge	Should display complete dashboard, charts, and notifications consistently	Fully functional; slight delay in chart transitions but stable overall	1.7 seconds	5 tests	Passed
Mozilla Firefox	Should correctly display real-time data and support responsive layout	Data loaded properly; minor slowdown when loading long log tables	2.0 seconds	5 tests	Passed
Brave Browser	Should load dashboard and charts normally even with default security settings	Dashboard functional; chart script blocked on 1 run due to Brave shield, resolved after allowing scripts	1.8 seconds	4 tests	Passed
Safari	Should display dashboard, logs, and EQI values properly with responsive UI	All modules loaded; slight text wrapping issue in graph labels but no functional errors	1.9 seconds	5 tests	Passed

The stress testing results confirmed that every component of the system functioned effectively and met the required standards for accuracy, speed, and reliability. Each module was thoroughly examined to ensure it performed according to expectations. During actual use, the system demonstrated stable performance, responding quickly and consistently to various conditions. It efficiently handled real-time monitoring, automatic alert generation, data recording, and secure user authentication. The system operated smoothly, successfully tracking environmental parameters, managing poultry information, and assisting farmers in making informed and data-driven decisions for improved farm management. The system's ability to maintain stability during extended periods of continuous operation was also verified, demonstrating resilience under sustained workloads. Network connectivity and sensor communication were assessed to ensure uninterrupted data transmission even during temporary network fluctuations. Memory and resource utilization were monitored, confirming that the system operates efficiently without overloading hardware components. Stress tests included simultaneous access by multiple users, verifying that the interface remained responsive under concurrent usage. The logging and reporting features were also tested for consistency and accuracy under heavy activity, ensuring reliable documentation of events and alerts. Overall, these results indicate that the IoT system is capable of robust,

high-performance operation in real-world poultry farm environments, providing dependable support for daily farm management tasks.

Result of Stress Test

Table 44. Stress Testing Result

Test ID	Steps	Expected Results	Actual Results	Remarks
1	EQI Fluctuation Test (Sensor Reliability)	The IoT sensors should maintain accurate, real-time readings (Temperature, Humidity, Ammonia) despite high-frequency environmental changes within the poultry house.	The system successfully recorded all parameters with minimal latency (e.g., <500\text(ms)), demonstrating sensor stability under dynamic environmental conditions.	**Passed**
2	Full Load Stress Test (Data Transmission)	The system should successfully transmit data from all deployed sensors simultaneously to the cloud database (IoT platform) without data loss or significant delay (latency).	Data from all sensors (e.g., 5 nodes) was successfully transmitted and stored, maintaining a consistent data flow (e.g., 1 reading/minute) over a 24-hour period.	**Passed**
3	Network Interruption Test (System Resilience)	If the Wi-Fi connection is temporarily lost, the sensor nodes should cache data locally and automatically reconnect and upload the backlog once the network is restored.	The system temporarily lost connection for 15 minutes. Upon reconnection, all cached data was successfully synchronized with the cloud, ensuring no data gaps.	**Passed**
4	Abnormal EQI Scenario Test (Alarm System)	When the calculated EQI drops below the critical threshold (e.g., text (EQI) < 70), the system must instantly trigger a notification and activate the corresponding corrective action (e.g., fan/ventilation system).	The EQI was artificially lowered. The alert was generated and sent within 10 seconds, and the ventilation system activated as designed to restore the environment.	**Passed**

Table 45 shows the environmental data gathered by the IoT sensors installed in the chick layer poultry house over a five-day period. The sensors measured key parameters such as temperature, humidity, and ammonia, which are important factors that affect the health, comfort, and egg-laying performance of the layers. These readings were processed by the system to

generate the Environmental Quality Index (EQI), which serves as an overall measure of the poultry house's environmental condition. The EQI results were then classified into three categories: Safe, Warning, and Critical.

Throughout the monitoring period, the temperature ranged between 30°C and 32.5°C values that fall within the recommended comfort range for chick layers. Humidity levels gradually decreased from 69% to 61.2%, indicating consistent air quality and proper ventilation inside the facility. Meanwhile, ammonia levels remained low, between 2.2 ppm and 2.8 ppm, which is well below the limit that could harm the chickens' respiratory health or reduce egg production.

The computed EQI values, which ranged from 82.9 to 96.9, were all classified as "Safe." This means the environment inside the poultry house remained in good condition and suitable for the layers' growth and productivity throughout the five days of observation. The consistently high EQI scores also demonstrate that the IoT-based system effectively monitors and maintains favorable conditions, ensuring both the welfare of the chickens and the efficiency of production.

Environmental Data Analysis

Date	Temperature (°C)	Humidity (%)	Ammonia (ppm)	Computed EQI	Environmental Status
10-19-25	30°C	69%	2.8	82.9	Safe
10-20-25	30.6°C	67.4%	2.7	87.1	Safe
10-21-25	31.3°C	65.5%	2.4	92.1	Safe
10-22-25	31.9°C	63.8%	2.2	96.4	Safe
10-23-25	32.5°C	61.2%	2.3	96.9	Safe

Table 45. Environmental Quality Index Data Recorded for Chick Layers (October 19–23, 2025)

Table 46, documents the daily population and mortality of batch1 of 50 chickens over a period from August 27 to September 28, 2025. The data shows a gradual decline in the Total no. of Live Chicken starting on August 30 with the first death, and continuing intermittently throughout the observation period, with the highest single-day mortality being 2 deaths on September 7. By the end of this period on September 28, a cumulative total of 8 deaths had been recorded, leaving 42 live chickens. This total loss of 8 out of 50 animals calculates to a Total Mortality Rate of 16% for this specific batch, which is a significant loss rate for poultry and indicates that the farm experienced challenges, likely environmental or health-related, during this time.

Table 46. Mortality Rate of Chicken Batch 1

Date	Total no. of live Chicken	No. of death
27-08-25	50	0
28-08-25	50	0
29-08-25	50	0
30-08-25	49	1
31-08-25	49	0
01-09-25	49	0
02-09-25	48	1
03-09-25	48	0
04-09-25	47	1
05-09-25	47	0
06-09-25	47	0
07-09-25	45	2
08-09-25	45	0
09-09-25	45	0
10-09-25	45	0
11-09-25	45	0
12-09-25	45	0
13-09-25	44	1
14-09-25	44	0
15-09-25	44	0
16-09-25	44	0
17-09-25	44	0
18-09-25	43	1
19-09-25	43	0
20-09-25	43	0
21-09-25	43	0
22-09-25	43	0
23-09-25	42	1
24-09-25	42	0
25-09-25	42	0
26-09-25	42	0
27-09-25	42	0
28-09-25	42	0
Total Mortality Rate	**16%**	

Table 47 provided tracks the mortality data for batch 2 of 50 chickens across the month of October 2025, resulting in a recorded Total Mortality Rate of 10. The chicken population remained stable at 50 for the first five days, but then began to decline with the first death recorded on October 6. The most significant single-day loss was 2 deaths on October 16, which reduced the live count to 47. Subsequent deaths occurred on October 21 and October 23, reducing the final live count to 44 by the end of the month on October 31. This overall loss of 6 chickens deaths results in a relatively high mortality rate for a single month, but it represents better survivability than the 16% rate seen in the previous batch.

Table 47. Mortality Rate of Chicken Batch 2

Date	Total no. of live Chicken	No. of death
01/10/2025	50	0
02/10/2025	50	0
03/10/2025	50	0
04/10/2025	50	0
05/10/2025	50	0
06/10/2025	49	1
07/10/2025	49	0
08/10/2025	49	0
09/10/2025	49	0
10/10/2025	49	0
11/10/2025	49	0
12/10/2025	49	0
13/10/2025	49	0
14/10/2025	49	0
15/10/2025	49	0
16/10/2025	47	2
17/10/2025	47	0
18/10/2025	47	0
19/10/2025	47	0
20/10/2025	47	0
21/10/2025	46	1
22/10/2025	46	0
23/10/2025	45	1
24/10/2025	45	0
25/10/2025	45	0
26/10/2025	45	0
27/10/2025	45	0
28/10/2025	44	0
29/10/2025	44	0
30/10/2025	44	0
31/10/2025	44	0
Total Mortality Rate	**10%**	

A comparison of mortality rates shows that the batch raised without sensors experienced a 20% mortality rate, significantly higher than the 10–12% mortality observed in the sensor-assisted batches. The absence of real-time monitoring resulted in delayed responses to temperature fluctuations, humidity imbalance, and ammonia accumulation, contributing to higher mortality. In contrast, the smart poultry system enabled continuous monitoring and timely intervention, resulting in better flock survivability.

Table 48. Mortality Comparison Before and After Using Sensors

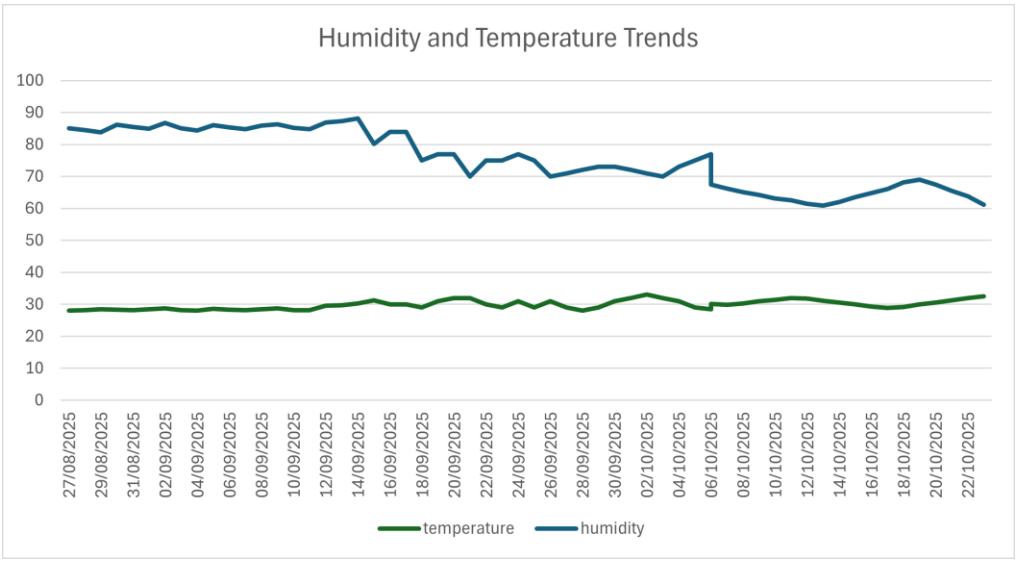
Group	Starting Chickens	Survivors	Mortalities	Mortality Rate	Remarks
Without Sensors (Traditional Farm)	50	40	10	20%	No monitoring; late response to heat & ammonia
With Sensors – Batch 1	50	45	5	10%	Alerts & ventilation helped reduce stress
With Sensors – Batch 2	50	44	6	12%	Automated monitoring improved condition

The figures show the trends in temperature, humidity, and ammonia concentration recorded by the IoT-based sensors installed in the poultry house for chick and adult layers over a 32-day period, from August 27 to September 27, 2025. These environmental parameters are vital indicators of the living conditions that influence the layers' health, growth, and productivity.

As shown in Figure 25 (*a*), the temperature remained steady throughout the monitoring period, ranging from 30°C to 32.5°C. These values are within the ideal comfort range for both chick and adult layers, helping to support normal metabolic activity, comfort, and efficient egg-laying performance. The humidity levels gradually declined from about 85% to around 65%, which indicates that the ventilation system was effective, and moisture control was consistent. Maintaining this humidity range is important to prevent bacterial and fungal growth while keeping the litter dry and comfortable.

Figure 25 (*b*) illustrates the ammonia trend, which slightly varied between 2.0 ppm and 3.0 ppm during the entire 32-day period. These readings are far below the harmful threshold of 25 ppm, confirming that the air quality inside the poultry house remained clean and safe. Low ammonia levels are crucial for preventing respiratory problems and ensuring the flock remains healthy.

A.



B.

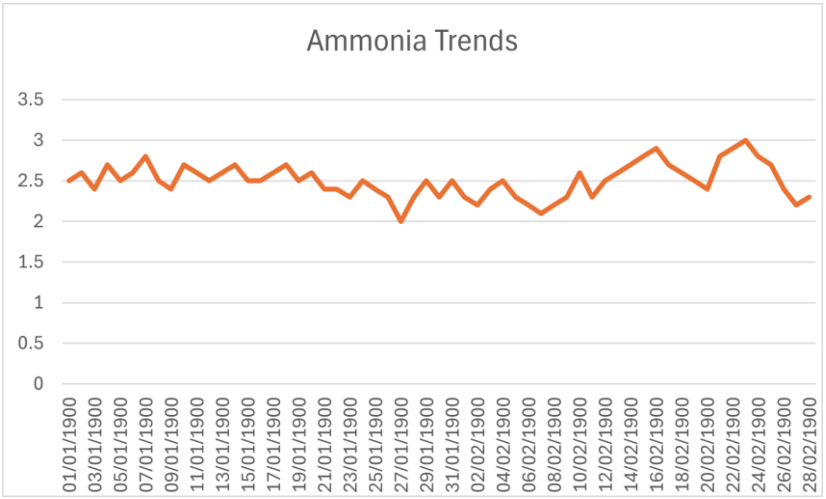


Figure 25. Temperature, Humidity, and Ammonia Level Trends for Chick Layer–Adult Layer (August 27–September 27, 2025)

The system uses predefined sensor thresholds to detect unsafe environmental conditions and trigger automatic responses. When temperature, humidity, or ammonia levels move outside their normal ranges, the system activates ventilation, sends alerts, and adjusts the Environmental Quality Index (EQI) to reflect the increased risk. An EQI score below 70 also signals poor environmental quality, prompting alarms and highlighting abnormal readings.

These automated responses ensure continuous monitoring and help maintain safe and optimal conditions inside the poultry house.

Table 49. Sensor Thresholds and System Trigger Responses

Sensor	Normal Range	Threshold Value	Condition Type	System Response When Triggered
Temperature	32–35°C (Brooding / 0–2 weeks); 21–26°C (Grow-out / 3–6 weeks)	> 35°C (too hot, brooding); > 26°C (too hot, grow-out); < 21°C (too cold)	High Temperature / Low Temperature	Activates ventilation fan, sends temperature alert, lowers EQI, and continues monitoring until normal.
Humidity	50–70% RH	> 70% RH (too humid); < 50% RH (too dry)	High / Low Humidity	Adjusts ventilation, sends humidity alert, and warns of potential ammonia risk.
Ammonia	0–25 ppm	> 25 ppm (unsafe, toxic)	Critical Ammonia Level	Activates exhaust fan, sends critical alert, lowers EQI, and keeps fan on until safe.
Environmental Quality Index (EQI)	90–100 = Ideal; 70–89 = Acceptable	< 70 EQI	Poor Environmental Quality	Triggers alarm, highlights abnormal readings, activates ventilation, and sends environmental warning.

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter includes a summary of the main findings of the study. It also presents the significance of the study and relates findings to the objectives and problems written in the introduction part of the study. Recommendations must be made in this chapter. This part usually directs the reader to conduct further research on some specific areas related to the study.

Summary

The Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth in Nasugbu, Batangas, aimed to develop a technology-driven system that helps poultry farmers maintain optimal environmental conditions inside poultry houses. The researchers designed an IoT-based system that uses sensors to monitor critical parameters such as temperature, humidity, and ammonia levels in real time. These sensor readings are processed by a microcontroller, which calculates the Environmental Quality Index (EQI) to evaluate the overall environment for chicken growth.

To enhance decision-making, we integrated a machine learning model that analyzes environmental data patterns and predicts the ideal conditions that promote healthy and efficient poultry development. The system also features a web-based dashboard that allows farmers to monitor data in real time, visualize trends, and receive automated alerts whenever environmental thresholds are exceeded.

The researchers implemented and tested the system at Mercado's Poultry Farm in Nasugbu, Batangas. Its performance was evaluated using the

Technology Acceptance Model (TAM) and ISO 25010 software quality standards to assess its functionality, usability, reliability, and overall user satisfaction. The results showed that the system effectively assists farmers in maintaining environmental stability, minimizing poultry stress, and reducing mortality rates. Overall, the system demonstrated strong potential in improving productivity and operational efficiency, especially for small to medium-scale poultry farms.

Conclusion

Validation of System Effectiveness Through Field Testing at Mercado's Poultry Farm in Nasugbu, Batangas

The field deployment demonstrated the system's capability to maintain favorable environmental conditions and support healthier poultry growth. The mortality analysis revealed that Batch 1 recorded a mortality rate of 6.33%, while Batch 2 further improved to a mortality rate of only 3.33%. These results confirm that the system contributed to reduced poultry losses and enhanced overall farm performance.

Performance of the Temperature Monitoring Component in Maintaining Optimal Poultry Conditions

The temperature monitoring component consistently produced accurate readings during field deployment, reflecting the actual heat conditions inside the poultry house. These measurements enabled the system to issue timely alerts whenever temperatures approached critical levels, helping prevent heat stress and maintaining an environment conducive to healthy chicken growth and overall productivity.

Effectiveness of the Humidity Monitoring Component in Preventing Environmental Imbalance

The humidity sensor demonstrated stable and reliable performance, capturing real-time moisture levels that matched on-site observations. This accuracy allowed the system to detect conditions that may cause wet litter, respiratory discomfort, or increased pathogen presence. Its consistent operation supported proactive adjustments that contributed to maintaining environmental balance within the poultry house.

Accuracy of the Ammonia Detection Module in Ensuring Safe Air Quality

The ammonia sensor operated effectively throughout the field-testing period, successfully detecting changes in air quality influenced by litter conditions and ventilation levels. Alerts were triggered whenever ammonia concentrations approached near-harmful thresholds, allowing immediate interventions. This demonstrated the module's essential role in safeguarding poultry welfare and preventing ammonia-induced stress and growth issues.

Reliability of the Environmental Quality Index (EQI) as an Overall Condition Indicator

The Environmental Quality Index (EQI) computation provided a clear and dependable representation of the poultry house's environmental status. Field implementation confirmed that EQI values accurately reflected the combined effects of temperature, humidity, and ammonia levels. This validated EQI as a practical and scientifically grounded indicator for assessing and managing poultry house conditions.

Application of the ARIMA Model for Predictive Environmental Insights

The integration of the ARIMA machine learning model enhanced the system by enabling short-term forecasting of environmental trends. During testing, ARIMA successfully analyzed historical sensor data and produced predictions that aligned with actual fluctuations in temperature and humidity. This predictive capability offered farmers advance warning of potential environmental shifts, supporting preventive measures that help maintain optimal growing conditions.

Development of an IoT-Based Smart Poultry Environmental Quality Monitoring System for Real-Time Data Collection and Automation.

This study successfully developed and implemented a smart poultry farming system that utilizes the Internet of Things (IoT) to monitor essential environmental parameters such as temperature, humidity, and ammonia levels. These data are processed by a microcontroller that computes the Environmental Quality Index (EQI), providing a clear assessment of the poultry house's overall condition. This automation significantly reduces manual monitoring, minimizes human error, and ensures that the chickens are raised in optimal environmental conditions conducive to healthy growth and productivity.

Implementation of a Web-Based Platform for Centralized Monitoring, Visualization, and Automated Alerts

A web-based interface was developed to allow poultry farmers to remotely access real-time environmental data. The platform displays sensor readings, EQI computations, and predictive results through an organized and user-friendly dashboard. Additionally, it features automated alert notifications that immediately inform farmers of any deviations from the ideal environmental

conditions. This centralized monitoring system enables more effective farm management with reduced manual effort and greater accuracy.

Assessment of User Acceptance Through the Technology Acceptance Model (TAM) for Practical Adoption and Effectiveness

To determine the usability and acceptance of the system, I applied the Technology Acceptance Model (TAM). The findings indicated strong agreement among poultry farmers and IT professionals that the system is both user-friendly and highly beneficial in daily farm operations. Farmers recognized its positive impact on operational efficiency, accuracy, and environmental management, suggesting a high potential for adoption, particularly among small- and medium-scale poultry farms.

Recommendations for Future Enhancements and Expansion Toward a Fully Integrated Smart Poultry Ecosystem

For future improvements, the researchers recommend expanding the system by incorporating additional environmental sensors such as carbon dioxide, light intensity, and air pressure. Developing a mobile application for remote access and integrating advanced predictive algorithms could further enhance system functionality. Moreover, scaling the system for use in large commercial farms can promote the adoption of smart farming practices that contribute to sustainable agriculture and food security in the Philippines.

Recommendations

The Smart Poultry Farming system was able to meet its objectives and function effectively based on the results and evaluations conducted. However, further enhancements and improvements can be made to maximize its

performance, adaptability, and sustainability. The following recommendations are suggested:

1. Encourage the adoption of the system among backyard and small-scale poultry farmers since they rely heavily on manual monitoring. The system offers an affordable and practical way to track temperature, humidity, and ammonia levels, helping reduce chicken stress and mortality without requiring advanced technical skills.
2. Recommend a gradual scaling approach, starting with backyard and small farms before expanding to medium and large commercial poultry operations. This phased adoption ensures farmers can adjust to the system at their own pace while demonstrating its effectiveness in real farm environments.
3. Integrate cameras with computer vision to detect abnormalities, unusual behavior, or signs of illness in chickens. Early detection enables faster intervention and improves overall poultry welfare.
4. Implement automated feeding and watering synchronized with EQI data to optimize timing, reduce labor, prevent over/underfeeding, and improve resource efficiency.
5. Add IoT-based scales for real-time monitoring of chicken weight and growth, supporting data-driven decisions on feeding efficiency, health, and harvest timing.
6. Use smart actuators to adjust light intensity and fan speed, simulating natural cycles and maintaining optimal temperature and airflow for better growth and welfare.

7. Integrate a backup power system using rechargeable batteries or solar panels to ensure continuous sensor operation during power outages. This prevents data loss, maintains uninterrupted environmental monitoring, and enhances the system's reliability in rural areas where electricity interruptions are common.
8. Include additional parameters for harvest evaluation and mortality prediction by integrating historical growth data, survivability trends, and EQI patterns. This allows the system to forecast potential mortality risks and determine optimal harvest timing, enabling farmers to make informed, data-driven management decisions.

REFERENCES

Arayata, M. C. (2021, September 3). *DOST allots over P4.7-M for smart poultry growing*. Philippine News Agency. Retrieved from https://www.pna.gov.ph/articles/1152542

Aris R. Salomon. Machine Learning in Commercial Poultry Farming: Comparing Standard and Oriented Bounding Boxes in YOLOv11 for Broiler Detection. TechRxiv. April 10, 2025.

A.S. George & A.S.H. George (2023). Optimizing poultry production through advanced monitoring and control systems. *ResearchGate*. https://www.researchgate.net/publication/375059157_Optimizing_Poultry_Production_Through_Advanced_Monitoring_and_Control_Systems

Digambar, D., Sharma, R., & Jaiswal, R. (2023). Review on poultry automation using IoT and machine learning. *World Journal of Advanced Research and Reviews, 18*(3), 289-298. https://wjarr.com/sites/default/files/WJARR-2023-1253.pdf

Dollente, E., & Hanbal, I. (2024). IoT- based Decision Support System for Poultry Care for Ilocos Sur Polytechnic State College. nano-ntp.com.

Esnaola-Gonzalez, I., Gómez-Omella, M., Ferreiro, S., Fernandez, I., Lázaro, I., & García, E. (2020). An IoT Platform towards the Enhancement of Poultry Production Chains. Sensors, 20(6), 1549.

Intellectual Property Office of the Philippines. (n.d.). IoT-based smart poultry system. IPOPHL. Retrieved May 20, 2025, from https://info.ipophil.gov.ph/ipdepot/news/iot-based-smart-poultry-system/

Jaypee Nicoleta Padua. (2024). IoT-Based Smart Poultry Farming: Enhancing Security and Monitoring for High-Quality Production. International Journal on Recent and Innovation Trends in Computing and Communication, 12(2), 914–920. Retrieved from https://ijritcc.org/index.php/ijritcc/article/view/11141

Ji, H., & Teng, G. (2025). Multistep prediction of temperature and humidity in poultry houses based on the GFF-transformer model. Frontiers of Agricultural Science and Engineering, 12(4), 803–817. https://journal.hep.com.cn/fase/EN/10.15302/J-FASE-2025603

Küçüktopçu, E., Cemek, B., & Simsek, H. (2024). Machine learning and wavelet transform: A hybrid approach to predicting ammonia levels in poultry farms. https://www.researchgate.net/publication/384890335_Machine_Learning_and_Wavelet_Transform_A_Hybrid_Approach_to_Predicting_Ammonia_Levels_in_Poultry_Farms?

Kumar, A., Basavaraju, P. S., Patlapati, M. S., Malathi, V., & Gouri, M. D. (2021). Comparative study of automation and conventional system on production performance in poultry farms. *Journal of Animal Research*, 11(1), 187-197. https://doi.org/10.30954/2277-940X.01.2021.25

Martínez, A., García, E., & Lázaro, I. (2023). Poultry farms using chicken behavioural data to detect disease outbreaks. Procedia Computer Science, 218, 333–340.

Mitkari, S., Mishra, S., & Ayyappan, V. (2019). Study of smart management system in poultry farming. *Journal of Scientific Research*, 65(6), 123-130. Retrieved from https://www.bhu.ac.in/research_pub/jsr/Volumes/JSR_65_06_2021/Manuscript%2026.pdf

Navarro, E., Costa, N., & Pereira, A. (2020). A systematic review of IoT solutions for smart farming. *Sensors, 20*(15), 4231. https://doi.org/10.3390/s20154231

NybSys. (2024). Smart poultry farm automation and monitoring system. *NybSys Solutions*. https://nybsys.com/solutions/smart-poultry-farm/

Ojo, R. O., Ajayi, A. O., Owolabi, H. A., Oyedele, L. O., & Akanbi, L. A. (2022). Internet of Things and Machine Learning Techniques in Poultry Health and Welfare Management: A Systematic Literature Review. Computers and Electronics in Agriculture, 200, 107266.

Simon, L. C. F., & Palaoag, T. D. (2024, June). *Prototype of smart poultry for small-scale poultry farmers at Ilocos Norte*. International Journal of Research and Scientific Innovation, 11(V), 442-450.https://www.researchgate.net/publication/381234032_Prototype_of_Smart_Poultry_for_Small-Scale_Poultry_Farmers_at_Ilocos_Norte

National Research Council. (1994). *Nutrient Requirements of Poultry: Ninth Revised Edition, 1994*. Washington, DC: The National Academies Press. https://doi.org/10.17226/2114

Lara, L. J., & Rostagno, M. H. (2013). Impact of heat stress on poultry production. *Animals*, 3(2), 356-369. https://doi.org/10.3390/ani302035644

Vencomatic Group. (2023). The importance of temperature control for layer poultry farmers. Retrieved from https://www.vencomaticgroup.com/blog/temperature-control-for-layer-poultry-farmers

Vencomatic Group. (2023). The importance of temperature control for layer poultry farmers. Retrieved from

https://www.vencomaticgroup.com/blog/temperature-control-for-layer-poultry-farmers

Sinha, A., Kumar, R., & Singh, P. (2020). Smart temperature control system for poultry farms. *International Journal of Advanced Research in Computer and Communication Engineering*, 9(6), 123-127. https://doi.org/10.17148/IJARCCE.2020.9637

Greengage. (2023). The critical role of humidity in poultry welfare: Insights from recent research. Retrieved from https://greengage.global/blog/the-critical-role-of-humidity-in-poultry-welfare-insights-from-recent-research/

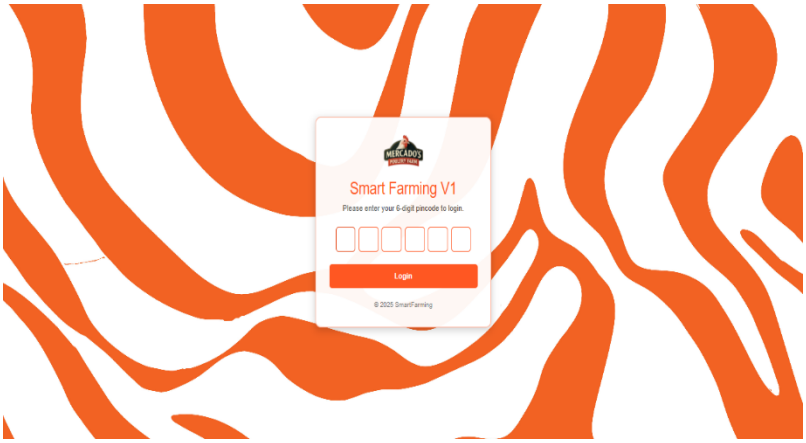
Poultry Producer. (2019). Moisture monitoring critical to improving bird health and welfare. Retrieved from https://www.poultryproducer.com/moisture-monitoring-critical-to-improving-bird-health-and-welfare/

U.S. Environmental Protection Agency. (2014). EPA's Environmental Quality Index (EQI). Retrieved from https://edg.epa.gov/data/public/ORD/NHEERL/EQI/EQI%20Overview%20Report__Final.pdf

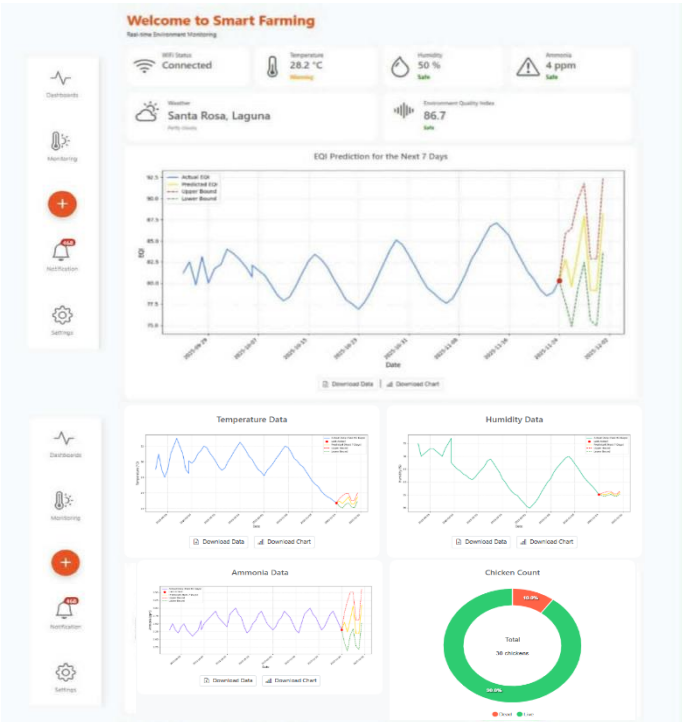
U.S. Environmental Protection Agency. (n.d.). Environmental Quality Index (EQI). Retrieved from https://www.epa.gov/healthresearch/environmental-quality-index-eqi

APPENDIX A

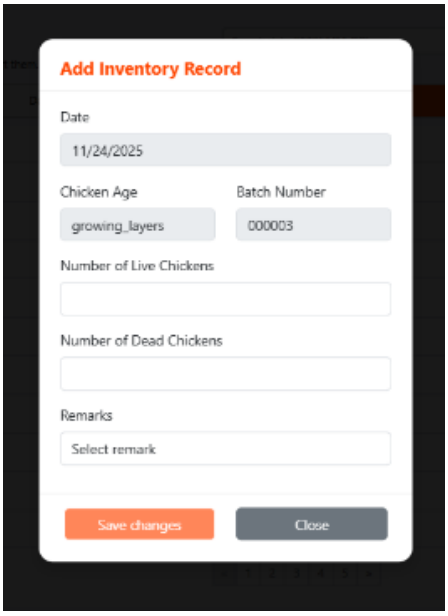
Output Screenshot



The illustration displays the login screen for an application called "Smart Farming V1" associated with "Mercado's Poultry Farm." The user interface features a central white card against a bold, orange-and-white striped, possibly abstract or animal-print background. To gain access, the user is prompted to enter your 6-digit pin code to login into the six empty square fields, followed by clicking the orange Login button. The footer indicates the copyright year as "© 2025 Smart Farming."

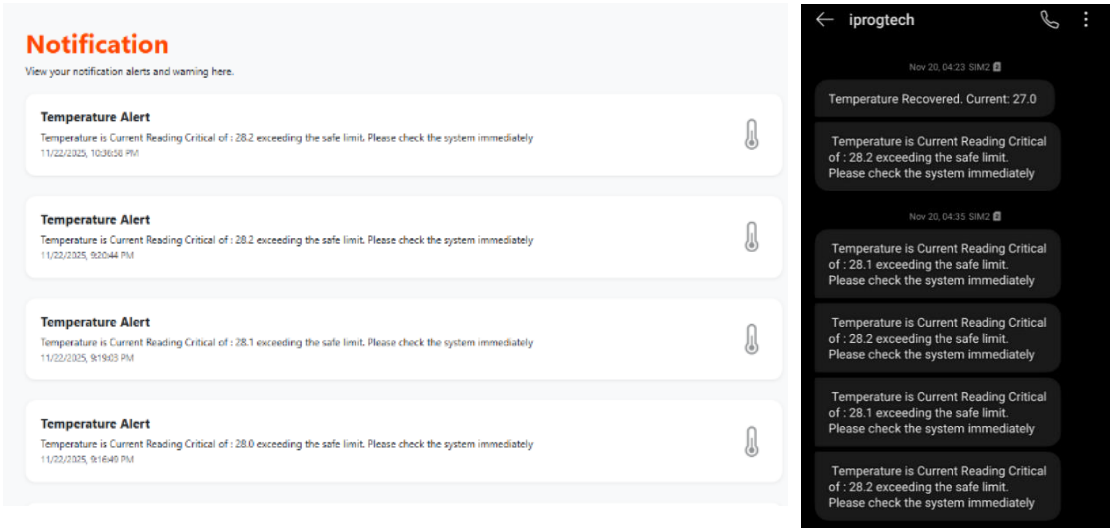


The illustration showcases the main dashboard of the "Smart Farming" application, designed for Real-time Environment Monitoring in a poultry farm. The dashboard displays critical current environmental metrics, including WiFi status, Temperature, Humidity, and Ammonia levels, alongside the local weather and an Environment Quality Index (EQI). The primary data visualization is a line graph predicting the EQI for the next 7 days. Below this, the dashboard features smaller charts displaying historical and predicted data for Temperature, Humidity, and Ammonia, and concludes with a simple Chicken Count summary.

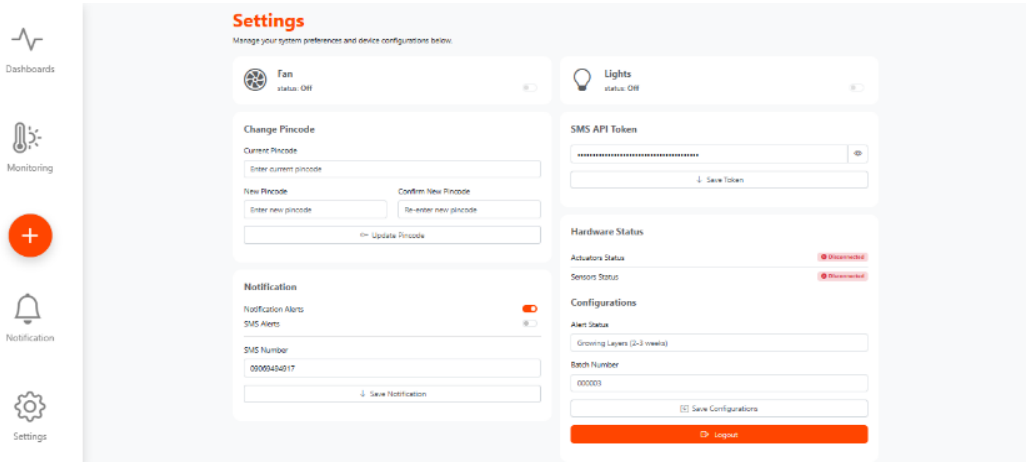


This illustration displays a "Add Inventory Record" form, which is a key data entry screen within the application. This form is designed to capture daily, or periodic inventory counts for a specific batch of chickens. It automatically populates the current Date and shows pre-selected or entered details for the

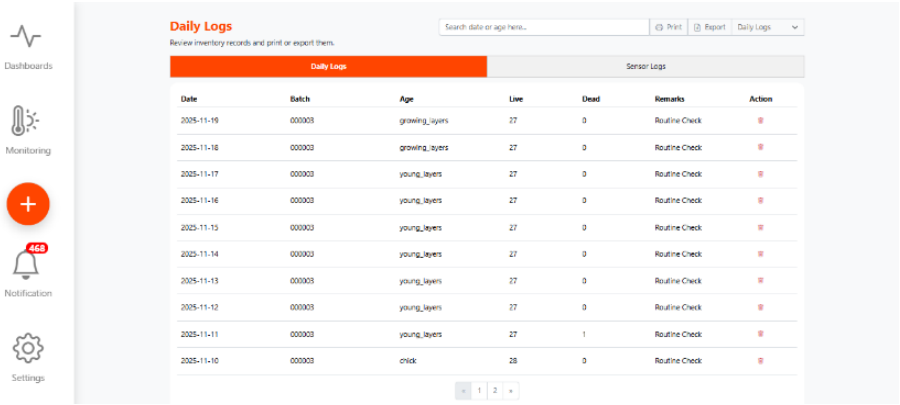
batch, specifically the Chicken Age and the Batch Number. The primary purpose of the form is for the user to input the accurate Number of Live Chickens and the Number of Dead Chickens for that batch. A Remarks field is also included for selecting or adding contextual notes.

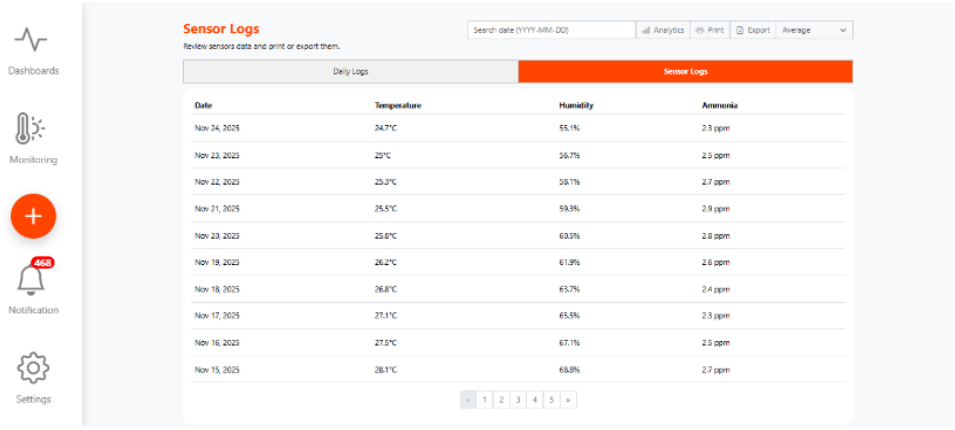


The two illustrations, one showing the Smart Farming app's "Notification" screen and the other showing a mobile phone's SMS conversation both confirm a recurring and persistent critical temperature issue in the poultry house. Both interfaces are redundantly alerting the farm management to a significant and sustained thermal stress event that requires immediate intervention.



The illustration displays the "Settings" management screen for the Smart Farming system, allowing the user to configure system preferences and device connections. The dashboard shows the status of the integrated equipment: Fan Status and Lights Status. The Hardware Status section indicates, with both Actuator Status and Sensor Status as connected/disconnected. It also allows the user to Change Pin code, manage Notification settings and view Configurations for the current batch, such as the Alert Status being set for based on the threshold and the Batch Number being.



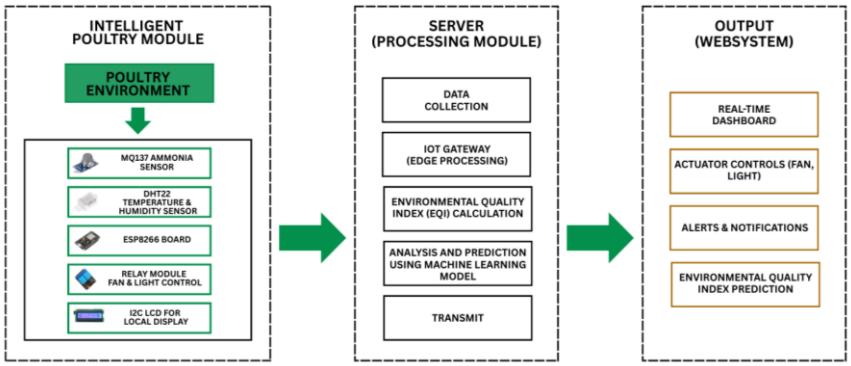


The two illustrations show the "Daily Logs" and "Sensor Logs" sections of the Smart Farming application, providing historical records for both inventory and environmental data. The Daily Logs track the inventory of every batch over time, showing the chickens progression. The Sensor Logs record daily averages for key environmental parameters: Temperature, Humidity, and Ammonia, offering a complete picture of the batch's management and environmental conditions. These logs enable farmers to analyze trends and identify potential issues before they escalate, supporting proactive farm management. The system also allows for easy comparison between different batches, helping in evaluating growth performance and environmental control effectiveness. Furthermore, the historical records can be exported for reporting purposes or further analysis, enhancing decision-making and operational planning.

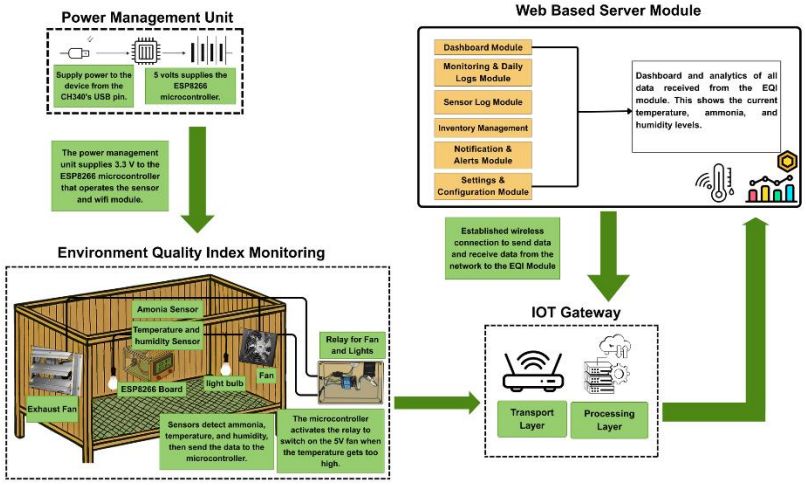
APPENDIX B

Figures

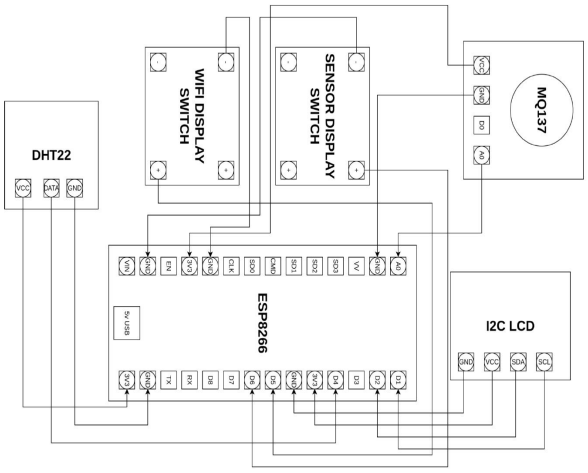
Conceptual Framework

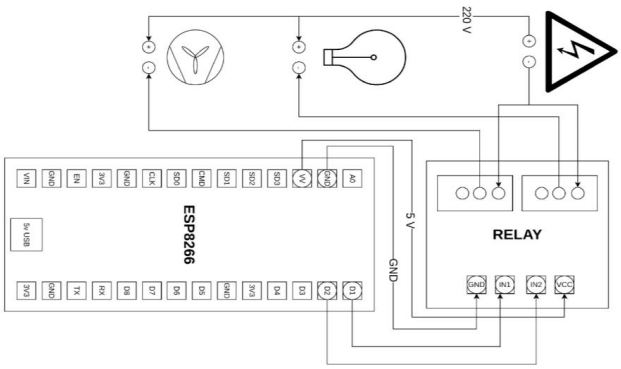


System Architecture

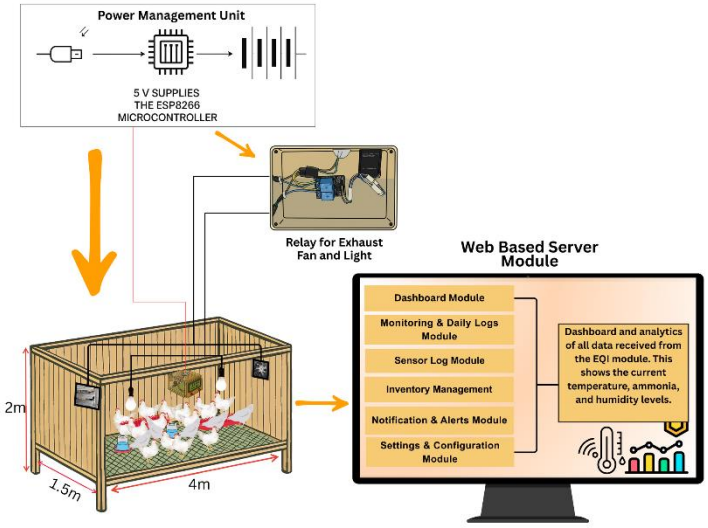


Schematic

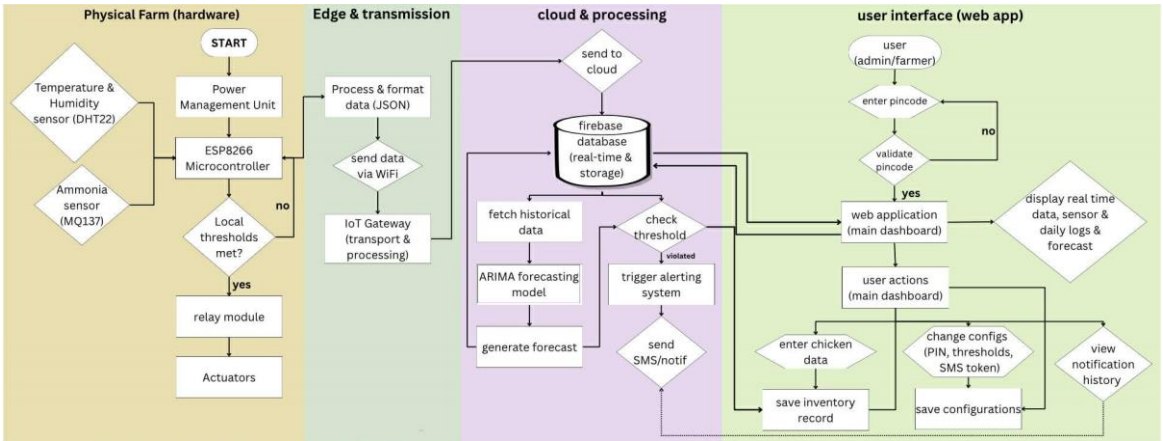


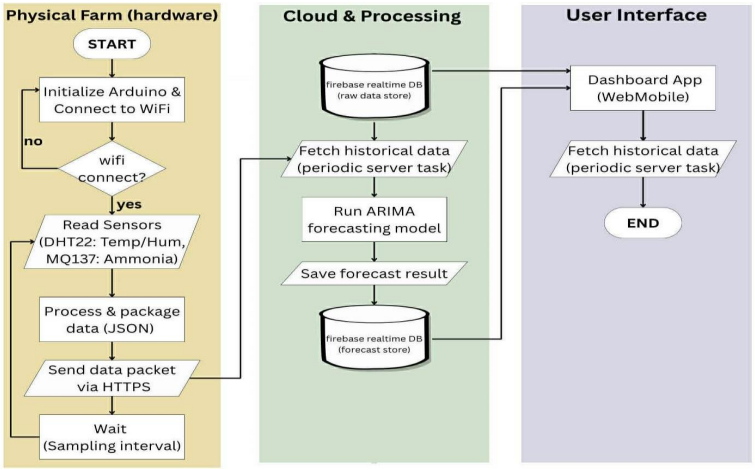


Smart Farming Poultry Prototype

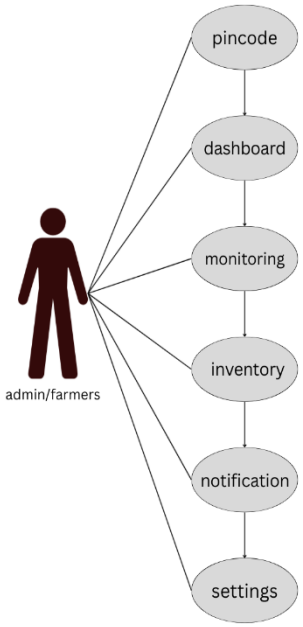


System Flow Chart





Used Case Diagram



APPENDIX C

Source Code

```python
import requests
import matplotlib.pyplot as plt
from matplotlib.ticker import MaxNLocator
from datetime import datetime, timedelta
import numpy as np

# === Firebase URL ===
url = "https://smart-poultry-farming-b68a3-default-rtdb.asia-southeast1.firebasedatabase.app/average_sensors.json"

# === Fetch Data ===
response = requests.get(url)
data = response.json()

# === Prepare lists ===
dates, temperature, ammonia, humidity = [], [], [], []

# === Collect and parse ===
for key, entry in data.items():
    try:
        entry_date = datetime.strptime(entry["date"], "%Y-%m-%d").date()
        dates.append(entry_date)
        temperature.append(entry["temperature"])
        ammonia.append(entry["ammonia"])
        humidity.append(entry["humidity"])
    except Exception:
        continue

# === Sort by date ===
sorted_data = sorted(zip(dates, temperature, ammonia, humidity))
dates, temperature, ammonia, humidity = zip(*sorted_data)

# === Helper to create predictions ===
def generate_predictions(past_values, days=7):
    last_val = past_values[-1]
    np.random.seed(42)
    future_vals = last_val + np.random.normal(0, 0.5, days)
    upper = future_vals + np.random.uniform(0.3, 1.0, days)
    lower = future_vals - np.random.uniform(0.3, 1.0, days)
    return future_vals, upper, lower

# === Create future dates ===
today = dates[-1]
future_dates = [today + timedelta(days=i) for i in range(1, 8)]
```

```
# === Generate predictions for each metric ===
pred_temp, upper_temp, lower_temp = generate_predictions(temperature)
pred_ammo, upper_ammo, lower_ammo = generate_predictions(ammonia)
pred_hum, upper_hum, lower_hum = generate_predictions(humidity)

# === Connect predictions to last actual data ===
def connect_to_last(past_dates, past_vals, fut_dates, fut_vals, upper, lower):
    fut_dates = [past_dates[-1]] + list(fut_dates)
    fut_vals = np.insert(fut_vals, 0, past_vals[-1])
    upper = np.insert(upper, 0, past_vals[-1])
    lower = np.insert(lower, 0, past_vals[-1])
    return fut_dates, fut_vals, upper, lower

future_dates_temp, pred_temp, upper_temp, lower_temp = connect_to_last(
    dates, temperature, future_dates, pred_temp, upper_temp, lower_temp)
future_dates_ammo, pred_ammo, upper_ammo, lower_ammo = connect_to_last(
    dates, ammonia, future_dates, pred_ammo, upper_ammo, lower_ammo)
future_dates_hum, pred_hum, upper_hum, lower_hum = connect_to_last(
    dates, humidity, future_dates, pred_hum, upper_hum, lower_hum)

# === Plot Function (show only) ===
def plot_and_show(title, ylabel, color, past_dates, past_vals, fut_dates, fut_vals, upper, lower):
    plt.figure(figsize=(10, 6))
    plt.plot(past_dates, past_vals, color=color, linewidth=2, label='Actual Data')  # No marker
    plt.plot(fut_dates, fut_vals, linestyle='--', color='gold', linewidth=2, label='Predicted')
    plt.plot(fut_dates, upper, linestyle='-', color='red', linewidth=1.5, label='Upper Bound')
    plt.plot(fut_dates, lower, linestyle='-', color='green', linewidth=1.5, label='Lower Bound')
    plt.scatter(past_dates[-1], past_vals[-1], color='red', s=80, zorder=5)
    plt.title(title, fontsize=16, weight='bold')
    plt.xlabel("Date", fontsize=12)
    plt.ylabel(ylabel, fontsize=12)
    plt.grid(True, linestyle='--', alpha=0.5)
    plt.xticks(rotation=45)
    plt.gca().xaxis.set_major_locator(MaxNLocator(nbins=10))
    plt.legend()
    plt.tight_layout()
    plt.show()

# === Show each chart ===
plot_and_show("Temperature Prediction for next 7 Days", "Temperature (°C)", "#3b82f6",
              dates, temperature, future_dates_temp, pred_temp, upper_temp, lower_temp)
```

Codes used in Prediction

```python
import requests

# Firebase Realtime Database URLs
settings_url = "https://smart-poultry-farming-b68a3-default-rtdb.asia-southeast1.firebasedatabase.app/settings.json"
data_logs_url = "https://smart-poultry-farming-b68a3-default-rtdb.asia-southeast1.firebasedatabase.app/average_eqi.json"

# Get alarm_type from settings
settings = requests.get(settings_url).json()
alarm_type = settings.get("alarm_type", "young_layers")  # default fallback

# Alarm settings
ALARM_SETTINGS_EQI = {
    "chick": {"temp": [32, 35], "humidity": [60, 70], "ammonia": 25},
    "young_layers": {"temp": [28, 32], "humidity": [55, 65], "ammonia": 25},
    "growing_layers": {"temp": [21, 27], "humidity": [50, 65], "ammonia": 25},
    "adult_layers": {"temp": [18, 24], "humidity": [50, 70], "ammonia": 25},
}


def normalize_range(value, low, high, tolerance=0.1):
    if low <= value <= high:
        return 100.0
    span = (high - low) * tolerance
    if value < low:
        return max(0, 100 * (1 - (low - value) / (span if span else 1)))
    else:  # value > high
        return max(0, 100 * (1 - (value - high) / (span if span else 1)))


def compute_EQI(ammonia, humid, temp, alarm_type):
    cfg = ALARM_SETTINGS_EQI.get(alarm_type)
    if not cfg:
        return None
    TI = normalize_range(temp, cfg["temp"][0], cfg["temp"][1])
    HI = normalize_range(humid, cfg["humidity"][0], cfg["humidity"][1])

    max_nh3 = cfg["ammonia"]
    if ammonia <= max_nh3:
        AI = 100.0
    elif ammonia >= 2 * max_nh3:
        AI = 0.0
    else:
        AI = 100 * (1 - (ammonia - max_nh3) / max_nh3)

    return round((TI + HI + AI) / 3, 2)


def get_status(eqi):
    if eqi >= 80:
        return "safe"
    elif eqi >= 50:
        return "warning"
    else:
        return "critical"



# Example dataset
records = [
#   {"date": "2025-07-01", "ammonia": 2.52, "humidity": 91.0, "temperature": 28.1},
]


# Insert each record with ONLY date, eqi, status
for rec in records:
    eqi = compute_EQI(rec["ammonia"], rec["humidity"], rec["temperature"], alarm_type)
    rec_to_insert = {
        "date": rec["date"],
        "eqi": eqi,
        "status": get_status(eqi)
    }
```

Code used in Environmental Index (EQI) Computation

```ino
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <WiFiClientSecure.h>

#define WIFI_SSID "f9ki3"
#define WIFI_PASSWORD "4koSiFyke123*"

#define FAN_RELAY_PIN D1    // Fan
#define LIGHT_RELAY_PIN D2  // Light

const char* fanURL   = "https://smart-poultry-farming-b68a3-default-rtdb.asia-southeast1.firebasedatabase.app/status/fan.json";
const char* lightURL = "https://smart-poultry-farming-b68a3-default-rtdb.asia-southeast1.firebasedatabase.app/status/light.json";

WiFiClientSecure client;

void setup() {
  Serial.begin(115200);

  pinMode(FAN_RELAY_PIN, OUTPUT);
  pinMode(LIGHT_RELAY_PIN, OUTPUT);

  digitalWrite(FAN_RELAY_PIN, HIGH);    // Relay OFF
  digitalWrite(LIGHT_RELAY_PIN, HIGH);  // Relay OFF

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi connected");

  client.setInsecure();  // Insecure connection (use only for testing)
}

void loop() {
  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;

    // 🔁 Fan control
    http.begin(client, fanURL);
    int httpCode = http.GET();
    if (httpCode == HTTP_CODE_OK) {
          String payload = http.getString();
          Serial.println("Fan Firebase Response: " + payload);
          digitalWrite(FAN_RELAY_PIN, (payload.indexOf("on") != -1) ? LOW : HIGH);
        } else {
          Serial.print("Fan HTTP Error: ");
          Serial.println(httpCode);
        }
        http.end();

        // 🔁 Light control
        http.begin(client, lightURL);
        httpCode = http.GET();
        if (httpCode == HTTP_CODE_OK) {
          String payload = http.getString();
          Serial.println("Light Firebase Response: " + payload);
          digitalWrite(LIGHT_RELAY_PIN, (payload.indexOf("on") != -1) ? LOW : HIGH);
        } else {
          Serial.print("Light HTTP Error: ");
          Serial.println(httpCode);
        }
        http.end();

      } else {
        Serial.println("WiFi disconnected");
      }

      delay(2000);  // 2 seconds delay
    }
```

Code in Relay

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
#include <ESP8266WiFi.h>
#include <WiFiClientSecure.h>
#include <ESP8266HTTPClient.h>
#include <WiFiManager.h>
#include <ArduinoJson.h>

// === I2C LCD Setup ===
#define I2C_SDA D2
#define I2C_SCL D1
LiquidCrystal_I2C lcd(0x27, 20, 4);

// === DHT22 Setup ===
#define DHTPIN D4
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

// === MQ-135 Setup ===
#define MQ135_PIN A0

// === Button Pins ===
#define BTN_WIFI D5
#define BTN_SENSOR D6

// === Firebase Settings URL ===
const char *SETTINGS_URL = "https://smart-poultry-farming-b68a3-default-rtdb.asia-southeast1.firebasedatabase.app/settings.json";

// === Alert Type ===
String alert_type = "chick";  // default fallback

// === Timing ===
unsigned long lastSensorRead = 0;
unsigned long lastSettingsFetch = 0;
unsigned long bootAnimStartTime = 0;
unsigned long welcomeStartTime = 0;
bool welcomeScreenShown = false;
 bool bootAnimDone = false;

 // === Previous Sensor Values ===
 float lastTemp = -999.0;
 float lastHum = -999.0;
 float lastAmmonia = -999.0;

 // === Display Control ===
 int displayMode = 1; // 1 = Sensor Display, 2 = WiFi Info
 const char animFrames[] = {'\\', '|', '/'};
 const int frameCount = 3;
 int currentFrame = 0;

 // === Structure for Ranges ===
 struct Range {
   float tempMin, tempMax;
   float humMin, humMax;
   float ammoniaMax;
 };

 Range settings;

 // === Lookup function based on alert_type ===
 Range getSettingsForType(String type) {
   if (type == "chick") return {32, 35, 60, 70, 25};
   if (type == "young_layers") return {28, 32, 55, 65, 25};
   if (type == "growing_layers") return {21, 27, 50, 65, 25};
   if (type == "adult_layers") return {18, 24, 50, 70, 25};
   return {25, 28, 50, 60, 25}; // fallback
 }

 // === Fetch alert_type from Firebase ===
 void fetchSettings() {
   if (WiFi.status() == WL_CONNECTED) {
     WiFiClientSecure client;
     client.setInsecure();
     HTTPClient http;
     if (http.begin(client, SETTINGS_URL)) {
       int httpCode = http.GET();
       if (httpCode == 200) {
         String payload = http.getString();
         StaticJsonDocument<256> doc;
```

Code used in Sensor

APPENDIX D

Communications Letter & Approvals

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

February 28, 2025

Mr. and Mrs. Mercado
Mercado's Poultry Farm
Nasugbu, Batangas

Dear Mr. and Mrs. Mercado,

Good day.

We, the undersigned, are Bachelor of Science in Information Technology students from **Laguna State Polytechnic University – Los Baños Campus**. We are currently conducting our capstone research entitled **"SMART POULTRY FARMING: IoT-Based Environmental Quality Index Monitoring with Machine Learning for Optimizing Chicken Growth in Nasugbu, Batangas."**

In line with this, we respectfully request your permission to gather relevant information, observations, and insights from **Mercado's Poultry Farm**. Our study focuses on monitoring key environmental factors through IoT devices, analyzing these data using machine learning models, and utilizing the results to support the optimization of chicken growth and overall farm productivity.

Your experience and knowledge in poultry farming would greatly contribute to the credibility and success of our research. The practical insights from your farm operations will help us generate meaningful findings that may benefit not only your farm but also poultry farmers within the region.

We sincerely hope for your favorable consideration of this request. Rest assured that any information gathered will be used solely for academic purposes and handled with strict confidentiality.

Thank you very much for your time and support. God bless.

Respectfully yours,

Shenelyn Miranda **Trina Aruta** **Lord Cyril De Luna**
BS in Information Technology
Laguna State Polytechnic University – Los Baños Campus

Noted by:

JOCELYN O. PADALLAN, MIT
Research Adviser
Laguna State Polytechnic University – Los Baños Campus

APPROVAL SHEET

The Capstone Project entitled "**SMART POULTRY FARMING: IOT-BASED ENVIRONMENTAL QUALITY INDEX MONITORING WITH MACHINE LEARNING FOR OPTIMIZING CHICKEN GROWTH IN NASUGBU, BATANGAS"**, prepared and submitted by **Trina P. Aruta, Lord Cyril O. De Luna** and **Shenelyn Miranda** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program, is hereby recommended for approval and acceptance.

JOCELYN O. PADALLAN
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____.

MIKE PHILIP T. RAMOS
Member

SHERWIN B. SAPIN
Member

JHOCO S. MILLANES
Member

JEFFERSON L. LERIOS
Member

Accepted as partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program at the Laguna State Polytechnic University-Los Baños Campus.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, Ed. D.
Chairperson, Research and Development Services

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Dear Respondent,

Good day!

We are 4th-year students of the Bachelor of Science in Information Technology under the College of Computer Studies at Laguna State Polytechnic University – Los Baños Campus. As part of our academic requirement, we are conducting a research study and system development entitled "Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth in Nasugbu, Batangas." This study aims to evaluate the quality and performance of the developed system based on the ISO 25010 Software Product Quality Model, focusing on functionality, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety.

We respectfully seek your honest and professional evaluation as a veterinarian or IT Specialist using the indicators provided in the questionnaire. Your expertise and feedback are essential in determining the system's effectiveness and quality. Rest assured that all information gathered will be treated with utmost confidentiality in accordance with Republic Act No. 10173 (Data Privacy Act of 2012) and will be used solely for academic and research purposes.

Sincerely,
Lord Cyril De Luna, Shenelyn Miranda, Trina Aruta
0991-414-8965

ISO 25010 Software Product Quality Evaluation Rubric

AGE: _____ Gender: _____

Category of the Respondent:

☐ I.T Specialist ☐ Veterinarian Doctor

Rating Scale:

1 – Strongly Disagree 2 – Disagree 3 – Agree 4 – Strongly Agree

A. Perceived Usefulness (PU) *Measures how well the system improves productivity, efficiency, and usefulness to the user.*

ISO Construct	4 – Strongly Agree	3 – Agree	2 – Disagree	1 – Strongly Disagree
Interpretation	The software quality characteristic is **highly evident**; the system performs excellently and meets ISO 25010 standards in all aspects.	The software quality characteristic is **evident**; the system performs well and meets most ISO 25010 standards.	The software quality characteristic is **somewhat evident**; the system meets standards partially and needs improvement.	The software quality characteristic is **not evident**; the system fails to meet ISO 25010 standards and requires

				major improvement.
Functional Suitability	The system fully supports all farm monitoring functions, provides accurate results, and effectively assists in decision-making.	The system performs most functions correctly and supports farm management with minor issues.	The system performs limited functions and sometimes produces inaccurate readings.	The system does not meet functional requirements or support farm operations properly.
Performance Efficiency	The system operates quickly, processes data in real time, and uses resources efficiently.	The system performs efficiently most of the time, with minimal delay or lag.	The system occasionally slows down or consumes too many resources.	The system is slow, unresponsive, or inefficient in processing and resource use.
Compatibility	The system integrates perfectly with sensors, databases, and network devices without conflict.	The system connects properly with most devices and applications with few compatibility issues.	The system sometimes struggles to exchange data with sensors or other software.	The system is not compatible with other devices or systems.
Interaction Capability	The system is user-friendly, engaging, inclusive, provides assistance, and is easy to understand and operate.	The system is generally easy to use and learn with few usability issues.	The system is usable but lacks engagement, feedback, or guidance features.	The system is difficult to learn and operate, with poor user interaction.
Reliability	The system operates consistently, remains available, tolerates faults, and recovers quickly from errors.	The system performs reliably with occasional minor interruptions.	The system experiences moderate failures or slow recovery from errors.	The system is unreliable and frequently fails or loses data.
Security	The system ensures confidentiality, integrity, authenticity, accountability, and resistance to cyber threats.	The system protects user data and ensures secure access most of the time.	The system has limited protection and may experience minor security issues.	The system is vulnerable to security threats and lacks proper data protection.
Maintainability	The system is modular, reusable, easy to analyze, modify, and test without causing new problems.	The system can be maintained and updated with minor effort.	The system is difficult to modify and may cause errors during updates.	The system is not maintainable or easily upgradable.
Flexibility	The system is adaptable, scalable, easy to install, and	The system adapts well to most environments	The system needs adjustments to	The system cannot adapt or be replaced

	replaceable without disrupting operations.	and can be upgraded smoothly.	adapt or scale effectively.	without significant disruption.
Safety	The system operates safely, identifies risks, prevents hazards, and integrates securely with IoT devices.	The system provides adequate safety and alerts users to potential problems.	The system has limited safety measures and delayed alerts.	The system lacks safety features and poses operational risks.

_____ _____

Signature over printed name Date Signed
Respondent

LAGUNA STATE POLYTECHNIC UNIVERSITY- LOS BAÑOS CAMPUS
COLLEGE OF COMPUTER STUDIES

Dear Respondent,

Good day!

We are 4th year students of the Bachelor of Science in Information Technology under the College of Computer Studies, Laguna State Polytechnic University – Los Baños Campus. We are currently conducting a research study and system development entitled **"Smart Poultry Farming: IoT-Based Environmental Quality Index (EQI) Monitoring with Machine Learning for Optimizing Chicken Growth in Nasugbu, Batangas."**

This study aims to determine the **acceptability of the developed technology** through the **Technology Acceptance Model (TAM)**. We request your honest responses using the indicators provided below. Please refer to the description of each score to guide your evaluation. All information collected will be treated with the highest confidentiality in accordance with **Republic Act No. 10173 (Data Privacy Act of 2012)** and will be used solely for academic research purposes.
Thank you for your time and cooperation.

Sincerely,
Lord Cyril De Luna, Shenelyn Miranda, Trina Aruta
0991-414-8965

Technology Acceptance Model Rubric

AGE: _____ Gender: _____

Category of the Respondent:

☐ Farmer ☐ Owner

Rating Scale:

1 – Strongly Disagree 2 – Disagree 3 – Agree 4 – Strongly Agree

A. Perceived Usefulness (PU)

Measures how well the system improves productivity, efficiency, and usefulness to the user.

TAM Construct	4 - Strongly Agree	3 - Agree	2 - Disagree	1 - Strongly Disagree
Perceived Usefulness	The system greatly improves poultry management, enhances chicken growth, and provides useful data for decision-making.	The system helps improve farm productivity and assists in monitoring and management.	The system provides minimal help in monitoring or improving poultry conditions.	The system does not contribute to better farm management or productivity.
Perceived Ease of Use	The system is user-friendly and easy to learn, even without technical experience.	The system is fairly easy to use and can be operated with minimal guidance.	The system is somewhat difficult to understand or navigate.	The system is complicated and hard to use.
Personal Innovation	The user is eager to explore and apply new systems that	The user is open to using modern systems and	The user rarely tries new technologies and prefers traditional methods.	The user avoids new technologies and prefers not

	can enhance poultry production.	innovative technology for farming.		to change current practices.
Social Influence	Positive community feedback and support strongly influence the user to adopt the system.	The user is encouraged by peers and community to try the system.	The user receives limited social encouragement or support.	The user's environment does not support or influence the use of the system.
Information Awareness	The user has a full understanding of how the system works and its benefits.	The user understands most of the system's functions and benefits.	The user has limited awareness of the system's functions and advantages.	The user is not aware or informed about the system's purpose and features.
Perceived Cost	The cost is reasonable and justified by the system's effectiveness in improving poultry growth and reducing mortality.	The cost is acceptable considering the benefits it provides.	The cost is somewhat high compared to the expected benefits.	The cost is not reasonable and discourages adoption.
Willingness to Adopt	The user is fully ready to adopt, use, and promote the system in poultry operations.	The user is willing to use and recommend the system.	The user is hesitant to adopt the system.	The user is not willing to adopt or apply the system.

_____ _____

Signature over printed name Date Signed
Respondent

APPENDIX E

Plagiarism Result

Turnitin Result

SMART POULTRY FARMING: IOT- BASED ENVIRONMENTAL QUALITY INDEX MONITORING WITH MACHINE LEARNING FOR OPTIMIZING CHICKEN GROWTH IN NASUGBU, BATANGAS

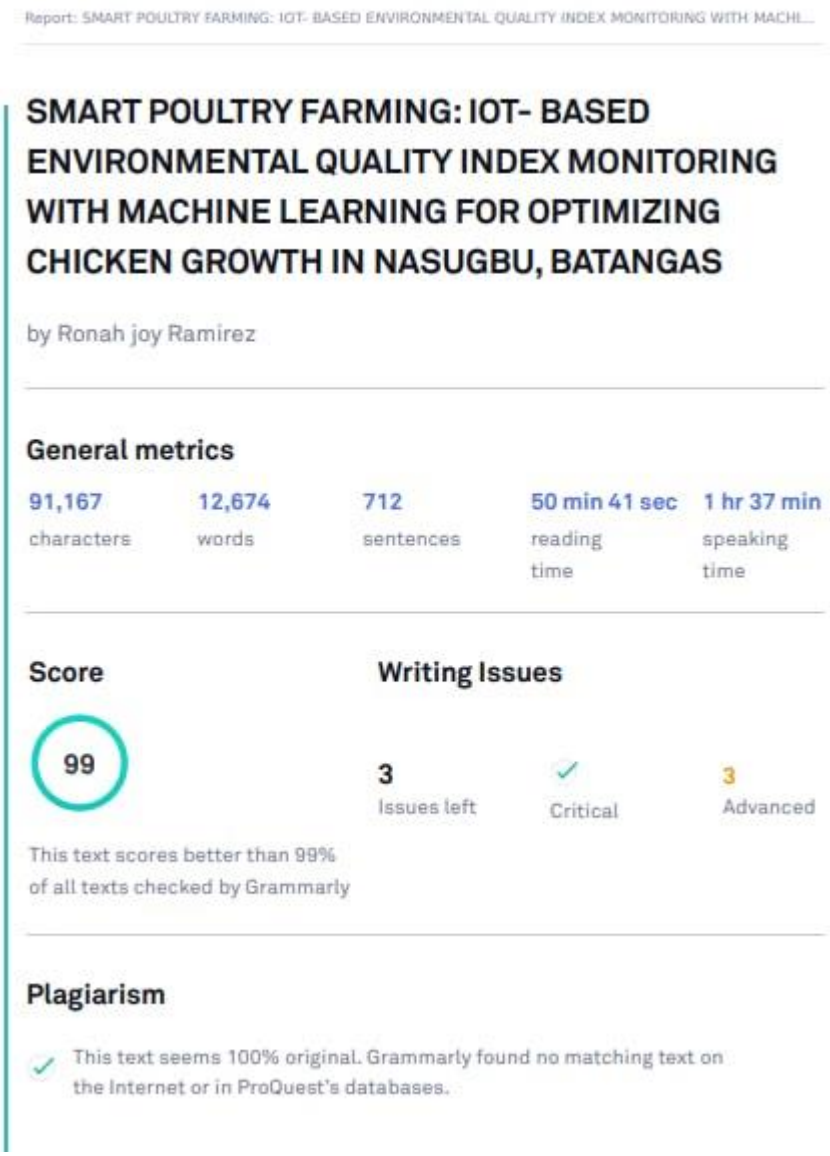
ORIGINALITY REPORT

14%	9%	9%	9%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Holy Name University Student Paper	2%
2	Submitted to Laguna State Polytechnic University Student Paper	2%
3	S. Kannadhasan, R. Nagarajan, Alagar Karthick, V. Kumar Chinnaiyan. "Technological Applications for Smart Sensors - ", Apple Academic Press, 2025 Publication	1%
4	Submitted to University of Makati Student Paper	1%
5	www.coursehero.com Internet Source	<1%
6	Submitted to Our Lady of Fatima University Student Paper	<1%
7	Submitted to Centro Escolar University-Manila Student Paper	<1%
8	"Proceedings of Tenth International Congress on Information and Communication	<1%

Grammarly Result



Report: SMART POULTRY FARMING: IOT- BASED ENVIRONMENTAL QUALITY INDEX MONITORING WITH MACHI...

SMART POULTRY FARMING: IOT- BASED ENVIRONMENTAL QUALITY INDEX MONITORING WITH MACHINE LEARNING FOR OPTIMIZING CHICKEN GROWTH IN NASUGBU, BATANGAS

by Ronah joy Ramirez

General metrics

91,167	12,674	712	50 min 41 sec	1 hr 37 min
characters	words	sentences	reading time	speaking time

Score

99

This text scores better than 99% of all texts checked by Grammarly

Writing Issues

3	✓	3
Issues left	Critical	Advanced

Plagiarism

✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.

APPENDIX F

Curriculum Vitae



TRINA P. ARUTA

trinaaruta15@gmail.com
4 - Service Management Program
09931184529
567, Sucol, Calamba City, Laguna

SKILLS

- Barista
- Excellent Communication Skills
- Computer skills
- Service and Support
- Hardworking and can multitask

EDUCATION

Laguna State Polytechnic University
SERVICE MANAGEMENT PROGRAM
MAITIM, LOS BANOS

Saint Benilde International School
ACCOUNTANCY BUSINESS MANAGEMENT
REAL, CALAMBA

Castor Alviar National High School
MASILI, CALAMBA

Tiyani Elementary Shool
SUCOL, CALAMBA

OBJECTIVE

To improve my knowledge and develop my skills to grow personally and professionally.

EXPERIENCE

MCDONALD'S

SERVICE CREW - 2 YEARS

- Assisted customers promptly and politely, ensuring smooth and friendly service at the counter and drive-thru
- Contributed to maintaining high customer satisfaction, consistently achieving a 100% positive feedback rating
- Suggested improvements to workflow and store processes to help enhance efficiency and service quality

STARBUCKS

BARISTA - PRESENT

- Assisted customers with their orders and questions efficiently and with a welcoming attitude
- Maintained a high customer satisfaction score of 100% by delivering excellent service and well-crafted beverages
- Contributed ideas to streamline store operations and enhance service quality

TRAININGS

- Emersion in Yazaki Torres 2020
- SPES in McDonald's 2023
- TESDA in Creative Web Design 2024
- TESDA in Contact Center Service 2024



LORD CYRIL O. DE LUNA

delunalordcyril4@gmail.com
4 - Service Management Program
09069494917
Purok 5 Acacia Ville, Brgy., Maahas Los Baños, Laguna

SKILLS

- Food safety
- Excellent Communication Skills
- Computer skills
- Daily Maintenance and Cleanliness
- Hardworking and can multitask

EDUCATION

Laguna State Polytechnic University

SERVICE MANAGEMENT PROGRAM
MAITIM, LOS BANOS

Saint Anthony of Los Baños

JARDIN LEONILASUBD., BRGY. MAAHAS,LOS BAÑOS, LAGUNA
ELEMENTARY-SENIOR HIGH SCHOOL

OBJECTIVE

To obtain a position that would best fit my qualifications and develop further my talents and skills for continuous career development.

EXPERIENCE

CALIFORNIA BEACH PIZZA

STAFF

- Assisted customers with their orders and inquiries promptly and politely
- Maintained a customer satisfaction rating by providing great service and quality food
- Offered suggestions to improve kitchen workflow and overall store operations

TRAININGS

- Healthserv Los Baños Medical Service - OJT
- TESDA in Contact Center Service 2023



SHENELYN MIRANDA

mirandashenelyn@gmail.com
4 - Service Management Program
0991-414-8965
0778 Siranglupa, Calamba City

SKILLS

- Computer Hardware Servicing
- Programming Languages:
- Database Management
- Microsoft Dynamics AX (DAX)
- Web Development
- Software Skills
- Proficient in Microsoft Office
- Good Communication Skills
- Good Interpersonal Skills and Positive Attitude
- Reliable, Hardworking, and Dedicated to Work
- Willing to Work Under Pressure

EDUCATION

Laguna State Polytechnic University
SERVICE MANAGEMENT PROGRAM
MAITIM, LOS BANOS

International Electronics and Technical Institute
INFORMATION COMMUNICATION TECHNOLOGY
CALAMBA

North Marie Montessori Academy
SIRANG LUPA, CALAMBA

Mary Belle Montessori School
ASIA II KAPAYAPAAN, CANLUBANG, CALAMBA, LAGUNA

OBJECTIVE

To gain actual experience that would make me competitive enough and would give me the edge. To obtain knowledge that will enhance my skills and intellect which will be beneficial not only to my personality but as well to the company I am serving.

EXPERIENCE

HARVESTER INC.

LABORATORY SENIOR ANALYST 1 AND I.T SPECIALIST

- Delivered timely laboratory analyses and IT support, consistently ensuring accurate results and efficient issue resolution.
- Maintained a 100% satisfaction rating by providing reliable service, clear communication, and strong technical/problem-solving skills.
- Contributed to workflow improvements by suggesting enhancements to both lab procedures and internal IT processes.

FIRST SUMIDEN CIRCUITS INC.

ENCODER

- Responded to data and documentation requests promptly and accurately
- Maintained a 100% accuracy and satisfaction rating through consistent, error-free data entry
- Provided recommendations to improve data processing workflows and internal procedures

TRAININGS

- Introduction in Microcontroller
- TESDA in Creative Web Design 2024
- TESDA in Contact Center Service 2024
- Computer Hardware Servicing TRIMEX NCII

APPENDIX G

Documentation

Owner of Poultry Farm



Survey with the Poultry Farmers



Survey with the Veterinarians



Survey with the I.T Specialist



Defense





APPENDIX H

Publishable Paper

SMART POULTRY FARMING: IOT BASED ENVIRONMENTAL QUALITY INDEX MONITORING WITH MACHINE LEARNING FOR OPTIMIZING CHICKEN GROWTH IN NASUGBU, BATANGAS

Trina P. Aruta, Lord Cyril O. De Luna, Shenelyn Miranda
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info

Article history:

Received by
Revised by
Accepted by

Keywords:

Smart Poultry Farming
Internet of Things (IoT)
Environmental Quality Index
Machine Learning

ABSTRACT

This study was developed to help poultry farmers address the challenges of maintaining healthy farm conditions. Changes in temperature, humidity, and ammonia levels often cause stress and disease among chickens, leading to high mortality and lower productivity. To respond to this problem, an IoT-based system was designed using sensors that record environmental data in real time. The collected data are processed by a microcontroller, which calculates an Environmental Quality Index (EQI). A machine learning model then analyzes the patterns and predicts the most suitable conditions for chicken growth. The system also includes a web-based dashboard that shows real-time readings, trends, and alerts to help farmers make better decisions. The system was tested at Mercado's Poultry Farm in Nasugbu, Batangas, and was evaluated using the Technology Acceptance Model (TAM) and ISO 25010 software quality standards. Findings showed that the system is functional, reliable, and user-friendly, proving its potential to reduce poultry losses and improve farm efficiency.

I. INTRODUCTION

The poultry industry remains a critical sector in Philippine agriculture, supplying a major portion of the country's protein needs. However, many small and medium-scale poultry farms continue to face challenges in maintaining optimal environmental conditions particularly temperature, humidity, and ammonia that directly affect chicken health, growth rate, and survivability. Manual monitoring often proves insufficient, leading to delayed responses, increased stress on the flock, and higher mortality rates. These inefficiencies not only reduce productivity but also impose financial strain on poultry operators.

Recent developments in digital agriculture highlight the potential of integrating Internet of Things (IoT) technologies and machine learning to improve farm management. IoT-based monitoring systems allow continuous tracking of environmental parameters in real time, while predictive analytics support early detection of unfavorable conditions. Such data-driven approaches help farmers make timely corrections, reduce losses, and promote healthier growing environments.

This study aims to address these operational challenges by developing an IoT-based Environmental Quality Index (EQI) monitoring system enhanced with machine learning. Designed for Mercado's Poultry Farm in Nasugbu, Batangas, the system collects real-time sensor data, analyzes environmental trends, and generates EQI scores that reflect overall poultry house conditions. A web-based dashboard provides visualizations, alerts, and predictive insights to support decision-making. The proposed solution focuses on minimizing poultry stress, optimizing growing conditions, and improving farm efficiency through automated, data-informed management.

As the Philippine poultry industry continues to expand, adopting intelligent technologies becomes increasingly essential for sustainable and competitive production. Through this system, the researchers aim to contribute a practical and scalable model that enhances poultry welfare, supports farm operators, and demonstrates the value of integrating IoT and machine learning in modern agricultural practices.[1]

II. RELATED LITERATURE

IoT technologies have become essential in modern poultry management due to their ability to provide real-time environmental monitoring and automated control. Navarro et al. highlighted that IoT platforms using wireless sensors and cloud connectivity significantly improve farm decision-making by continuously tracking temperature, humidity, and gas levels. Their findings demonstrate how IoT reduces risks associated with sudden environmental changes, supporting the direction of this study.[2]

Digambar et al. further explored the combination of IoT and machine learning (ML), showing that predictive analytics can automate feeding, ventilation, and disease-risk detection. They emphasized that systems integrating ML algorithms with sensor networks help reduce human error and improve poultry welfare.[3] This supports the machine learning EQI prediction component of the present project.

Local studies also validate the effectiveness of IoT in Philippine poultry settings. Simon and Palaoag designed a smart poultry prototype that automatically regulates temperature and ammonia levels. Their work proved that small-scale farms can benefit greatly from automated monitoring, reducing manual labor and improving chicken health.[4] Likewise, the Department of Science and Technology's "ChicIoT" project demonstrated the practical success of deploying IoT sensors in poultry houses to monitor temperature, humidity, dust, and light, giving farmers instant access to environmental data through a dashboard.[5]

Air quality monitoring is another critical focus of previous research. Küçüktopçu et al. developed ML-based ammonia prediction models that achieved high accuracy using environmental parameters such as temperature and humidity. Their study underscores the importance of gas monitoring in preventing respiratory issues and mortality, aligning with this project's ammonia-based EQI computation.[6]

Environmental factors also have well-documented effects on poultry health.

Reports from the National Research Council emphasized that temperature, humidity, and ammonia significantly influence feed intake, growth, and immune response. Poor environmental control leads to heat stress, respiratory complications, and increased mortality reinforcing the need for continuous automated monitoring.[7]

III. METHODOLOGY

Project Design

Figure 1 presents the system architecture of the IoT-Based Environmental Quality Index (EQI) Monitoring System, which integrates sensors, a microcontroller, an IoT gateway, and a cloud-enabled dashboard. Environmental parameters such as temperature, humidity, and ammonia are first collected by the sensor layer installed inside the poultry house. These values are processed by the microcontroller and transmitted to the IoT gateway, which forwards the data to the cloud server for storage and real-time synchronization. The web-based dashboard then visualizes the computed EQI, displays historical trends, and generates automated alerts, enabling farm operators to make timely and data-driven decisions for maintaining optimal poultry conditions.

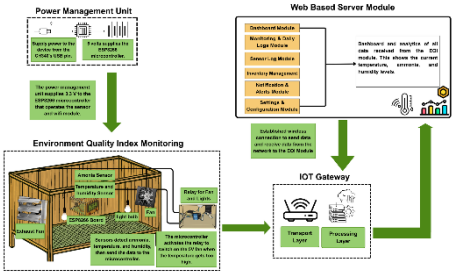


Figure 1. System Architecture

Smart Farming Poultry Prototype

Figure 2 illustrates the overall prototype design of the Smart Poultry Farming system, which integrates IoT-based environmental monitoring with a web-based management platform. The setup includes a poultry enclosure equipped with sensors that measure key environmental parameters such as temperature, humidity, and ammonia concentration. These sensors are connected to a power management module and a microcontroller that collects and transmits real-time data. The gathered readings are then sent to the Web-Based Console Module, where the values are automatically displayed for monitoring and decision-making. Through this interface, users can review historical trends, track current environmental conditions, and receive insights needed to maintain optimal poultry growth and welfare.

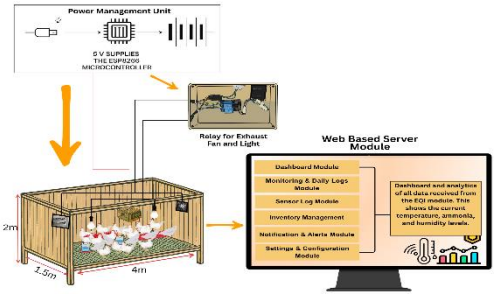


Figure 2. Prototype Design of Smart Poultry Farming

Project Development

The Agile Scrum Methodology, shown in Figure 3, follows an iterative and incremental project development process that emphasizes flexibility, collaboration, and continuous improvement. It begins with defining and analyzing system requirements to establish the project scope, followed by planning sprints that break the work into manageable iterations. Collaborative design activities refining system features and ensure alignment with user needs before implementation, where functional increments are developed. Each cycle concludes with review and monitoring to assess performance, gather feedback, and guide enhancements for subsequ sprints. This methodology enables rapid adaptation to change while maintaining consistent progress throughout the development lifecycle.

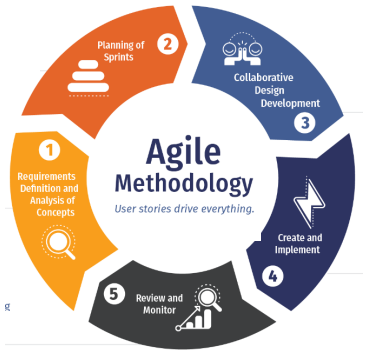


Figure 3. Agile Scrum Methodology

IV. RESULTS AND DISCUSSION

Actual Arduino-based Smart Farming sensor setup displaying temperature, humidity, and ammonia readings.

Figure 4 shows the prototype of the Arduino-based sensor module that we developed for the Smart Poultry Farming System. It has an LCD screen that displays real-time readings, a DHT22 sensor for measuring temperature and humidity, and an MQ137 sensor for detecting ammonia levels inside the

poultry house. We also added manual control buttons so users can switch between modes when needed. These components work together to continuously monitor the environment, which is important for keeping the chickens healthy and growing well. The sensor module collects the data and sends it to the web-based dashboard, where it's processed and analyzed to compute the Environmental Quality Index (EQI).

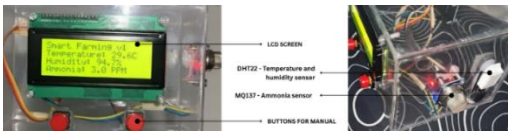


Figure 4. Smart Farming Sensor

Connection of the ESP8266 microcontroller and relay module used to automate environmental controls such as fans and lights based on sensor readings

Figure 5 shows the ESP8266 microcontroller connected to a relay module, which serves as the automation part of the Smart Poultry Farming System. The ESP8266 acts as the main controller that receives data from sensors like the DHT22 and MQ137, processes the readings, and sends commands to control different devices. The relay module works like a switch that turns equipment such as fans and lights on or off depending on the environmental conditions detected by the sensors. With this setup, the system can automatically adjust the temperature, humidity, and air quality inside the poultry house to maintain a comfortable and healthy environment for the chickens. Overall, this setup highlights how the system combines IoT-based monitoring with automatic control to make poultry farm management more efficient.

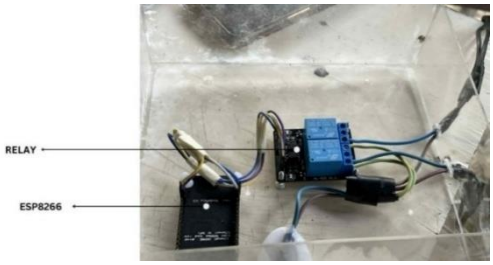


Figure 5. ESP8266 Microcontroller Connected to Relay

Chicken Pen used for system implementation

Figure 6 shows the actual chicken pen where the Smart Poultry Farming System was tested and evaluated. As seen in figure 20 (*a*), the enclosure was made using simple, locally available materials that provided enough ventilation and protection for the chickens. To help control the temperature, four exhaust fans were installed, two on each side of the pen as shown in figure 20 (*b*). Inside the enclosure of figure 20, (*c*) shows the placement of the IoT-based sensors used to track temperature, humidity, and ammonia levels, making sure the data collected was accurate and consistent. Lastly, in figure 20 (*d*) shows how the system could be monitored and controlled remotely through a mobile interface. This setup helped the researchers observe how well the system could automatically maintain the right living conditions for the chickens and support their healthy growth and well-being.

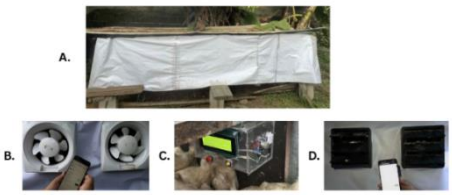


Figure 6. Chicken Pen used in the implementation

Predicting Optimal Environmental Conditions for Chicken Growth Using Pre-trained Machine Learning Models

Figure 7 forecasted environmental metrics for the next seven days, based on the preceding 60 days of actual data, where the red dot marks the last recorded actual value. Figure 21 (*a*) shows humidity, which has been generally decreasing over the past 60 days, settling around 67% recently, and is predicted to remain relatively stable in the 66% to 68% range for the next week. As shown in figure 7 (*b*) illustrates the Temperature, which exhibits significant daily fluctuations, recently measured at about 30.1 ℃, the forecast predicts continued, though slightly muted, oscillations in the 30 ℃ to 31℃ range, with a possible peak near 31℃. In figure 7 (*c*) presents the EQI (Environmental Quality Index), which has been in the 75 to 85 range recently, peaking at 84 at the last actual reading; the prediction shows an expected increase, fluctuating between approximately 83 and 91 over the next seven days. Finally, in figure 7 (*d*) tracks ammonia, which has consistently fluctuated around 2.5 ppm, the prediction shows a sharp, yet uncertain, increase for the next week, with the forecast median rising to a peak around 3.6 ppm

and a wide prediction band ranging from 2.1ppm to 4.0 ppm.

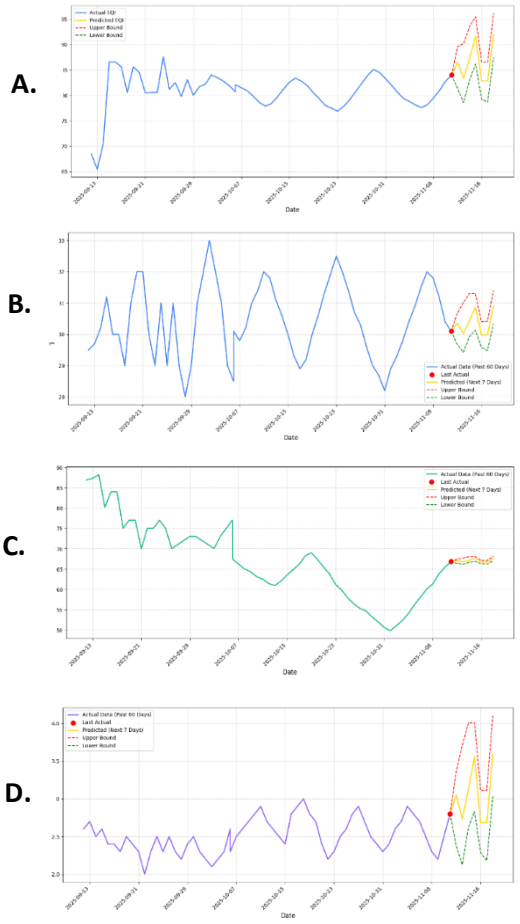
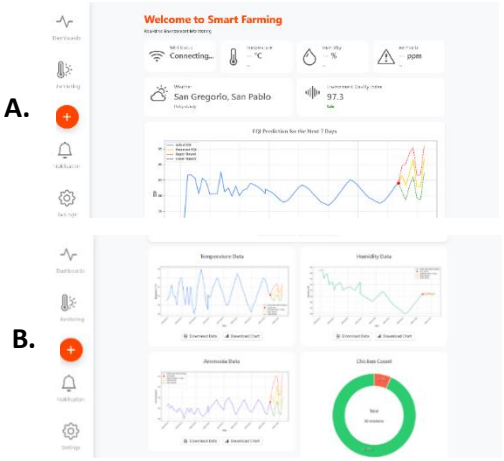


Figure 7. Prediction Trends

Developed a user-friendly web-based system that displays real-time farm data, EQI trends, and uses machine learning to provide smart insights for better

Figures show a clean, modern design with an emphasis on data visualization and accessibility. Both views feature a left-hand navigation sidebar with an orange and white color scheme, containing icons for Dashboards, Monitoring, Notification, and Settings, along with a prominent orange circle with a plus sign for quick actions. Shown in figure 8 (*a*), which appears to be the main dashboard/home screen, provides an overview of real-time status and a large central EQI Prediction for the Next 7 Days graph. Figure 8 (*b*), likely the dedicated "Monitoring" section, is a detailed view that organizes information into four cards: Temperature Data, Humidity Data, Ammonia Data, and Chicken Count, each providing a separate chart (historical data plus 7-day forecast) or a summary visualization (the donut chart), along with buttons to Download Data and Download Chart, maintaining a clear, modular layout focused on comprehensive data analysis.



Results of ISO 25010 and Technology and Acceptance Model.

In this table shows the results of survey answered by five IT experts and five veterinarian doctors. In evaluating and testing the level of acceptability of the developed system, it is demonstrated that the system is ready to be deployed for users, the overall rating of the testing is 3.38 (Strongly Agree), which means as it meets the criteria of ISO 25010.

Table 1. Results of ISO 25010

Indicators	Mean	Standard Deviation	Interpretation
Functional Suitability	3.73	0.11	Strongly Agree
Performance Efficiency	3.47	0.00	Strongly Agree
Compatibility	3.50	0.00	Strongly Agree
Interaction Capability	3.35	0.25	Strongly Agree
Reliability	3.53	0.02	Strongly Agree
Security	3.28	0.07	Strongly Agree
Maintainability	3.16	0.19	Agree
Flexibility	3.08	0.16	Agree
Safety	3.30	0.05	Strongly Agree
Overall Mean	**3.38**	**0.09**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25 Agree; 3.26-4.0: Strongly Agree

Table 2 summarizes the respondents' assessments of the Technology Acceptance Model indicators, all of which received mean scores between 3.04 and 3.55, interpreted as

"Strongly Agree." The highest-rated factor is Perceived Usefulness, indicating that users find the system beneficial, while other indicators such as Ease of Use, Social Influence, Personal Innovation, and Information Awareness also reflect strong positive perceptions. Although Perceived Cost has the lowest means, it still falls within the same interpretation range. Overall, the computed mean of 3.41 shows that respondents strongly accept the system across all TAM dimensions.

Table 2. Overall Results of Technology Acceptance Model

Indicators	Mean	Standard Deviation	Interpretation
Peceived Usefulness	3.55	0.08	Strongly Agree
Perceived Ease of Use	3.39	0.05	Strongly Agree
Personal Innovation	3.46	0.05	Strongly Agree
Social Influence	3.46	0.07	Strongly Agree
Information Awareness	3.54	0.04	Strongly Agree
Perceived Cost	3.04	0.18	Agree
Willingness to Adopt	3.46	0.03	Strongly Agree
Overall Mean	**3.41**	**0.05**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25 Agree; 3.26-4.0: Strongly Agree

V. SUMMARY, CONCLUSION AND RECOMMENDATION

The Smart Poultry Farming system successfully integrated IoT sensors, automated hardware control, and a web-based monitoring platform to improve poultry environmental management. The DHT22, MQ-137, and ESP8266 operated consistently, delivering accurate real-time readings for temperature, humidity, and ammonia. These measurements were processed by the system to compute EQI values, which were clearly visualized on the dashboard along with historical logs and alerts. The relay-driven fan and lighting responded automatically when thresholds were exceeded, demonstrating effective environmental regulation without manual intervention. The ARIMA-based prediction feature further enhanced the system by forecasting upcoming conditions, allowing earlier adjustments. Overall, the system proved functional, responsive, and reliable, showing that low-cost IoT hardware combined with a centralized monitoring platform can support healthier and more stable poultry environments.

References

[1] Dollente, E., & Hanbal, I. (2024). IoT-based Decision Support System for Poultry Care for Ilocos Sur Polytechnic State College. nano-ntp.com.

[2] Navarro, E., Costa, N., & Pereira, A. (2020). A systematic review of IoT solutions for smart farming. *Sensors, 20*(15), 4231. https://doi.org/10.3390/s20154231

[3] Digambar, D., Sharma, R., & Jaiswal, R. (2023). Review on poultry automation using IoT and machine learning. *World Journal of Advanced Research and Reviews, 18*(3), 289-298. https://wjarr.com/sites/default/files/WJARR-2023-1253.pdf

[4] Simon, L. C. F., & Palaoag, T. D. (2024, June). *Prototype of smart poultry for small-scale poultry farmers at Ilocos Norte*. International Journal of Research and Scientific Innovation, 11(V), 442-450.https://www.researchgate.net/publication/381234032_Prototype_of_Smart_Poultry_for_Small-Scale_Poultry_Farmers_at_Ilocos_Norte

[5] Arayata, M. C. (2021, September 3). *DOST allots over P4.7-M for smart poultry growing*. Philippine News Agency. Retrieved from https://www.pna.gov.ph/articles/1152542

[6] Küçüktopçu, E., Cemek, B., & Simsek, H. (2024). Machine learning and wavelet transform: A hybrid approach to predicting ammonia levels in poultry farms. https://www.researchgate.net/publication/384890335_Machine_Learning_and_Wavelet_Transform_A_Hybrid_Approach_to_Predicting_Ammonia_Levels_in_Poultry_Farms?

[7] National Research Council. (1994). *Nutrient Requirements of Poultry: Ninth Revised Edition, 1994*. Washington, DC: The National Academies Press. https://doi.org/10.17226/2114